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MTH 112 — Past Questions & Materials

by Renance

Course: MTH 112

Institution: Federal University of Technology, Minna

Papers in this pack: 1

Levels: 100

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Paper 1 of 1 — MAT112 PQ

Level 100 • First Semester • PastQuestion

MAT 112

SCANNED PAST QUESTIONS, FUTMINNA

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MAT 112

SCANNED PAST QUESTIONS

SCANNED PAST QUESTIONS Mathematics

SCANNED PAST QUESTIONS Mathematics

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Content Developer

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CONIC

SECTIONS

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CIRCLE

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Question 1 (***)

A circle is given parametrically by the equations

$$x = 4 + 3\cos \theta, y = 3 + 3\sin \theta, 0 \leq \theta < 2\pi.$$

- Find a Cartesian equation for the circle.
- Find the equations of the two tangents to the circle, which pass through the origin O.

$$(x - 4)^2 + (y - 3)^2 = 9, y = 0 \text{ and } y = 24$$

7

x

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Question 2 (****)

The points A, B and C have coordinates (6,6), (0,8) and (-2,2), respectively.

- Find an equation of the perpendicular bisector of AB.

The points A, B and C lie on the circumference of a circle whose centre is located at the point D.

- Determine the coordinates of D.

$$MP1-W, y = 3x - 2, D(2, 4)$$

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Question 3 (****)

y

C1 C2

O C3

x

The figure above shows three circles C1, C2 and C3.

The coordinates of the centres of all three circles are positive.

- The circle C1 has centre at (6,6) and touches both the x axis and the y axis.
- The circle C2 has the same size radius as C1 and touches the x axis.
- The circle C3 touches the x axis and both C1 and C2 .

Determine an equation of C3 .

2

2

C2V , (x - 12) + y - 3

2) = 94

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Question 4 (****+)

A circle C has equation

$$x^2 + y^2 + 2x - 4y + 1 = 0$$

The straight line L with equation $y = mx$ is a tangent to C .

Find the possible values of m and hence determine the possible coordinates at which L meets C .

MP1-Y , m = 0, m = 4 , (-1,0) ,

3 (53 , 54)

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Question 5 (****+)

A circle C has equation

$$x^2 + y^2 + 4x - 10y + 9 = 0 .$$

a) Find the coordinates of the centre of C and the size of its radius.

A tangent to the circle T , passes through the point with coordinates (0, -1) and has gradient m , where $m < 0$.

b) Show that m is a solution of the equation

$$2m^2 - 3m - 2 = 0 .$$

The tangent T meets C at the point P .

c) Determine the coordinates of P .

(-2,5) , r = 20 , P (-4,1)

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Question 6 (****+)

A circle has equation

$$x^2 + y^2 - 4x - 2y = 13 .$$

a) Find the coordinates of the centre of the circle and the size of its radius.

The points A and B lie on the circle such that the length of AB is 6 units.

b) Show that $\angle ACB = 90^\circ$, where C is the centre of the circle.

A tangent to the circle has equation $y = k - x$, where k is a constant.

c) Show clearly that

$$2x^2 + 2(1 - k)x + k^2 - 2k - 13 = 0.$$

d) Determine the possible values of k.

$$(2, 1), r = 18, k = -3, 9$$

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Question 7 (****+)

y

B

C

A

O

x

l

The figure above shows a circle whose centre is located at the point C (k , h), where k and h are constants such that $2 < h < 5$.

The points A (3, 2) and B (7, 8) lie on this circle.

The straight line l passes through C and the midpoint of AB.

Given that the radius of the circle is 26, find an equation for l, the value of k and the value of h.

$$MP1-S, 2x + 3y = 25, k = 8, h = 3$$

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Question 8 (****+)

The straight line passing through the points P (1, 9) and Q (5, 5) is a tangent to a circle with centre at C (6, 8).

Determine, in exact surd form, the radius of the circle.

In this question you may not use ...

... a standard formula which determines the shortest distance of a point from
iii a straight line.

... any form of calculus.

$$r = 8$$

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Question 9 (****+)

The straight line with equation $y = 2x - 3$ is a tangent to a circle with centre at the point C (2, -3) .

Determine, in exact surd form, the radius of the circle.

In this question you may not use ...

... a standard formula which determines the shortest distance of a point from

iiii a straight line.

... any form of calculus.

MP1-U , $r = 4\sqrt{5}$

5

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Question 10 (*****)

The figure below shows the circle with centre at C (5,10) and radius 5.

y

$y = 3x$

4

C (5,10)

P

Q

O x

The straight lines with equations, $x = 0$ and $y = 3x$ are tangents to the circle at the

4

points P and Q respectively.

Show that the area of the triangle PCQ is 10 square units.

SP-B , proof

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Question 11 (*****)

A circle passes through the points A (x_1 , y_1) and A (x_2 , y_2) .

Given that AB is a diameter of the circle, show that the equation of the circle is

$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$.

SP-T , proof

Question 12 (*****)

A circle passes through the points P (18,0) and Q (32,0) . A tangent to this circle passes through the point S (0,199) and touches the circle at the point T .

Given that the y axis is a tangent to this circle, determine the coordinates of T

SP-T , (49,31)

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Question 13 (*****)

The circle C1 has equation

$$x^2 + y^2 - 4x - 4y + 6 = 0 .$$

The circle C2 has equation

$$x^2 + y^2 - 10x - 10y + k = 0 ,$$

where k is a constant.

Given that C1 and C2 have exactly two common tangents, determine the range of possible values of k .

SP-Y , $18 < k < 42$

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Question 14 (*****) non calculator

y

A

C D

x

O

B

The figure above shows the circle with equation

$$x^2 + y^2 - 4x - 8y = 205 ,$$

with centre at the point C and radius r .

The straight line AB is parallel to the y axis and has length 24 units.

The tangents to the circle at A and B meet at the point D .

Find the length of AD and hence deduce the area of the kite CADB .

SP-E , AD = 20 , area = 300

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Question 15 (*****)

y

C2

O x

P

C1

The figure above shows a circle C1 with equation

$$x^2 + y^2 - 18x + ky + 90 = 0,$$

where k is a positive constant.

a) Determine, in terms of k, the coordinates of the centre of C1 and the size of its radius.

Another circle C2 has equation

$$x^2 + y^2 - 2x - 2y = 34.$$

b) Given that C1 and C2 are touching externally at the point P, find ...

i. ... the value of k.

ii. ... the coordinates of P.

2

() (

$$C_2: x^2 + y^2 - 2x - 2y = 34, \quad r = k - 9, \quad k = 10, \quad P(29, -13)$$

2 4 5 5)

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Question 16 (*****)

The curve C has equation

$$y = x^2 - 4x + 7.$$

The points P (-1,12) and Q (4,7) lie on C.

The point R also lies on C so that $\angle PRQ = 90^\circ$.

Determine, as exact surds, the possible coordinates of R.

$$C_2: (5 + 221, 17 + 221) \text{ or } (5 - 221, 17 - 221)$$

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Question 17 (*****)

A circle C is centred at (a, a) and has radius a, where a is a positive constant.

The straight line L has equation

$$4x - 3y + 4 = 0.$$

Given that L is tangent to C at the point P, determine ...

a) ... an equation of C.

b) ... the coordinates of P.

You may not use a formula which determines the shortest distance of a point from a straight line in this question.

2 2

SP-W , $(x - 1) + (y - 1) = 1$, P 1 , 8

5 5

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Question 18 (*****)

A curve in the x-y plane has equation

$$x^2 + y^2 + 6x \cos \theta - 18y \sin \theta + 45 = 0 ,$$

where θ is a parameter such that $0 \leq \theta < 2\pi$.

Given that curve represents a circle determine the range of possible values of θ .

$$\text{SP-I , } \{ 14\pi < \theta < 43\pi \} \cup \{ 54\pi < \theta < 74\pi \}$$

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Question 19 (*****)

The circles C1 and C2 have respective equations

$$(x + 1)^2 + (y + 2)^2 = 94 \text{ and } (x - 5)^2 + (y - 6)^2 = 36 .$$

The point P lies on C2 so that the distance of P from C1 is least.

Determine the exact coordinates of P .

SP-P , P 7 , 6

5 5

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Question 20 (*****)

The straight line L and the circle C , have respective equations

$$L : y = \lambda (x - a) + a \lambda^2 + 1 \text{ and } C : x^2 + y^2 = 2ax ,$$

where a is a positive constant and λ is a parameter.

Show that for all values of λ , L is a tangent to C .

SPX-L , proof

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Question 21 (*****)

The straight line with equation

$$y = t(x - 2) ,$$

where t is a parameter,

crosses the circle with equation

$$x^2 + y^2 = 1$$

at two distinct points A and B .

a) Show that the coordinates of the midpoint of AB are given by

$$\left(\frac{2t^2 + 2t}{1 + t^2}, -\frac{2t}{1 + t^2} \right)$$

$$\left(\frac{2t^2 + 2t}{1 + t^2}, -\frac{2t}{1 + t^2} \right)$$

$$\left(\frac{2t^2 + 2t}{1 + t^2}, -\frac{2t}{1 + t^2} \right)$$

$$\left(\frac{2t^2 + 2t}{1 + t^2}, -\frac{2t}{1 + t^2} \right)$$

$$\left(\frac{2t^2 + 2t}{1 + t^2}, -\frac{2t}{1 + t^2} \right)$$

b) Hence show that the locus of M as t varies is a circle, stating its radius and the coordinates of its centre.

$$(x - 1)^2 + y^2 = 1$$

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Question 22 (*****)

Two parallel straight lines, L1 and L2, have respective equations

$$y = 2x + 5 \text{ and } y = 2x - 1.$$

L1 and L2, are tangents to a circle centred at the point C.

A third line L3 is perpendicular to L1 and L2, and meets the circle in two distinct points, A and B.

Given that L3 passes through the point (9, 0), find, in exact simplified surd form, the coordinates of C.

$$\text{SP-Q, } C \left(\frac{1}{5} + \frac{6}{1}, \frac{1}{5} + \frac{6}{1} \right)$$

$$\left(\frac{1}{5}, \frac{1}{5} \right)$$

$$\left(\frac{1}{5}, \frac{1}{5} \right)$$

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Question 23 (*****)

Two circles, C1 and C2, have respective radii of 4 units and 1 unit and are touching each other externally at the point A.

The coordinate axes are tangents to C1, whose centre P lies in the first quadrant.

The x-axis is a tangent to C2, whose centre Q also lies in the first quadrant.

The straight line l1, passes through P and Q, and meets the x-axis at the point R.

The straight line l2 has negative gradient, passes through R and is a common tangent to C1 and C2.

Determine, in any order and in exact form where appropriate, the coordinates of A, the length of PR and an equation of l2.

$$\left(\frac{1}{5}, \frac{1}{5} \right)$$

$$\text{SPX-J, } A \left(\frac{36}{5}, \frac{8}{5} \right), PR = 20, 24x + 7y = 224$$

$$\left(\frac{36}{5}, \frac{8}{5} \right)$$

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Question 24 (*****)

A family of circles is passing through the points with coordinates (2,1) and (4,5)

Show that the equation of every such circle has equation

$$x^2 + y^2 + 2x(2k - 9) + 2ky = 6k - 41,$$

where k is a parameter.

SP-H , proof

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Question 25 (*****)

Three circles, C1 , C2 and C3 , have their centres at A , B and C , respectively, so that

$AB = 5$, $AC = 4$ and $BC = 3$.

The positive x and y axis are tangents to C1 .

The positive x axis is a tangent to C2 .

C1 and C2 touch each other externally at the point M .

Given further that C3 touches externally both C1 and C2 , find, in exact simplified form, an equation of the straight line which passes through M and C .

$$5y - 106x + 36 + 306 = 0$$

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Question 26 (*****)

The point A (6, -1) lies on the circle with equation

$$x^2 + y^2 - 4x + 6y = 7.$$

The tangent to the circle at A passes through the point P , so that the distance of P from the centre of the circle is 65 .

Another tangent to the circle, at some point B , also passes through P .

Determine in any order the two sets of the possible coordinates of P and B .

$$MP1-T, P(3,5) \cap B(-1, -18)$$

$$13, 13) \cup P(9, -7) \cap B(13, 13)$$

$$30, -97$$

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PARABOLA

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Question 1 (**)

()

The general point $P(9t^2, 18t)$, where t is a parameter, lies on the parabola with Cartesian equation

$$y^2 = 36x.$$

a) Show that the equation of a tangent at the point P is given by

$$x - ty + 9t^2 = 0.$$

The tangent to the parabola $y^2 = 36x$ at the point $Q(1, 6)$ crosses the directrix of the parabola at the point D .

b) Find the coordinates of D .

$$D(-9, -24)$$

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Question 2 (**)

y

$$y^2 = 28x$$

$$P(7, 14)$$

O

x

F Q

not drawn to scale

The figure above shows the graph of the parabola with equation

$$y^2 = 28x, \quad x \in \mathbb{R}, \quad x \geq 0.$$

The point $P(7, 14)$ lies on the parabola.

a) Find an equation of the normal to the parabola at P .

This normal meets the x axis at the point Q and F is the focus of the parabola.

b) Determine the area of the triangle PQF .

$$x + y = 21, \quad \text{area} = 98$$

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Question 3 (**+)

A parabola H has Cartesian equation

$$y^2 = 12x, \quad x \geq 0.$$

()

The point $P(3t^2, 6t)$, where t is a parameter, lies on H .

a) Show that the equation of a tangent to the parabola at P is given by

$$y = x + 3t^2.$$

The tangent to the parabola at P meets the y axis at the point Q and the point S is the focus of the parabola.

b) Show further that ...

i. ... PQ is perpendicular to SQ .

2 (

ii. ... the area of the triangle PQS is $9t + t^2$.)

FP3-M , proof

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Question 4 (***)

()

The general point P $(3t^2, 6t)$ lies on a parabola.

a) Show that the equation of a tangent at P is given by

$$y = x + 3t^2.$$

The point Q $(-12, 9)$ does not lie on the parabola.

b) Find the equations of the two tangents to the parabola which pass through Q and deduce the coordinates of their corresponding points of tangency.

$$x + y + 3 = 0, (3, -6), 4y = x + 48, (48, 24)$$

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Question 5 (***)

()

The general point P $(2t^2, 4t)$ lies on a parabola.

a) Show that the equation of a normal at P is given by

$$y + tx = 4t + 2t^3.$$

b) Find the equation of each of the three normals to the parabola that meet at the point with coordinates $(12, 0)$.

$$y = 0, y + 2x = 24, y - 2x = -24$$

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Question 6 (***)

A parabola is defined parametrically by

$$x = at^2, y = 2at, t \in \mathbb{R},$$

where a is a positive constant and t is a parameter.

a) Show that an equation of a normal to the parabola at the point P , where $t = p$,

$p \neq 0$, is given by

$$y + px = 2ap + ap^3.$$

The normal at P meets the x axis at the point Q.

The point R, lies on the x axis, so that PR is parallel to the y axis.

b) Show that the distance QR remains constant for all values of the parameter, and state this distance.

$$QR = 2a$$

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Question 7 (***)

()

The point P $(4p^2, 8p)$, $p \geq 0$, lies on the parabola with equation

$$y^2 = 16x, x \geq 0.$$

a) Show that the equation of the tangent to the parabola at P is given by

$$yp = x + 4p^2.$$

The tangent to the parabola at P meets the directrix of the parabola at the point A and the x axis at the point B. The point F is the focus of the parabola.

b) Given that the y coordinate of A is 42, find the area of the triangle FBP.

5

$$FP3-J, \text{ area} = 290$$

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Question 8 (***)

()

The point P $(3p^2, 6p)$, $p > 0$, lies on the parabola with equation

$$y^2 = 12x, x \geq 0.$$

a) Show that the equation of the normal to the parabola at P is given by

$$y + px = 6p + 3p^3.$$

The normal to the parabola at P meets the x axis at the point A and the directrix of the parabola at the point B. The point C is the point of intersection of the directrix of the parabola with the x axis.

b) Given that the coordinates of A are $(\frac{223}{3}, 0)$, find as an exact simplified fraction the area of the triangle BCP.

403

area =

27

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Question 9 (***)

A parabola has its focus at T (1,1) and its directrix has equation $x - 7 = 0$.

a) Find an equation for the parabola.

b) Sketch the parabola and show that its latus rectum is 12 units.

$$(y - 1)^2 = 12(4 - x)$$

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Question 10 (***)

A parabola is given parametrically by the equations

$$x = 4 - t^2, y = 1 - t, t \in \mathbb{R}.$$

a) Show that an equation of the normal at the general point on the parabola is

$$y + 2tx = 1 + 7t - 2t^3.$$

The normal to parabola at P (3,0) meets the parabola again at the point Q.

b) Find the coordinates of Q.

Q 7,5

4 2

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Question 11 (***)

The point P has coordinates

()

P at $2a, 2at$,

where a is a positive constant and t is a real parameter.

The point P traces a parabola.

a) Show that the equation of a normal at the point P is given by

$$y + tx = 2at + at^3.$$

b) Show that the straight line with equation

$$y = 2x - 12a,$$

is the only normal to the parabola passing through the point Q (3a, -6a).

c) Determine the coordinates of the two points of intersection between this normal and the parabola, indicating clearly which point of intersection represents the point of normality.

(9a, 6a), normal at (4a, -4a)

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Question 12 (***)

A straight line L is a tangent to the parabola with equation

$$y^2 = Ax$$

where A is a positive constant.

Given that L does not pass through the origin O , show that the product of the gradient and the y intercept of L equals the x coordinate of the focus of the parabola.

proof

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Question 13 (***)

y

$$y^2 = 4ax$$

P

Q

O x

F

L

The figure above shows the sketch of the parabola with equation

$$y^2 = 4ax,$$

where a is a positive constant.

The straight line L and the point F are the directrix and the focus of the parabola, respectively.

The point $P(8, y)$, $y > 0$, lies on the parabola. The point Q lies on L , so that QP is parallel to the x axis.

Given further that $PF = 10$, determine the area of the triangle FPQ .

$$\text{area} = 40$$

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Question 14 (***)

()

The point T at $2, 2at$, lies on the parabola with equation

$$y^2 = 4ax, a > 0, x \geq 0.$$

a) Show clearly that an equation of a normal to the parabola at the point

()

$P(ap^2, 2ap$, $p \neq 0$, can be written as

$$y + px = 2ap + ap^3.$$

The normal at P meets the x axis at the point Q .

The midpoint of PQ is M .

b) Show that the locus of M as p varies is the parabola with equation

$$y^2 = a (x - a) .$$

c) Find the coordinates of the focus of $y^2 = a (x - a)$.

$$(\frac{5}{4} a, 0)$$

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Question 15 (****)

A parabola has its focus at the point with coordinates (4,1) and its directrix has equation $x = -1$.

Determine a Cartesian equation of the parabola.

$$(y - 1)^2 = 10 (x - 32)$$

Question 16 (****)

A straight line L passes through the focus of the parabola with equation

$$y^2 = 12 x .$$

Given further that the shortest distance of L from the origin O is 12 , determine an

5

equation for L .

$$3 y = 4 x - 12$$

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Question 17 (****)

A parabola P has Cartesian equation

$$y^2 - 4 y - 8 x + 28 = 0 .$$

a) Determine ...

i. ... the coordinates of the vertex of P .

ii. ... the coordinates of the focus of P .

iii. ... the equation of the directrix of P .

The line with equation

$$y = mx + 1 , \text{ where } m \text{ is a constant,}$$

is a tangent at some point of P .

b) Find the possible values of m .

vertex at (3,2) , focus at (5,2) , directrix $x = 1$, $m = -2 , 1$

3

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Question 18 (****)

$$C: y = x^2$$

$$y = 12$$

L: A

P

O

x

Q

$$12$$

The figure above shows the parabola C with equation $y = x^2$.

$$12$$

The dotted line in the figure is the reflection of C in the line $y = x$.

a) Find the exact distance between the focus of C and the focus of its reflection.

The parabola intersects its reflection at the origin and at the point A.

b) Determine the coordinates of A.

The straight line L is a common tangent to both C and the reflection of C.

L touches C at the point P and the reflection of C at the point Q.

c) Determine the coordinates of P and Q.

$$P(3, 3), A(12, 12), P(-6, 3), Q(3, -6)$$

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Question 19 (****)

()

The point T at $(2, 2a)$, lies on the parabola with equation

$$y^2 = 4ax, a > 0, x \geq 0.$$

a) Show clearly that an equation of a normal to the parabola at the point

()

$P(ap^2, 2ap, p \neq 0)$, can be written as

$$y + px = 2ap + ap^3.$$

()

The normal at P re-intersects the parabola at the point Q $(aq^2, 2aq)$.

b) Show that

$$p^2 + 2$$

$$q = -\frac{1}{p}.$$

p

c) Given that the midpoint of PQ has coordinates $(5a, -2a)$, find the value of p.

$$p = 1$$

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Question 20 (****)

()

The point $P(2t^2, 4t)$ lies on the parabola with equation

$$y^2 = 8x, x \geq 0.$$

a) Show that an equation of a tangent to the parabola at P , can be written as

$$yt = x + 2t^2, t \neq 0.$$

The tangent to the parabola at P meets the y axis at the point A . The perpendicular bisectors of the straight line segments AP and OA , meet at the point B .

b) Find the coordinates of B , in terms of t .

c) Sketch the locus of B as t varies.

$$B(t^2 + 2, t)$$

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Question 21 (****)

A parabola C has Cartesian equation

$$y^2 + 4y - 16x + 36 = 0.$$

a) Describe the transformations that map the graph of the curve with equation

$$y^2 = 16x$$

onto the graph of C .

b) Determine the coordinates of the focus of C .

c) Show that ...

()

i. ... the point $P(4t^2 + 2, 8t - 2)$ lies on the parabola.

ii. ... the equation of a tangent to the parabola at the point P , is

$$yt = x + 4t^2 - 2t - 2.$$

d) Hence show that the gradients of the two tangents from the origin to the parabola have gradients -2 and 1 .

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

translation by vector $\begin{pmatrix} 6 \\ -2 \end{pmatrix}$

$$\begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

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Question 22 (****)

Sketch the graph of the parabola with equation

$$y(y - 2) = 4x + 3.$$

The sketch must include the coordinates of any intersections with the axes and the coordinates of the vertex of the parabola.

graph

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Question 23 (****)

()

The point P is $(2p, 2ap)$, where p is a parameter, lies on the parabola, with Cartesian equation

$$y^2 = 4ax,$$

where a is a positive constant.

The point F is the focus of the parabola and O represents the origin.

The straight line which passes through P and F meets the directrix of the parabola at

the point Q, so that the area of the triangle OPQ is $\frac{a^2}{4}$.

4

Show that one of the possible values of p is 3 and find in exact surd form the other 2 possible values.

8 (

FP3-M, $p = 1 \pm \sqrt{89}$)

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Question 24 (****+)

A parabola P has focus S $(6, 0)$ and directrix the line $x = 0$.

a) Show that a Cartesian equation for P is $y^2 = 12(x - 3)$.

b) Verify that the parametric equations of P are

$$x = 3t^2 + 3, y = 6t.$$

c) Show that the equation of the tangent at the point Q $(3q^2 + 3, 6q)$ is

$$qy + 3 = x + 3q^2$$

The diagram below shows the parabola and its tangents at the points Q and R. The point R lies on the parabola so that QSR is a straight line.

y

Q

$$y^2 = 12(x - 3)$$

x

O S

R

d) Show that the tangents to the parabola at Q and at R , meet on the y axis.

proof

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Question 25 (****+)

A parabola C has parametric equations

$$x = -2t^2, y = 4t$$

a) Determine the coordinates of the focus and the equation of directrix of C .

()

b) Show that an equation of the tangent to C , at the general point T $-2t^2, 4t$ is

$$yt + x = 2t^2$$

c) By considering the product of the roots of a suitable quadratic equation, show that any two tangents that meet on the directrix of C , are perpendicular.

$$F(-2, 0), x = 2$$

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Question 26 (****+)

()

The point P at $2, 2at$ lies on the parabola with equation

$$y^2 = 4ax,$$

where a is a positive constant and t is a real parameter.

()

The normal to the parabola at P , meets the parabola again at the point Q as $2, 2as$.

Show that

$$16a^2s^3$$

4 (

$$PQ = t^2 + 1) .$$

t

proof

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Question 27 (****+)

A parabola C has Cartesian equation

$$y^2 = 4x, x \in \mathbb{R}, x \geq 0 .$$

() ()

The points $P(p^2, 2p)$ and $Q(q^2, 2q)$ are distinct and lie on C .

The tangent to C at P and the tangent to C at Q meet at $R(-1, 15/4)$.

Calculate as an exact simplified fraction the area of the triangle PQR .

area = $\frac{4913}{128}$

128

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Question 28 (****+)

A parabola C has Cartesian equation

$$y^2 = 4ax,$$

where a is a positive constant.

() ()

The points $P(ap^2, 2ap)$ and $Q(aq^2, 2aq)$ are distinct and lie on C .

The tangent to C at P and the tangent to C at Q meet at the point R .

Show that

$$2$$

$$SR = SP \cdot SQ,$$

where S is the focus of the parabola.

SPX-F, proof

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Question 29 (****+)

A parabola has Cartesian equation

$$y = x^2, x \in \mathbb{R}.$$

A chord of the parabola is defined as the straight line segment joining any two distinct points on the parabola.

Find the equation of the locus of the midpoints of parallel chords of the parabola whose gradient is m .

$$SPX-D, x = \frac{1}{m}$$

$$2$$

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Question 30 (*****)

A parabola has Cartesian equation

$$y^2 = 12x, x \geq 0.$$

The point P lies on the parabola and the point Q lies on the directrix of the parabola

so that PQ is parallel to the x axis.

The area of the triangle PQF is $8\frac{2}{3}$ square units, where the point F represents the focus of the parabola.

Determine the coordinates of P, given further that the y coordinate of P is a positive integer.

3 ()

SPX-I, P 4, 4

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Question 31 (*****)

()

The point $P(2p, p^2)$, where p is a parameter, lies on the parabola, with Cartesian equation

$$x^2 = 4y.$$

The point F is the focus of the parabola and O represents the origin.

The tangent to the parabola at P forms an angle θ with the positive x axis.

The straight line which passes through P and F forms an acute angle ϕ with the tangent to the parabola at P.

Show that $\theta + \phi = \frac{1}{2}\pi$ and hence state the coordinates of P if $\theta = \phi$.

2

SPX-I, P (2,1)

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Question 32 (*****)

A parabola has Cartesian equation

$$y = \frac{1}{2}x^2, \quad x \in \mathbb{R}.$$

2

The points P and Q both lie on the parabola so that POQ is a right angle, where O is the origin.

The point M represents the midpoint of PQ.

Show that as the position of P varies along the parabola, the locus of M is the curve with equation

$$y = x^2 - 2.$$

FP3-T, proof

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Question 33 (*****)

The cubic equation

$$x^3 + px + q = 0,$$

has 2 distinct real roots.

a) Show that $27q^2 + 4p^3 < 0$.

A parabola has Cartesian equation

$$y = x^2, x \in \mathbb{R}.$$

Three distinct normals to this parabola pass through the point, which does not lie on the parabola, whose coordinates are (a, b) .

b) Show further that

$$2$$

$$2 \left(\frac{b}{a} \right)$$

$$b > 1 + 3 \frac{1}{a^3}.$$

$$4$$

SPX-M, proof

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Question 34 (*****)

A parabola is given parametrically by

$$x = 1 - t^2, y = 2t, t \in \mathbb{R}.$$

$$3 \frac{3}{4}$$

The normal to the parabola at the point P meets the parabola again at the point Q.

Show that the minimum value of PQ is $\frac{1}{2}$.

FP3-S, proof

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ELLIPSE

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Question 1 (**)

An ellipse has Cartesian equation

$$\frac{x^2}{16} + \frac{y^2}{9} = 1.$$

$$+ = 1.$$

$$16 \frac{9}{16}$$

a) Find the coordinates of its foci.

b) Sketch the ellipse.

$(\pm 7, 0)$

Question 2 (**)

An ellipse has parametric equations

$$x = 4\cos\theta, y = 7\sin\theta, 0 \leq \theta < 2\pi.$$

a) Find the coordinates of its foci.

b) Sketch the ellipse.

$(\pm 3, 0)$

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Question 3 (**)

25

An ellipse has a focus at $(4, 0)$ and the associated directrix has equation $x =$.

4

Determine a Cartesian equation of the ellipse.

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

$$+ = 1$$

25 9

Question 4 (**)

$$\frac{x^2}{4} + \frac{y^2}{1} = 1$$

$$+ y^2 = 1$$

4

The ellipse with Cartesian equation above and a parabola with vertex at the origin share the same focal point.

Find the possible Cartesian equation for the parabola.

$$y^2 = \pm 48x$$

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Question 5 (**+)

An ellipse E is given parametrically by the equations

$$x = \cos t, y = 2\sin t, 0 \leq t < 2\pi.$$

a) Show that an equation of the normal to E at the general point P $(\cos t, 2\sin t)$ can be written as

$$2y \cos t - x \sin t = 3$$

$$- = 3$$

$$\sin t \cos t$$

The normal to E at P meets the x axis at the point Q . The midpoint of PQ is M .

b) Find the equation of the locus of M as t varies.

$$x^2 + y^2 = 1$$

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Question 6 (***)

$$x^2 + y^2$$

$$= 1.$$

$$9 + 4$$

The ellipse with the Cartesian equation given above, has foci S and S'.

- Find the coordinates S and S'.
- Sketch the ellipse.
- Show that for every point P on this ellipse,
 $SP + S'P = 6$.

$$(\pm 5, 0)$$

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Question 7 (***)

An ellipse E has Cartesian equation

$$x^2 + y^2$$

$$= 1.$$

$$289 + 64$$

- Find the coordinates of the foci of E, and the equations of its directrices.
- Sketch the ellipse.

The point P lies on E so that PS is vertical, where S is the focus of the ellipse with positive x coordinate.

- Show that the tangent to the ellipse at the point P meets one of the directrices of the ellipse on the x axis.

$$(\pm 15, 0), x = \pm 289$$

$$15$$

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Question 8 (***)

The point P ($5\cos\theta$, $4\sin\theta$) lies on the ellipse E with Cartesian equation

$$16x^2 + 25y^2 = 400.$$

- Find the coordinates of the foci of E.
- Show that an equation of the normal to the ellipse at P is
 $4y\cos\theta - 5x\sin\theta + 9\sin\theta\cos\theta = 0.$

The normal to the ellipse intersects the coordinate axes at the points A and B , and the point M is the midpoint of AB .

c) Show that the locus of M , as θ varies, is the ellipse with equation

$$100x^2 + 64y^2 = 81 .$$

$$(\pm 3, 0)$$

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Question 9 (***)

The ellipse E has parametric equations

$$x = a \cos \theta , y = b \sin \theta , 0 \leq \theta < 2\pi$$

where a and b are positive constants.

a) Show that an equation of the tangent at a general point on E is

$$bx \cos \theta + ay \sin \theta = ab .$$

This tangent to E intersects the coordinate axes at the points A and B , and the point M is the midpoint of AB .

b) Find a Cartesian locus of M , as θ varies.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2}$$

$$= 1$$

$$= 1$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2}$$

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Question 10 (***)

An ellipse has Cartesian equation

$$\frac{x^2}{16} + \frac{y^2}{4} = 1 .$$

$$\frac{x^2}{16} + \frac{y^2}{4} = 1 .$$

$$\frac{x^2}{16} + \frac{y^2}{4} = 1 .$$

The general point P ($4 \cos \theta$, $2 \sin \theta$) lies on the ellipse.

a) Show that the equation of the normal to the ellipse at P is

$$2x \sin \theta - y \cos \theta = 6 \sin \theta \cos \theta .$$

The normal to the ellipse at P meets the x axis at the point Q and O is the origin.

b) Show clearly that as θ varies, the maximum area of the triangle OPQ is $4 \frac{1}{2}$.

$$4 \frac{1}{2}$$

proof

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Question 11 (***)

An ellipse has Cartesian equation

$$2x^2 + 3y^2 - 4x + 12y + 8 = 0.$$

Determine ...

- a) ... the coordinates of the centre of the ellipse.
- b) ... the eccentricity of the ellipse.
- c) ... the coordinates of the foci of the ellipse.
- d) ... the equations of the directrices of the ellipse.

$$(1, -2), e = \frac{3}{4}, (0, -2), (2, -2), x = -2, x = 4$$

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Question 12 (***)

An ellipse has Cartesian equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2}$$

$$= 1,$$

$$a > b$$

$$a > 0$$

where a is a positive constant.

The straight line with equation $x = 8$ is a directrix for the ellipse.

Determine the possible set of coordinates for the foci of the ellipse.

$$(\pm 2, 0) \text{ or } (\pm 6, 0)$$

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Question 13 (***)

An ellipse has equation

$$x^2 - 8x + 4y^2 + 12 = 0.$$

- a) Determine the coordinates of the foci and the equations of the directrices of the ellipse.

A straight line with positive gradient passes through the origin O and touches the ellipse at the point A.

- b) Find the coordinates of A.

$$(\frac{1}{2}, \frac{1}{2})$$

$$F_1(-3, 0), F_2(3, 0), x = -4, x = 4, \frac{1}{2}, \frac{1}{2}$$

$$\frac{1}{2}, \frac{1}{2}$$

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Question 14 (***)

A point P lies on the ellipse with Cartesian equation

$$\frac{x^2}{64} + \frac{y^2}{16} = 1.$$

$$\frac{x^2}{64} + \frac{y^2}{16} = 1.$$

The point Q is the foot of the perpendicular from the point P to the straight line with equation $x = 10$.

a) Sketch in the same diagram the ellipse, the straight line with equation $x = 12$ and the straight line segment PQ.

The point M is the midpoint of PQ.

b) Determine a Cartesian equation for the locus of M as the position of P varies, further describing this locus geometrically.

2

$$\text{FP3-N, } (x - 5)^2 + y^2 = 16$$

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Question 15 (***)

An ellipse E has Cartesian equation

$$\frac{x^2}{16} + \frac{y^2}{4} = 1.$$

$$\frac{x^2}{16} + \frac{y^2}{4} = 1.$$

$$\frac{x^2}{16} + \frac{y^2}{4} = 1.$$

a) Show that an equation of the tangent to E at the point A $(4\cos\theta, 2\sin\theta)$ is given by

$$2y\sin\theta + x\cos\theta = 4.$$

The point B $(4\cos\theta, 4\sin\theta)$ lies on the circle with Cartesian equation

$$x^2 + y^2 = 16.$$

The tangent to the circle at the point B meets the tangent to the ellipse at the point A at the point P.

b) Determine the coordinates of P, in terms of θ .

c) Describe mathematically the locus of P as θ varies.

$$\text{FP3-Q, } P(4\sec\theta, 0), \text{ the } x \text{ axis, so that } x \in (-\infty, -4] \cup [4, \infty)$$

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Question 16 (****)

An ellipse has Cartesian equation

$$\frac{x^2}{4} + \frac{y^2}{1} = 1.$$

$$\frac{x^2}{4} + \frac{y^2}{1} = 1.$$

2

A straight line L has equation $y = mx + c$, where m and c are positive constants.

a) Show that the x coordinates of the points of intersection between L and the ellipse satisfy the equation

$$(2m^2 + 1)x^2 + 4mcx + 2(c^2 - 1) = 0.$$

b) Given that L is a tangent to the ellipse, show that $c^2 = 2m^2 + 1$.

The line L meets the negative x axis and the positive y axis at the points X and Y respectively. The point O is the origin.

c) Find the area of the triangle OXY, in terms of m

d) Show that as m varies, the minimum area of the triangle OXY is 2.

e) Find the x coordinate of the point of tangency between the line L and the ellipse when the area of the triangle is minimum.

1

$$\text{area} = m + 1, x = -1$$

2m

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Question 17 (****)

The point P (x, y) lies on an ellipse with foci at A (2,0) and B (6,0) .

Given further that

$$AP + BP = 10,$$

determine a simplified Cartesian equation for the ellipse, giving the final answer in the form

$$f(x, y) = 1.$$

$$(x - 4)^2 + y^2 = 1$$

25 21

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Question 18 (****)

An ellipse has Cartesian equation

$$x^2 + y^2$$

$$+ = 1.$$

25 9

The general point P ($5\cos \theta, 3\sin \theta$) lies on the ellipse.

a) Show that the equation of the normal to the ellipse at P is

$$3y \cos \theta - 5x \sin \theta + 16 \sin \theta \cos \theta = 0.$$

The normal to the ellipse at P meets the x axis at the point Q and R is one of the

foci of the ellipse.

b) Show clearly that

QR

=e,

PR

where e is the eccentricity of the ellipse.

proof

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Question 19 (****+)

An ellipse is given, in terms of a parameter θ , by the equations

$$x = 3 \cos \theta, y = 4 \sin \theta, 0 \leq \theta < 2\pi.$$

a) Determine ...

i. ... the coordinates of the foci of the ellipse.

ii. ... the equations of the directrices of the ellipse.

b) Show that an equation of the tangent at a general point on the ellipse is

$$y \sin \theta + x \cos \theta$$

$$= 1.$$

$$4 \cos^2 \theta$$

A straight line passes through the origin and meets the general tangent whose equation is given in part (b), at the point P.

c) Show that, as θ varies, P traces the curve with equation

$$2$$

$$(x^2 + y^2) = 2(9x^2 + 8y^2).$$

$$()$$

$$F \pm 2, 0, x = \pm 9$$

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Question 20 (****+)

An ellipse has a focus at $(5, -3)$ and directrix with equation $y = 2x - 7$.

$$5$$

Given that the eccentricity of the ellipse is $\frac{1}{2}$, find the coordinates of the points of

$$10$$

intersection of the ellipse with the straight line with equation $y = -3$.

$$FP3-U, (234, -3), (92, -3)$$

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Question 21 (****+)

The equation of an ellipse is given by

$$\frac{x^2}{4} + \frac{y^2}{9} = 1.$$

$$\frac{x^2}{4} + \frac{y^2}{9} = 1.$$

$$\frac{x^2}{4} + \frac{y^2}{9} = 1.$$

a) Determine the coordinates of the foci of the ellipse, and the equation of each of its two directrices.

b) Show that

$$SP + TP = 4.$$

A chord of the ellipse is defined as the straight line segment joining any two distinct points on the ellipse.

c) Find the equation of the locus of the midpoints of parallel chords of the ellipse whose gradient is 2.

$$(\pm 1, 0), x = \pm 4, y = -18x$$

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Question 22 (*****)

The straight line L with equation $y = mx + c$, where m and c are constants, passes through the point (25, 25).

Given further that L is a tangent to the ellipse with equation

$$\frac{x^2}{25} + \frac{y^2}{9} = 1,$$

$$\frac{x^2}{25} + \frac{y^2}{9} = 1,$$

$$\frac{x^2}{25} + \frac{y^2}{9} = 1,$$

determine the possible equations of L.

$$y = 4x + 5, y = 77x - 85$$

$$y = 4x + 5, y = 77x - 85$$

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Question 23 (*****)

The point P lies on an ellipse whose foci are on the x axis at the points S and T.

Given further that the triangle STP is right angled at T, show that

$$1 - \tan^2 \theta$$

$$e = 2,$$

$$1 + \tan^2 \theta$$

$$2$$

where e is the eccentricity of the ellipse, and θ is the angle PST.

SPX-A, proof

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Question 24 (*****)

The point P lies on the ellipse with polar equation

$$r(5 - 3\cos\theta) = 8, 0 \leq \theta < 2\pi.$$

The ellipse has foci at O (0,0) and at T (3, 0).

Show that OP + PT is constant for all positions of P.

SPX-D, OP + PT = 5

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RECTANGULAR

HYPERBOLA

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Question 1 (**)

The rectangular hyperbola H has Cartesian equation

$$xy = 9, x \neq 0, y \neq 0.$$

$$\begin{pmatrix} 3 \\ t \end{pmatrix}$$

The point P $\begin{pmatrix} 3t \\ t \end{pmatrix}$, $t \neq 0$, where t is a parameter, lies on H.

$$\begin{pmatrix} 3 \\ t \end{pmatrix}$$

a) Show that the equation of a normal to H at P is given by

$$yt - xt^3 = 3 - 3t^4.$$

The normal to H at the point where $t = -3$ meets H again at the point Q.

b) Determine the coordinates of Q.

()

Q 1,81

9

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Question 2 (**)

The tangents to the hyperbola with equation $xy = 9$, at two distinct points A and B,

1

have gradient $-\frac{1}{16}$.

16

Determine in any order ...

a) ... the coordinates of A and B.

b) ... the equation of each of the two tangents.

() ()

A 12, 3 , B -12, - 3 , $x + 16 y = 24$, $x + 16 y = -24$

4 4

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Question 3 (**+)

$\square 4 \square$

The general point P $\square 4t$, $\square$, $t \neq 0$, where t is a parameter, lies on a hyperbola H .

$\square t \square$

a) Show that the equation of a tangent at the point P is given by

$$x + t^2 y = 8t .$$

b) Find the equation of each of the two tangents to H which pass through the point Q (-12, 7) , and further deduce the coordinates of their corresponding points of tangency.

$$x + 4 y = 16, (8, 2) , 49 x + 36 y + 336 = 0, (- 247 , - 143)$$

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Question 4 (***)

$\square c \square$

The general point P $\square ct$, $\square$, $c > 0$, $t > 0$, lies on a hyperbola H with Cartesian

$\square t \square$

equation

$$xy = c^2 .$$

The tangent to H at P meets the coordinate axes at the points A and B .

Given the area of the triangle BOA is 72 square units, find the value of c .

$$c=6$$

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Question 5 (***)

$\square 3 \square$

The general point P $\square 3t$, $\square$, $t \neq 0$, where t is a parameter, lies on a hyperbola H .

$\square t \square$

a) Show that the equation of a tangent at the point P is given by

$$x + t^2 y = 6t .$$

The tangents to the hyperbola at points A and B intersect at the point Q (-1, 7) .

b) Determine in any order ...

i. ...the coordinates of A and B .

ii. ... the equation of each of the two tangents.

()

A (3,3) , B - 3 , -21 , $x + y = 6$, $49x + y + 42 = 0$

7

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Question 6 (***)

a

The point $P(a, p)$ lies on the rectangular hyperbola H , with Cartesian equation

p

$xy = a^2$,

where a is a positive constant and p is a parameter.

a) Show that the equation of a tangent at the point P is given by

$x + p^2 y = 2ap$.

a

The point $Q(aq, q)$ also lies on H , where q is a parameter, so that $q \neq p$.

q

The tangent at P and the tangent at Q intersect at the point R .

b) Find simplified expressions for the coordinates of R .

The values of p and q are such so that $p = 3q$.

c) Find a Cartesian locus of R as p varies.

$2apq$ $2a$ 3

R , $xy = 4a$

$p+q$ $p+q$

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Question 7 (****)

c

The general point $P(cp, p)$, $p \neq 0$, where p is a parameter, lies on the rectangular

p

hyperbola, with Cartesian equation

$xy = c^2$,

where c is a positive constant.

a) Show that an equation of the tangent to the hyperbola at P is given by

$yp^2 + x = 2cp$.

c

Another point $Q(cq, q)$, $p \neq \pm q$ also lies on the hyperbola.

□ q□

The tangents to the hyperbola at P and Q meet at the point R .

b) Show that the coordinates of R are given by

□ 2cpq 2c □

□ , □.

□ p+q p+q□

c) Given that PQ is perpendicular to OR , show that

$p^2 + q^2 = 1$.

proof

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Question 8 (****)

The distinct points A , B and C lie on the hyperbola with equation

$xy = p^2$,

where p is a positive constant.

Given that ABC is a right angle, show that the tangent to the hyperbola at B , is perpendicular to AC .

proof

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Question 9 (****+)

□ 5□

The general point P □ 5t , □ , t ≠ 0 , where t is a parameter, lies on the hyperbola, with

□ t□

Cartesian equation

$xy = 25$.

a) Show that an equation of the normal to the hyperbola at the point P is

5

$y = t^2 x + -5t^3$

t

The normal to the hyperbola at P meets the hyperbola again at the point Q .

b) Show that the coordinates of Q are given by

□ 5 3□

□ - 3 , -5t □ .

□ t □

c) Show that the Cartesian form of the locus of the midpoint of PQ , as t varies, is given by

2

 $\square y x \square$

$$4xy + 25\square - \square = 0.$$

 $\square x y \square$

proof

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Question 10 (****+)

 $\square c \square$

The general point $P \square cp, \square$, $p \neq 0$, where p is a parameter, lies on the rectangular

 $\square p \square$

hyperbola, with Cartesian equation

$$xy = c^2,$$

where c is a positive constant.

The normal to the hyperbola at P meets the hyperbola again at the point Q .

Show that the coordinates of Q are

 $\square c^3 \square$ $\square - 3, -cp \square$. $\square p \square$

FP3-X, proof

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Question 11 (****+)

 $\square p^1 \square$

The general point $P \square \square, \square$, $p \neq 0$, where p is a parameter, lies on the rectangular

 $\square 2^2 p \square$

hyperbola, with Cartesian equation

$$4xy = 1.$$

The normal to the hyperbola at P meets the hyperbola again at the point Q .

Show that the Cartesian form of the locus of the midpoint of PQ , as p varies, is

2

$$(y^2 - x^2) + 16x^3y^3 = 0.$$

FP3-V, proof

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Question 12 (****+)

 $\square 2 \square \square 2 \square$

Two distinct points $P \square 2p, \square$ and $Q \square 2q, \square$, lie on the hyperbola with Cartesian

$p \neq q$

equation $xy = 4$.

The tangents to the hyperbola at the points P and Q, meet at the point R.

a) Show that the coordinates of the point R are given by

$4pq$

$x = \frac{4}{p}, y = \frac{4}{q}$.

$p+q, p-q$

b) Given that the point R traces the rectangular hyperbola $xy = 3$, find the two possible relationships between p and q, in the form $p = f(q)$

$p = 3q, p = \frac{1}{3}q$

3

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Question 13 (****+)

$t \neq 0$

The general point P $(2t, \frac{1}{t})$, $t \neq 0$, where t is a parameter, lies on the rectangular

$\frac{1}{t}$

hyperbola, with Cartesian equation

$xy = 4$.

a) Find an equation of the normal to the hyperbola at the point P.

The normal to the hyperbola at P meets the hyperbola again at the point Q.

The point M is the midpoint of PQ.

b) Find an equation of the locus of M, as t varies.

Give a simplified answer in the form $f(x, y) = 0$.

2

FP3-O, $ty - 2 = t^3x - 2t^4, (y^2 - x^2) + x^3y^3 = 0$

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Question 14 (*****)

$p \neq 0$

The point P $(p + \frac{1}{p}, p - \frac{1}{p})$, $p \neq 0$, lies on the rectangular hyperbola, with Cartesian

$\frac{1}{p}$

equation

$x^2 - y^2 = 4$.

The normal to the hyperbola at P meets the y axis at the point Q $(0, -k)$, $k > 0$.

The area of the triangle OPQ, where O is the origin, is 15.

4

Determine the two possible sets of coordinates for P .

SPX-C , (52 , - 32) , (52 , - 32)

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Question 15 (*****)

The points P and Q are two distinct points which lie on the curve with equation

1

$y = \frac{1}{x}, x \in \mathbb{R}, x \neq 0$.

x

P and Q are free to move on the curve so that the straight line segment PQ is a normal to the curve at P .

The tangents to the curve at P and Q meet at the point R .

Show that R is moving on the curve with Cartesian equation

2

$(y^2 - x^2) + 4xy = 0$.

SP-K , proof

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Question 16 (*****)

The variable point P lies on the rectangular hyperbola, with Cartesian equation

$xy = a^2$,

where a is a positive constant.

The normal to the hyperbola at P meets the hyperbola again at the point Q .

The point M is the midpoint of PQ .

Determine, in the form $f(x, y) = 0$, an equation of the locus of M , for all the possible positions of P .

2

SPX-E , $a^2(y^2 - x^2) + 4x^3y^3 = 0$

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HYPERBOLA

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Question 1 (**)

A hyperbola H has foci at the points with coordinates (-10,0) and (10,0) , and its Cartesian equation is given by

$x^2 - y^2$

2

$- = 1,$

a 36

where a is a positive constant.

a) Find the value of a .

b) Deduce the equations of the directrices of H .

$a = 8, x = \pm 32$

5

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Question 2 (**)

A hyperbola H has foci at the points with coordinates $(\pm 13, 0)$ and the equations of its

144

directrices are $x = \pm$.

13

Determine a Cartesian equation for H .

$x^2 y^2$

$- = 1$

144 25

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Question 3 (***)

A hyperbola is given parametrically by

$3 \leq 1 \leq 5 \leq 1$

$x = \frac{1}{t} + \frac{1}{t}, y = \frac{1}{t} - \frac{1}{t}, t \neq 0.$

$2 \leq t \leq 2 \leq t$

a) Show that the Cartesian equation of the hyperbola can be written as

$x^2 y^2$

$- = 1.$

9 25

b) Find ...

i. ... the equations of its asymptotes.

ii. ... the coordinates of its foci.

iii. ... the equations of its directrices.

c) Sketch the hyperbola indicating any intersections with the coordinate axes, as well as the information stated in part (b).

$y = \pm 5x$

$$3 (\pm 34, 0), x = \pm 34$$

9

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Question 4 (***)

The point $P (a \sec \theta, b \tan \theta)$ lies on the hyperbola with equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

2

$$- = 1,$$

$$a^2 - b^2$$

where a and b are positive constants.

a) Show that an equation of the normal at P is given by

()

$$by + ax \sin \theta = a^2 + b^2 \tan \theta.$$

The normal to the hyperbola meets the coordinate axes at the points A and B .

b) Show that, as θ varies, the Cartesian locus of the midpoint of AB is given by

2

() (

$$4a^2x^2 - b^2y^2 = a^2 + b^2).$$

FP3-L, proof

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Question 5 (***)

A hyperbola has Cartesian equation

$$2x^2 - 4x - y^2 - 4y = 4.$$

Find the coordinates of its foci and the equations of its directrices.

$$(1 + 3, -2), (1 - 3, -2), x = -13 + 1, x = 13 + 1$$

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Question 6 (****)

The general point $P (a \sec \theta, b \tan \theta)$ lies on the hyperbola with Cartesian equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

2

$$- = 1,$$

$$a^2 - b^2$$

where a and b are positive constants.

a) Show that an equation of the normal at P is given by

()

$$by + ax \sin \theta = a^2 + b^2 \tan \theta .$$

The normal to the hyperbola meets the x axis at the point X .

The eccentricity of the hyperbola is 3 and its foci are denoted by S and S' , where S

2

has a positive x coordinate.

b) Given that $OX = 3 OS$, find the possible values of θ for $0 \leq \theta \leq 2\pi$.

π 5π

FP3-R , $\theta =$,

3 3

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Question 7 (****)

The equation of a hyperbola H is given in terms of a parameter t by

$$x = \sinh t , y = \cosh t , t \in \mathbb{R} .$$

a) Sketch the graph of H , clearly marking the equation of each of its asymptotes.

The equation of the tangent to H at the point P ($\sinh t$, $\cosh t$) , meets each of the asymptotes at the points A and B .

b) Show that P is equidistant from A and B .

c) Show further that the area of the triangle OAB , where O is the origin, is exactly 1 square unit.

graph/proof

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Question 8 (****+)

A hyperbola H and a line L have the following Cartesian equations

$$x^2 - y^2$$

$$H: \frac{x^2}{a^2} - \frac{y^2}{b^2}$$

$$= 1$$

$$a, b$$

$$L : y = mx + c ,$$

where a , b , m and c are non zero constants.

a) Show that the x coordinates of the points of intersection between L and H satisfy the equation

$$(a^2 m^2 - b^2) x^2 + (2a^2 mc) x + a^2 (b^2 + c^2) = 0 .$$

$$2 \ 2 \ 2 \ 2 \ 2 \ 2 \ 2 \ 2$$

b) Given the line is a tangent to the hyperbola show that

$$a^2 + b^2 = c^2$$

c) Find the equations of the two tangents to the hyperbola with Cartesian equation

$$\frac{x^2}{25} - \frac{y^2}{16} = 1$$

$$- = 1$$

$$25 \ 16$$

that pass through the point $(1, 4)$, and for each tangent the coordinates of their point of tangency.

$$y = x + 3, -25 \ 16$$

$$) \ 4 \ 16$$

$$(25$$

$$3, -3, y = -3x + 3, 4, -3)$$

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Question 9 (****+)

A hyperbola H has Cartesian equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

$$2$$

$$- = 1,$$

$$a \ b^2$$

where a and b are positive constants.

The straight line T1 is the tangent to H at the point $(a \cosh \theta, b \sinh \theta)$.

T1 meets the x axis at the point P.

The straight line T2 is a tangent to the hyperbola at the point $(a, 0)$.

T1 and T2 meet each other at the point Q.

Given further that M is the midpoint of PQ, show that as θ varies, the locus of M traces the curve with equation

$$()$$

$$x^4 + y^2 + b^2 = ab^2.$$

SPX-H, proof

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PreCalculus

Ellipse Questions

Use the information provided to write the standard form equation of each ellipse,

$$1) 9x^2 + 4y^2 + 72x - 5y - 176 = 0 \quad 2) 16x^2 + y^2 - 64x + 4y + 4 = 0$$

$$3) \text{ Vertices: } (-6, 4), (-6, -24) \quad 4) \text{ Vertices: } (7, 20), (7, -4)$$

Co-vertices: (4,-10), (-16,-10) Co-vertices: (15, 8), (-1, 8)

5) Vertices: (-10, 5), (-10,-5) 6) Vertices: (-6, 12), (-6,-14)

Foci: (-10, 4), (-10,-4) Foci: (-6, 11), (-6,-13)

7) Foci: (-7, 4), (-7,-6) 8) Foci: (-3, 4), (-3,-6)

Co-vertices: (5,-1), (-19,-1) Co-vertices: (9, -1), (-15, -1)

9) Foci: (-3, 1), (-3,-5) 10) Foci: (0, 2), (0,-6)

Endpoints of major axis: (-3, 3), (-3,-7) Endpoints of major axis: (0, 3), (0,-7)

11) Foci: (-1, -5), (-1, -15) 12) Foci: (-4, 2), (-4,-22)

Endpoints of minor axis: (11,-10), (-13,-10) Endpoints of minor axis: (1,-10), (-9, -1 O)

13) Center: (3, 10) 14) Center: (-2, 1 O)

Vertex: (3, -3) Vertex: (-2, 5)

Focus: (3, 5) Focus: (-2, 6)

15) Center: (2, 10) 16) Center: (6, 10)

Focus: (2, 14) Focus' (18, 10)

Co-vertex: (5, 1 O) Co-vertex: (6, 5)

17) Endpoints of major axis: (16, 10), (-8, 10) 18) Endpoints of major axis: (-4, 4), (-14, 4)

Endpoints of minor axis: (4, 20), (4, 0) Endpoints of minor axis: (-9, 8), (-9, 0)

19) Major axis is horizontal 20) Major axis is horizontal

Center: (-5,-8) Center: (-5, 7)

Major axis is 20 units long Major axis is 16 units long

Minor axis is 2 units long Minor axis is 6 units long

21) 22)

y

'5 -4 -2

5 3

23) Eccentricity- 24) Eccentricity=-

' 13 5

Foci: (-1, 0), (-11, 0) Foci: (13,-1), (7,-1)

4 12

25) Eccentricity=- 26) Eccentricity-

5 13

Center: (-7,-1) Center: (-4,-1)

Vertex: (-12,-1) Vertex: (-17,-1)

Identify the center, vertices, co-vertices, foci, and eccentricity of each. Then sketch the graph.

27) $(x + 1)^2 + y^2$

=1

16 -1

28) +

y 36 16

-3-

Answers to Ellipse Questions

$$(X+4)^2 (y-1)^2$$

$$+ (X-2)^2 (y+2)^2 (x+6)^2 (y+10)^2$$

$$1) -1^2 + -1^3 + =1$$

$$36 \ 81 \ 4 \ 64 \ 100 \ 196$$

$$(X-7)^2 (y-8)^2 (y+1)^2$$

$$4)$$

$$64$$

$$144$$

$$-1 \ 5) (X+10)^2 y^2 + ---1$$

$$9 \ 25$$

$$6) (x^25$$

$$+ 6)$$

$$+ -1$$

$$169$$

$$(X+7)^2 (y+1)^2 (x+2)^2 (y+2)^2$$

$$7)$$

$$144$$

$$169$$

$$=1$$

$$144$$

$$8) (X+3) (y+$$

$$+ 1) -1 \ 9) (X+3) + (y+2) -1$$

$$169 \ 16 \ 25$$

$$2$$

$$X (y+2)^2 (x+1)^2 (y+10)^2 (x+4)^2 (y+10)^2$$

$$10) --+ -1 \ 11) + =1 \ 12) =1$$

$$9 \ 25 \ 144 \ 169 \ 25 \ 169$$

$$13)$$

$$(x-3) +$$

$$(y-10)=1 \ 14)$$

$$(X+2)^2 (y-10)^2$$

$$=1$$

$$(X-2)^2 (y-10)^2$$

$$+ 15) + =1$$

$$144 \ 169 \ 9 \ 25 \ 9 \ 25$$

$$(X-6)^2 (y-10)^2 (x-4)^2 (y-10)^2. (X+9)^2 (y-4)^2$$

$$16) + =1 \ 17) + =1 \ 18) + -1$$

$$169 \ 25 \ 144 \ 100 \ 25 \ 16$$

$$19) (x+5)^2 + (y-8)^2 = 1 (x+5)^2 (y-7)^2 X$$

2 2

$$20) + -1 \ 21) + Y \sim \sim \sim 1$$

100 64 9 49 36

2 2 2

$$X (y-4) (x + 6) 2 y (x - 10) 2 (y + 1)^2$$

$$22) \sim + -1 \ 23) + -1 \ 24) + = 1$$

36 9 169 144 25 16

$$(x + 7)^2 (y + 1)^2 (X + 4)^2 (y + 1)^2$$

$$25) + -1 \ 26) + -1$$

25 9 169 25

27) 28)

Y Center: (-1, 0) $\sim y$ Center: (21 $\sim$, -1)

Vertices: (3, 0)

(-5, 0) Vertices: ($\sim$, -1)

Co-vertices: (-1, 1)

(-1, -i) ($\sim \sim \sim$, -1)>.Foci: (-1 + "Q $\sim$ ", 0)

1

$$X (-1 - " \sim i7 -, 0)$$

Eccentricity: $\sim \sim \sim 0.968$ Eccentricity: $\sim \sim \sim \sim 0.745$

-4-

ID: A

Conic Sections Practice Test

1. Give the coordinates of the circle's center and it radius.

2 2

$$(x - 2) + (y + 9) = 1$$

___ 2. Find the equation of the circle graphed below.

2 2 2 2 2

$$A) x + y = 4 \ C) x + y = 16 \ E) x + y = 16$$

2 2 2 2

$$B) y = x + 16 \ D) x + y = 1$$

1

Name: _____ ID: A

___ 3. Graph the following equation.

2 2

$$x - 10x + y = -9$$

A) C)

B)

___ 4. Find the vertex and focus of the parabola.

$$y - 2x^2 + 16(x - 3) = 0$$

E -

A) vertex: $(-3, -2)$ focus: $(-3, 14)$ B) vertex: $(-3, -2)$ focus: $(-3, -18)$ C) vertex: $(-3, -2)$ focus: $(-7, -2)$ D) vertex: $(3, 2)$ focus: $(-1, 2)$

2

Name: _____ ID: A

___ 5. Find the standard form of the equation of the parabola with the given characteristic and vertex at

the origin.

focus: $(0, 7)$ A) $x^2 = 28y$ C) $x^2 = -7y$ E) $y^2 = 7x$ B) $x^2 = 7y$ D) $y^2 = 28x$

___ 6. Find the standard form of the equation of the parabola with the given characteristic and vertex at

the origin.

directrix: $x = 1$ A) $x^2 = -4y$ C) $x^2 = y$ E) $y^2 = -4x$ B) $x^2 = 4y$ D) $y^2 = x$

___ 7. Find the vertex and focus of the parabola.

$$y = -9x^2$$

2

y

8

$$y = 4x^2$$

A) vertex: $(0, -5)$ focus: $(-9, -9)$

$$y = 4x^2 - 8x$$

$$y = 4x^2$$

B) vertex: $(0, 0)$ focus: $(0, -9)$

$$y = 8x^2$$

$$y = 4x^2$$

C) vertex: $(0, 0)$ focus: $(-9, 0)$

$$y = 8x^2$$

$$y = 4x^2$$

D) vertex: $(0, 0)$ focus: $(-9, 0)$

$$y = 32x^2$$

___ 8. Find the equation of the parabola with vertex at $(5, 4)$ and focus at $(-3, 4)$.

2 2

$$A) (y - 4)^2 = -32(x - 5) \quad D) (y + 4)^2 = -32(x - 5)$$

2 2

$$B) (y - 4)^2 = 32(x - 5) \quad E) (y - 4)^2 = 8(x - 5)$$

2

$$C) (y + 4)^2 = 32(x + 5)$$

9. Find the equation of the parabola with vertex at (0, 0) and focus at (0, 5). Express the equation in standard form.

3

Name: _____ ID: A

____ 10. Find the center and vertices of the ellipse.

2 2

$$x^2 + y^2 = 1$$

49 4

$$A) \text{ center: } (7, 0) \text{ vertices: } (0, -2), (0, 2)$$

$$B) \text{ center: } (0, 0) \text{ vertices: } (-2, 0), (2, 0)$$

$$C) \text{ center: } (0, 0) \text{ vertices: } (0, -7), (0, 7)$$

$$D) \text{ center: } (0, 0) \text{ vertices: } (-7, 0), (7, 0)$$

____ 11. Find the center and foci of the ellipse.

2

$$(x + 5)^2 + (y + 9)^2 = 59$$

5 9

Ê

$$A) \text{ center: } (5, 9) \text{ foci: } (5, 7), (5, 11)$$

$$B) \text{ center: } (-5, -9) \text{ foci: } (-5, -11), (-5, -7)$$

$$C) \text{ center: } (-5, -9) \text{ foci: } (-7, -9), (-3, -9)$$

$$D) \text{ center: } (5, 9) \text{ foci: } (3, -9), (7, -9)$$

____ 12. Find the center and vertices of the ellipse.

2 2

$$4x^2 + 9y^2 - 24x + 72y + 144 = 0$$

$$A) \text{ center: } (-4, 3) \text{ vertices: } (-7, 3), (-1, 3)$$

$$B) \text{ center: } (-3, 4) \text{ vertices: } (-5, 4), (-1, 4)$$

$$C) \text{ center: } (3, -4) \text{ vertices: } (1, -4), (5, -4)$$

$$D) \text{ center: } (3, -4) \text{ vertices: } (0, -4), (6, -4)$$

$$E) \text{ center: } (-3, 4) \text{ vertices: } (-6, 4), (0, 4)$$

4

Name: _____ ID: A

____ 13. Identify the graph of the following ellipse.

2 2

$$x + y = 1$$

$$16 \ 4$$

A) C)

B)

___ 14. Find the center and vertices of the hyperbola.

$$2 \ 2$$

$$11x - 25y + 22x + 250y - 889 = 0$$

A) center: (1, -5), vertices: (1, -10), (1, 0)

B) center: (-1, 5), vertices: (-1, 0), (-1, 10)

C) center: (-1, 5), vertices: (-6, 5), (4, 5)

D) center: (1, -5), vertices: (-4, -5), (6, -5)

5

Name: _____ ID: A

___ 15. Find the vertices and asymptotes of the hyperbola.

$$2 \ 2$$

$$9y - 16x = 144$$

A) vertices: $(0, \pm 4)$ asymptote: $y = \pm 4x$

3

B) vertices: $(0, \pm 4)$ asymptote: $y = \pm 3x$

4

C) vertices: $(\pm 4, 0)$ asymptote: $y = \pm 4x$

3

D) vertices: $(\pm 4, 0)$ asymptote: $y = \pm 3x$

4

___ 16. Find the standard form of the equation of the hyperbola with the given characteristics.

vertices: $(0, \pm 6)$ foci: $(0, \pm 7)$

$$2 \ 2 \ 2 \ 2$$

$$y^2 - x^2 = 1$$

A) $-x = 1$ C)

$$36 \ 49 \ 36 \ 13$$

$$2 \ 2 \ 2 \ 2$$

$$y^2 - x^2 = 49$$

B) $-x = 1$ D)

$$36 \ 13 \ 36 \ 13$$

6

Name: _____ ID: A

___ 17. Find the graph of the following ellipse.

$$2 \ 2$$

$$9x^2 + 16y^2 - 36x - 64y - 44 = 0$$

A) C)

B) D)

___ 18. Write the equation of the ellipse that has its center at the origin with focus at (0, 4) and vertex at (0, 7).

$$\frac{x^2}{2^2} + \frac{y^2}{2^2} = 1$$

$$x^2 + y^2 = 1 \quad x^2 + y^2 = -1$$

A) C)

$$\frac{x^2}{49} + \frac{y^2}{33} = 1 \quad \frac{x^2}{33} + \frac{y^2}{49} = 1$$

$$\frac{x^2}{2^2} + \frac{y^2}{2^2} = 1$$

$$x^2 - y^2 = 1 \quad x^2 + y^2 = 1$$

B) D)

$$\frac{x^2}{33} + \frac{y^2}{49} = 1 \quad \frac{x^2}{49} + \frac{y^2}{33} = 1$$

___ 19. Find the center and vertices of the ellipse.

$$x^2 + 9y^2 + 16x - 54y + 136 = 0$$

$$x^2 + 9y^2 + 16x - 54y + 136 = 0$$

A) center: (3, -8) vertices: (0, -8), (6, -8)

B) center: (8, -3) vertices: (7, -3), (9, -3)

C) center: (-8, 3) vertices: (-9, 3), (-7, 3)

D) center: (-8, 3) vertices: (-11, 3), (-5, 3)

E) center: (8, -3) vertices: (5, -3), (11, -3)

7

Name: _____ ID: A

___ 20. Find the standard form of the equation of the ellipse with the following characteristics.

foci: $(\pm 4, 0)$ major axis of length: 12

$$\frac{x^2}{2^2} + \frac{y^2}{2^2} = 1$$

$$x^2 + y^2 = 1 \quad x^2 + y^2 = 1$$

A) D) $x^2 + y^2 = 1$

$$\frac{x^2}{36} + \frac{y^2}{20} = 1 \quad \frac{x^2}{144} + \frac{y^2}{16} = 1$$

$$\frac{x^2}{2^2} + \frac{y^2}{2^2} = 1$$

$$x^2 + y^2 = 1 \quad x^2 + y^2 = 1$$

B) E) $x^2 + y^2 = 1$

$$\frac{x^2}{36} + \frac{y^2}{16} = 1 \quad \frac{x^2}{144} + \frac{y^2}{128} = 1$$

$$\frac{x^2}{2^2} + \frac{y^2}{2^2} = 1$$

$$x^2 + y^2 = 1$$

C)

$$\frac{x^2}{16} + \frac{y^2}{36} = 1$$

___ 21. Find the standard form of the equation of the hyperbola with the given characteristics.

vertices: (-2, -4), (-2, 6) foci: (-2, -5), (-2, 7)

$$\hat{A} y - 1 \hat{\sim} 2 \hat{A} y - 2 \hat{\sim} 2 \hat{A}$$

$$\hat{E}^{-} (x + 2) \hat{E}^{-} (x + 1)$$

$$A) - = 1 \quad C) - = 1$$

$$25 \ 11 \ 11 \ 25$$

$$\hat{A} y + 1 \hat{\sim} 2 \hat{A} y - 1 \hat{\sim} 2 \hat{A}$$

$$\hat{E}^{-} (x - 2) \hat{E}^{-} (x + 2)$$

$$B) - = 1 \quad D) - = 1$$

$$25 \ 11 \ 25 \ 36$$

____ 22. Graph the hyperbola.

$$2 \ 2$$

$$9x - 9y = 81$$

$$A) \ C)$$

$$B) \ D)$$

$$8$$

Name: _____ ID: A

____ 23. Identify the conic by writing the equation in standard form.

$$2 \ 2$$

$$10y - 20x + 60y + 160x - 255 = 0$$

$$\hat{A} y - 3 \hat{\sim} 2 \hat{A}$$

$$\hat{E}^{-} (x - 4)$$

$$A) - = 1; \text{ hyperbola}$$

$$5 \ 5$$

$$2 \ 4$$

$$\hat{A} y + 3 \hat{\sim} 2 \hat{A}$$

$$\hat{E}^{-} (x - 4)$$

$$B) - = 1; \text{ hyperbola}$$

$$5 \ 5$$

$$2 \ 4$$

$$2$$

$$\hat{A} y + 3 \hat{\sim} 2 \hat{A}$$

$$\hat{E}^{-} (x - 4)$$

$$C) - = 1; \text{ hyperbola}$$

$$97 \ 97$$

$$2 \ 4$$

____ 24. Identify the conic by writing the equation in standard form.

$$2 \ 2$$

$$4x + 4y + 40x + 16y + 40 = 0$$

$$2$$

$$A) (x + 5) + \hat{A} \hat{E} y + 2 \hat{\sim} = 19; \text{ circle}$$

$$2$$

2

B) $(x + 5) + y^2 = 39$; circle

2

 $2x + y^2 = 2$ $(x + 5) + y^2 = 39$ C) $x^2 + y^2 = 1$; ellipse

11 11

4 4

9

ID: A

Conic Sections Practice Test

Answer Section

1. $(2, -9)$, $r = 1$

2. C

3. B

4. D

5. A

6. E

7. D

8. A

2

9. $x = 20y$

10. D

11. B

12. D

13. A

14. C

15. A

16. B

17. B

18. D

19. D

20. A

21. A

22. C

23. B

24. A

1

7-2 Ellipses and Circles

Graph the ellipse given by each equation.

$$1. + = 1$$

SOLUTION:

The ellipse is in standard form, where $h = -2$, $k = 0$, $a = 7$, $b = 3$, and $c = 6.3$. The

orientation is vertical because the y -term contains a $+$.

center: $(h, k) = (-2, 0)$

foci: $(h, k \pm c) = (-2, 6.3)$ and $(-2, -6.3)$

vertices: $(h, k \pm a) = (-2, 7)$ and $(-2, -7)$

covertices: $(h \pm b, k) = (1, 0)$ and $(-5, 0)$

Graph the center, vertices, foci, and axes. Then use a table of values to sketch the ellipse.

$$2. + = 1$$

SOLUTION:

The ellipse is in standard form, where $h = -4$, $k = -3$, $a = 3$, $b = 2$, and $c = 1.8$. The

orientation is horizontal because the x -term contains a $+$.

center: $(h, k) = (-4, -3)$

foci: $(h \pm c, k) = (-1.8, -3)$ and $(-6.2, -3)$

vertices: $(h \pm a, k) = (-1, -3)$ and $(-7, -3)$

covertices: $(h, k \pm b) = (-4, -1)$ and $(-4, -5)$

Graph the center, vertices, foci, and axes. Then use a table of values to sketch the ellipse.

$$3. x^2 + 9y^2 - 14x + 36y + 49 = 0$$

SOLUTION:

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Complete the square for each variable to write the equation in standard form.

$$x^2 + 9y^2 - 14x + 36y + 49 = 0$$

$$x^2 - 14x + 9y^2 + 36y + 49 = 0$$

7-2 Ellipses and Circles

$$3. x^2 + 9y^2 - 14x + 36y + 49 = 0$$

SOLUTION:

Complete the square for each variable to write the equation in standard form.

$$x^2 + 9y^2 - 14x + 36y + 49 = 0$$

$$x^2 - 14x + 9y^2 + 36y + 49 = 0$$

$$x^2 - 14x + 9y^2 + 36y + 49 = 0$$

$$x^2 - 14x + 49 + 9(y^2 + 4y + 4) = 0 + 36$$

$$(x - 7)^2 + 9(y + 2)^2 = 36$$

$$(x - 7)^2 + 9(y + 2)^2 = 36$$

$$(x - 7)^2 + 9(y + 2)^2 = 36$$

$$(x - 7)^2 + 9(y + 2)^2 = 36$$

$$(x - 7)^2 + 9(y + 2)^2 = 36$$

The ellipse is in standard form, where $h = 7$, $k = -2$, $a = 6$, $b = 2$, and $c = 5$. The orientation is horizontal because the x -term contains a larger denominator.

center: $(h, k) = (7, -2)$

foci: $(h \pm c, k) = (13, -2)$ and $(1, -2)$

vertices: $(h \pm a, k) = (13, -2)$ and $(1, -2)$

covertices: $(h, k \pm b) = (7, 0)$ and $(7, -4)$

Graph the center, vertices, foci, and axes. Then use a table of values to sketch the ellipse.

$$4. \quad 4x^2 + y^2 - 64x - 12y + 276 = 0$$

SOLUTION:

Complete the square for each variable to write the equation in standard form.

$$4x^2 + y^2 - 64x - 12y + 276 = 0$$

$$4x^2 - 64x + y^2 - 12y + 276 = 0$$

$$4x^2 - 64x + y^2 - 12y + 276 = 0$$

$$4x^2 - 64x + y^2 - 12y + 276 = 0$$

$$4x^2 - 64x + y^2 - 12y + 276 = 0$$

$$4x^2 - 64x + 256 + y^2 - 12y + 36 + 276 = 0 + 256 + 36$$

$$4x^2 - 64x + 256 + y^2 - 12y + 36 + 276 = 0 + 256 + 36$$

$$4(x^2 - 16x + 64) + y^2 - 12y + 36 = 0 + 256 + 36 - 276$$

$$4(x^2 - 16x + 64) + y^2 - 12y + 36 = 0 + 256 + 36 - 276$$

$$4(x - 8)^2 + (y - 6)^2 = 16$$

$$\frac{(x - 8)^2}{4} + \frac{(y - 6)^2}{4} = 1$$

The ellipse is in standard form, where $h = 8$, $k = 6$, $a = 2$, $b = 2$, and $c = 0$. The orientation is vertical because the y -term contains a larger denominator.

center: $(h, k) = (8, 6)$

foci: $(h, k \pm c) = (8, 9.5)$ and $(8, 2.5)$

vertices: $(h, k \pm a) = (8, 10)$ and $(8, 2)$

covertices: $(h \pm b, k) = (10, 6)$ and $(6, 6)$

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Graph the center, vertices, foci, and axes. Then use a table of values to sketch the ellipse.

7-2 Ellipses and Circles

$$4. \quad 4x^2 + y^2 - 64x - 12y + 276 = 0$$

SOLUTION:

Complete the square for each variable to write the equation in standard form.

$$4x^2 + y^2 - 64x - 12y + 276 = 0$$

$$4x^2 - 64x + y^2 - 12y + 276 = 0$$

$$4x^2 - 64x + y^2 - 12y + 276 = 0$$

$$4x^2 - 64x + y^2 - 12y + 276 = 0$$

$$4x^2 - 64x + y^2 - 12y + 276 = 0$$

$$\begin{aligned}
 4x^2 - 64x + 256 + y^2 - 12y + 36 + 276 &= 0 + 256 + 36 \\
 4(x^2 - 16x + 64) + y^2 - 12y + 36 &= 0 + 256 + 36 - 276 \\
 4(x - 8)^2 + (y - 6)^2 &= 16 \\
 \div 16 &= 1
 \end{aligned}$$

The ellipse is in standard form, where $h = 8$, $k = 6$, $a = 4$, $b = 2$, and $c = 3$. The orientation is vertical because the y -term contains a larger denominator.

center: $(h, k) = (8, 6)$

foci: $(h, k \pm c) = (8, 9.5)$ and $(8, 2.5)$

vertices: $(h, k \pm a) = (8, 10)$ and $(8, 2)$

covertices: $(h \pm b, k) = (10, 6)$ and $(6, 6)$

Graph the center, vertices, foci, and axes. Then use a table of values to sketch the ellipse.

$$5. \quad 9x^2 + y^2 + 126x + 2y + 433 = 0$$

SOLUTION:

Complete the square for each variable to write the equation in standard form.

$$\begin{aligned}
 9x^2 + y^2 + 126x + 2y + 433 &= 0 \\
 9(x^2 + 14x + 49) + y^2 + 2y + 1 + 433 - 9(49) - 1 &= 0 \\
 9(x + 7)^2 + (y + 1)^2 - 9 &= 0 \\
 9(x + 7)^2 + (y + 1)^2 &= 9 \\
 \div 9 &= 1
 \end{aligned}$$

The ellipse is in standard form, where $h = -7$, $k = -1$, $a = 3$, $b = 1$, and $c = 2$. The orientation is horizontal because the x -term contains a larger denominator.

center: $(h, k) = (-7, -1)$

foci: $(h, k \pm c) = (-7, 1.8)$ and $(-7, -3.8)$

vertices: $(h, k \pm a) = (-7, 2)$ and $(-7, -4)$

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covertices: $(h \pm b, k) = (-6, -1)$ and $(-8, -1)$

Page 3

Graph the center, vertices, foci, and axes. Then use a table of values to sketch the ellipse.

7-2 Ellipses and Circles

$$5. \quad 9x^2 + y^2 + 126x + 2y + 433 = 0$$

SOLUTION:

Complete the square for each variable to write the equation in standard form.

2 2

$$9x^2 + y^2 + 126x + 2y + 433 = 0$$

2 2

$$9x^2 + 126x + y^2 + 2y + 433 = 0$$

2 2

$$9(x^2 + 14x + 49) + y^2 + 2y + 1 + 433 - 9(49) - 1 = 0$$

2 2

$$9(x + 7)^2 + (y + 1)^2 - 9 = 0$$

2

$$(x + 7)^2 + = 1$$

The ellipse is in standard form, where $h = -7$, $k = -1$, $a = 3$, $b = 1$, and $c = 2$. The orientation is

2

vertical because the y -term contains a .

center: $(h, k) = (-7, -1)$

foci: $(h, k \pm c) = (-7, 1.8)$ and $(-7, -3.8)$

vertices: $(h, k \pm a) = (-7, 2)$ and $(-7, -4)$

covertices: $(h \pm b, k) = (-6, -1)$ and $(-8, -1)$

Graph the center, vertices, foci, and axes. Then use a table of values to sketch the ellipse.

$$6. x^2 + 25y^2 - 12x - 100y + 111 = 0$$

SOLUTION:

Complete the square for each variable to write the equation in standard form.

2 2

$$x^2 + 25y^2 - 12x - 100y + 111 = 0$$

2 2

$$x^2 - 12x + 25y^2 - 100y + 111 = 0$$

2 2

$$x^2 - 12x + 36 + 25y^2 - 100y + 100 + 111 = 0 + 36 + 100$$

2 2

$$x^2 - 12x + 36 + 25(y^2 - 4y + 4) + 111 = 136$$

2 2

$$(x - 6)^2 + 25(y - 2)^2 = 25$$

2

$$+ (y - 2)^2 = 1$$

The ellipse is in standard form, where $h = 6$, $k = 2$, $a = 5$, $b = 1$, and $c = 2$. The orientation is

2

horizontal because the x -term contains a .

center: $(h, k) = (6, 2)$

foci: $(h \pm c, k) = (10.9, 2)$ and $(1.1, 2)$

vertices: $(h \pm a, k) = (11, 2)$ and $(1, 2)$

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covertices: $(h, k \pm b) = (6, 3)$ and $(6, 1)$

Graph the center, vertices, foci, and axes. Then use a table of values to sketch the ellipse.

7-2 Ellipses and Circles

$$6. x^2 + 25y^2 - 12x - 100y + 111 = 0$$

SOLUTION:

Complete the square for each variable to write the equation in standard form.

$$x^2 + 25y^2 - 12x - 100y + 111 = 0$$

$$x^2 - 12x + 25y^2 - 100y + 111 = 0$$

$$x^2 - 12x + 36 + 25y^2 - 100y + 100 + 111 = 0 + 36 + 100$$

$$x^2 - 12x + 36 + 25(y^2 - 4y + 4) + 111 = 136$$

$$(x - 6)^2 + 25(y - 2)^2 = 25$$

$$(x - 6)^2 + 25(y - 2)^2 = 25$$

$$(x - 6)^2 + 25(y - 2)^2 = 25$$

$$(x - 6)^2 + 25(y - 2)^2 = 25$$

$$(x - 6)^2 + 25(y - 2)^2 = 25$$

$$(x - 6)^2 + 25(y - 2)^2 = 25$$

$$(x - 6)^2 + 25(y - 2)^2 = 25$$

$$(x - 6)^2 + 25(y - 2)^2 = 25$$

The ellipse is in standard form, where $h = 6$, $k = 2$, $a = 5$, $b = 1$, and $c = 3$. The orientation is horizontal because the x -term contains a larger denominator.

center: $(h, k) = (6, 2)$

foci: $(h \pm c, k) = (9, 2)$ and $(3, 2)$

vertices: $(h \pm a, k) = (11, 2)$ and $(1, 2)$

covertices: $(h, k \pm b) = (6, 3)$ and $(6, 1)$

Graph the center, vertices, foci, and axes. Then use a table of values to sketch the ellipse.

Write an equation for the ellipse with each set of characteristics.

7. vertices $(-7, -3)$, $(13, -3)$; foci $(-5, -3)$, $(11, -3)$

SOLUTION:

Because the y -coordinates of the vertices are the same, the major axis is horizontal, and the standard form of the equation is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

The center is the midpoint of the segment between the vertices, or $(3, -3)$. So, $h = 3$ and $k = -3$.

The distance between the vertices is equal to $2a$ units. So, $2a = 20$, $a = 10$, and $a^2 = 100$. The distance between the foci is equal to $2c$ units. So, $2c = 16$ and $c = 8$.

Use the values of a and c to find b .

$b^2 = a^2 - c^2 = 100 - 64 = 36$, so $b = 6$.

The equation of the ellipse is $\frac{(x-3)^2}{100} + \frac{(y+3)^2}{36} = 1$.

$$\frac{(x-3)^2}{100} + \frac{(y+3)^2}{36} = 1$$

$$c^2 = a^2 - b^2$$

$$2^2 = 10^2 - b^2$$

$$8 = 10^2 - b^2$$

$$b^2 = 36$$

$$b = 6$$

So, $b = 6$ and $b = 36$.

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Using the values of h , k , a , and b , the equation for the ellipse is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

7-2 Ellipses and Circles

Write an equation for the ellipse with each set of characteristics.

7. vertices $(-7, -3)$, $(13, -3)$; foci $(-5, -3)$, $(11, -3)$

SOLUTION:

Because the y -coordinates of the vertices are the same, the major axis is horizontal, and the standard form of the

equation is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

The center is the midpoint of the segment between the vertices, or $(3, -3)$. So, $h = 3$ and $k = -3$.

$$2$$

The distance between the vertices is equal to $2a$ units. So, $2a = 20$, $a = 10$, and $a^2 = 100$. The distance between the

foci is equal to $2c$ units. So, $2c = 16$ and $c = 8$.

Use the values of a and c to find b .

$$2^2 = 10^2 - b^2$$

$$c^2 = a^2 - b^2$$

$$2^2 = 10^2 - b^2$$

$$8 = 10^2 - b^2$$

$$b^2 = 36$$

$$b = 6$$

So, $b = 6$ and $b = 36$.

Using the values of h , k , a , and b , the equation for the ellipse is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

8. vertices $(4, 3)$, $(4, -9)$; length of minor axis is 8

SOLUTION:

Because the x -coordinates of the vertices are the same, the major axis is vertical, and the standard form of the

equation is $\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$.

The center is the midpoint of the segment between the vertices, or $(4, -3)$. So, $h = 4$ and $k = -3$.

$$2$$

The distance between the vertices is equal to $2a$ units. So, $2a = 12$, $a = 6$, and $a^2 = 36$. The length of the minor axis

$$2$$

is equal to $2b$. So, $2b = 8$, $b = 4$, and $b^2 = 16$.

Using the values of h , k , a , and b , the equation for the ellipse is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

9. vertices $(7, 2)$, $(-3, 2)$; foci $(6, 2)$, $(-2, 2)$

SOLUTION:

Because the y -coordinates of the vertices are the same, the major axis is horizontal, and the standard form of the

equation is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

The center is the midpoint of the segment between the vertices, or $(2, 2)$. So, $h = 2$ and $k = 2$.

2

The distance between the vertices is equal to $2a$ units. So, $2a = 10$, $a = 5$, and $a^2 = 25$. The distance

between the foci is equal to $2c$ units. So, $2c = 8$ and $c = 4$.

Use the values of a and c to find b .

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$c^2 = a^2 - b^2$

$2^2 = 5^2 - b^2$

$4 = 25 - b^2$

2

is equal to $2b$. So, $2b = 8$, $b = 4$, and $b^2 = 16$.

7-2 Using

Ellipses and of Circles

the values h , k , a , and b , the equation for the ellipse is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

9. vertices $(7, 2)$, $(-3, 2)$; foci $(6, 2)$, $(-2, 2)$

SOLUTION:

Because the y -coordinates of the vertices are the same, the major axis is horizontal, and the standard form of the

equation is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

The center is the midpoint of the segment between the vertices, or $(2, 2)$. So, $h = 2$ and $k = 2$.

2

The distance between the vertices is equal to $2a$ units. So, $2a = 10$, $a = 5$, and $a^2 = 25$. The distance

between the foci is equal to $2c$ units. So, $2c = 8$ and $c = 4$.

Use the values of a and c to find b .

$c^2 = a^2 - b^2$

$2^2 = 5^2 - b^2$

$4 = 25 - b^2$

$4 = 25 - b^2$

$b^2 = 21$

2

So, $b = 3$ and $b = 9$.

Using the values of h , k , a , and b , the equation for the ellipse is $+ = 1$.

10. major axis $(-13, 2)$ to $(1, 2)$; minor axis $(-6, 4)$ to $(-6, 0)$

SOLUTION:

Because the y -coordinates of the major axis are the same, the major axis is horizontal, and the standard form of the

equation is $+ = 1$.

2

The length of the major axis is equal to $2a$ units. So, $2a = 14$, $a = 7$, and $a = 49$. The length of the minor axis is

2

equal to $2b$ units. So, $2b = 4$, $b = 2$, and $b = 4$. The center of the ellipse is at the midpoint of the major axis, or $(-6, 2)$.

So, $h = -6$ and $k = 2$.

Using the values of h , k , a , and b , the equation for the ellipse is $+ = 1$.

11. foci $(-6, 9)$, $(-6, -3)$; length of major axis is 20

SOLUTION:

Because the x -coordinates of the foci are the same, the major axis is vertical, and the standard form of the equation

is $+ = 1$.

The center is located at the midpoint of the foci, or $(-6, 3)$. So, $h = -6$ and $k = 3$. The length of the major axis is $2a$

2

units. So, $2a = 20$, $a = 10$, and $a = 100$. The distance between the foci is equal to $2c$ units. So, $2c = 12$ and $c = 6$.

Use the values of a and c to find b .

$2^2 + 2^2 = 2^2$

$c^2 = a^2 - b^2$

$2^2 + 2^2 = 2^2$

$6 = 10 - b$

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$b = 8$

2

So, $b = 8$ and $b = 64$.

equal to $2b$ units. So, $2b = 4$, $b = 2$, and $b = 4$. -

So, $h = -6$ and $k = 2$.

7-2 Using

Ellipses and ofCircles

the values h , k , a , and b , the equation for the ellipse is $+ = 1$.

11. foci $(-6, 9)$, $(-6, -3)$; length of major axis is 20

SOLUTION:

Because the x-coordinates of the foci are the same, the major axis is vertical, and the standard form of the equation

is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

The center is located at the midpoint of the foci, or $(-6, 3)$. So, $h = -6$ and $k = 3$. The length of the major axis is $2a$

2

units. So, $2a = 20$, $a = 10$, and $a^2 = 100$. The distance between the foci is equal to $2c$ units. So, $2c = 12$ and $c = 6$.

Use the values of a and c to find b .

$a^2 = b^2 + c^2$

$c^2 = a^2 - b^2$

$6^2 = 10^2 - b^2$

$36 = 100 - b^2$

$b^2 = 64$

2

So, $b = 8$ and $b^2 = 64$.

Using the values of h , k , a , and b , the equation for the ellipse is $\frac{(x+6)^2}{100} + \frac{(y-3)^2}{64} = 1$.

12. co-vertices $(-13, 7)$, $(-3, 7)$; length of major axis is 16

SOLUTION:

Because the y-coordinates of the co-vertices are the same, the major axis is vertical, and the standard form of the

equation is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

The center is located at the midpoint of the co-vertices, or $(-8, 7)$. So, $h = -8$ and $k = 7$. The distance between the

2

co-vertices is equal to $2b$ units. So, $2b = 10$, $b = 5$, and $b^2 = 25$. The length of the major axis is equal to $2a$ units. So,

2

$2a = 16$, $a = 8$, and $a^2 = 64$.

Using the values of h , k , a , and b , the equation for the ellipse is $\frac{(x+8)^2}{64} + \frac{(y-7)^2}{25} = 1$.

13. foci $(-10, 8)$, $(14, 8)$; length of major axis is 30

SOLUTION:

Because the y-coordinates of the foci are the same, the major axis is horizontal, and the standard form of the

equation is $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$.

The midpoint is the distance between the foci, or $(2, 8)$. So, $h = 2$ and $k = 8$. The distance between the foci is equal

2

to $2c$ units. So, $2c = 24$ and $c = 12$. The length of the major axis is equal to $2a$. So, $2a = 30$, $a = 15$, and $a^2 = 225$.

Use the values of a and c to find b .

$$2^2 + 2^2 = 2^2$$

$$c^2 = a^2 - b^2$$

$$2^2 + 2^2 = 2^2$$

$$12 = 15 - b$$

$$b = 9$$

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So, $b = 9$ and $b = 81$.

2

$$2a = 16, a = 8, \text{ and } a = 64.$$

7-2 Using

Ellipses and of Circles

the values h , k , a , and b , the equation for the ellipse is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

13. foci $(-10, 8)$, $(14, 8)$; length of major axis is 30

SOLUTION:

Because the y -coordinates of the foci are the same, the major axis is horizontal, and the standard form of the

equation is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

The midpoint is the distance between the foci, or $(2, 8)$. So, $h = 2$ and $k = 8$. The distance between the foci is equal

2

to $2c$ units. So, $2c = 24$ and $c = 12$. The length of the major axis is equal to $2a$. So, $2a = 30$, $a = 15$, and $a^2 = 225$.

Use the values of a and c to find b .

$$2^2 + 2^2 = 2^2$$

$$c^2 = a^2 - b^2$$

$$2^2 + 2^2 = 2^2$$

$$12 = 15 - b$$

$$b = 9$$

2

So, $b = 9$ and $b = 81$.

Using the values of h , k , a , and b , the equation for the ellipse is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

Determine the eccentricity of the ellipse given by each equation.

$$14. \frac{x^2}{16} + \frac{y^2}{9} = 1$$

SOLUTION:

The ellipse is in standard form. So, $a = 4$ or about 8.49 and $b = 3$.

First, determine the value of c .

$$2^2 + 2^2 = 2^2$$

$$c^2 = a^2 - b^2$$

2

$$c = 72 - 54$$

$$c = \text{or about } 4.24$$

Use the values of a and c to find the eccentricity of the ellipse.

$$e =$$

$$e = \text{or about } 0.5$$

$$15. + = 1$$

SOLUTION:

The ellipse is in standard form. So, $a =$ or about 6.32 and $b =$.

First, determine the value of c .

$$2^2 + 2^2 = c^2$$

$$c^2 = a^2 - b^2$$

$$2^2$$

$$c = 40 - 12$$

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or about 5.29 Page 9

Use the values of a and c to find the eccentricity of the ellipse.

$$e =$$

$$e = \text{or about } 0.5$$

7-2 Ellipses and Circles

$$15. + = 1$$

SOLUTION:

The ellipse is in standard form. So, $a =$ or about 6.32 and $b =$.

First, determine the value of c .

$$2^2 + 2^2 = c^2$$

$$c^2 = a^2 - b^2$$

$$2^2$$

$$c = 40 - 12$$

$$c = \text{or about } 5.29$$

Use the values of a and c to find the eccentricity of the ellipse.

$$e =$$

$$e = \text{or about } 0.837$$

$$16. + = 1$$

SOLUTION:

The ellipse is in standard form. So, $a =$ or about 7.55 and $b =$.

First, determine the value of c .

$$2^2 + 2^2$$

$$c = a - b$$

$$2$$

$$c = 57 - 14$$

$$c = \text{or about } 6.56$$

Use the values of a and c to find the eccentricity of the ellipse.

$$e =$$

$$e = \text{or about } 0.869$$

$$17. + = 1$$

SOLUTION:

The ellipse is in standard form. So, $a =$ or about 5.74 and $b =$.

First, determine the value of c .

$$2^2 + 2^2$$

$$c = a - b$$

$$2$$

$$c = 33 - 27$$

$$c = \text{or about } 2.45$$

Use the values of a and c to find the eccentricity of the ellipse.

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$$e =$$

$$e = \text{or about } 0.426$$

$$e =$$

7-2 Ellipses

$e =$ and Circles

or about 0.869

$$17. + = 1$$

SOLUTION:

The ellipse is in standard form. So, $a =$ or about 5.74 and $b =$.

First, determine the value of c .

$$2^2 + 2^2$$

$$c = a - b$$

$$2$$

$$c = 33 - 27$$

$$c = \text{or about } 2.45$$

Use the values of a and c to find the eccentricity of the ellipse.

$$e =$$

$$e = \text{or about } 0.426$$

$$18. + = 1$$

SOLUTION:

The ellipse is in standard form. So, $a =$ or about 3.46 and $b = 3$.

First, determine the value of c .

$$c^2 = a^2 - b^2$$

$$c^2 = 12 - 9$$

$$c^2 = 3$$

$$c = \sqrt{3}$$

$$c = \text{or about } 1.73$$

Use the values of a and c to find the eccentricity of the ellipse.

$$e = \frac{c}{a}$$

$$e = \text{or about } 0.5$$

$$19. + = 1$$

SOLUTION:

The ellipse is in standard form. So, $a =$ or about 4.8 and $b =$.

First, determine the value of c .

$$c^2 = a^2 - b^2$$

$$c^2 = 23 - 17$$

$$c^2 = 6$$

$$c = \sqrt{6}$$

$$c = \text{or about } 2.45$$

Use the values of a and c to find the eccentricity of the ellipse.

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$$e = \frac{c}{a}$$

$$e = \text{or about } 0.511$$

$$e =$$

7-2 Ellipses

$e =$ and Circles

or about 0.5

$$19. + = 1$$

SOLUTION:

The ellipse is in standard form. So, $a =$ or about 4.8 and $b =$.

First, determine the value of c .

$$c^2 = a^2 - b^2$$

$$c^2 = 23 - 17$$

$$c^2 = 6$$

$$c = \sqrt{6}$$

$$c = \text{or about } 2.45$$

Use the values of a and c to find the eccentricity of the ellipse.

$$e =$$

$$e = \text{or about } 0.511$$

$$20. + = 1$$

SOLUTION:

The ellipse is in standard form. So, $a = \text{or about } 6.16$ and $b =$.

First, determine the value of c .

$$c^2 = a^2 - b^2$$

$$c^2 = 38 - 13$$

$$c^2 = 25$$

$$c = 5$$

$$c = \text{or } 5$$

Use the values of a and c to find the eccentricity of the ellipse.

$$e = \frac{c}{a}$$

$$e = \text{or about } 0.811$$

$$21. + = 1$$

SOLUTION:

The ellipse is in standard form. So, $a = \text{or about } 3.16$ and $b =$.

First, determine the value of c .

$$c^2 = a^2 - b^2$$

$$c^2 = 10 - 8$$

$$c^2 = 2$$

$$c = \sqrt{2}$$

$$c = \text{or } 1.41$$

Use the values of a and c to find the eccentricity of the ellipse.

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$$e = \frac{c}{a}$$

$$e = \text{or about } 0.447$$

$$e =$$

7-2 Ellipses

$e =$ and Circles

or about 0.811

$$21. + = 1$$

SOLUTION:

The ellipse is in standard form. So, $a = \text{or about } 3.16$ and $b =$.

First, determine the value of c .

$$c^2 = a^2 - b^2$$

$$c^2 = 10 - 8$$

$$c^2 = 2$$

$$c = \sqrt{2}$$

$$c = \text{or } 1.41$$

Use the values of a and c to find the eccentricity of the ellipse.

$$e =$$

$$e = \text{or about } 0.447$$

22. RACING The design of an elliptical racetrack with an eccentricity of 0.75 is shown.

- What is the maximum width w of the track?
- Write an equation for the ellipse if the origin x is located at the center of the racetrack.

SOLUTION:

2

a. The length of the track is 1000 feet. So, $2a = 1000$, $a = 500$, and $a = 250,000$. Because the eccentricity is given,

you can use the eccentricity equation to find c .

$$e =$$

$$0.75 =$$

$$375 = c$$

Use the values of a and c to find b .

$$2 \ 2 \ 2$$

$$b^2 = a^2 - c^2$$

$$2 \ 2 \ 2$$

$$b^2 = 500^2 - 375^2$$

$$b^2 =$$

$$b^2 \approx 330.72$$

So, the maximum width of the track is $2(330.72)$ or about 661.44 feet.

- Use the values of a and b to write the equation of the ellipse.

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$$e =$$

7-2 Ellipses

$e =$ and Circles

or about 0.447

22. RACING The design of an elliptical racetrack with an eccentricity of 0.75 is shown.

- What is the maximum width w of the track?
- Write an equation for the ellipse if the origin x is located at the center of the racetrack.

SOLUTION:

2

a. The length of the track is 1000 feet. So, $2a = 1000$, $a = 500$, and $a = 250,000$. Because the eccentricity is given,

you can use the eccentricity equation to find c .

$$e =$$

$$0.75 =$$

$$375 = c$$

Use the values of a and c to find b.

$$b^2 = c^2 - a^2$$

$$b^2 = 500 - 375$$

$$b^2 = 125$$

$$b = \sqrt{125}$$

$$b \approx 330.72$$

So, the maximum width of the track is $2(330.72)$ or about 661.44 feet.

b. Use the values of a and b to write the equation of the ellipse.

23. CARPENTRY A carpenter has been hired to construct a sign for a pet grooming business. The plans for the sign

call for an elliptical shape with an eccentricity of 0.60 and a length of 36 inches.

a. What is the height of the sign?

b. Write an equation for the ellipse if the origin is located at the center of the sign.

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SOLUTION:

7-2 Ellipses and Circles

23. CARPENTRY A carpenter has been hired to construct a sign for a pet grooming business. The plans for the sign

call for an elliptical shape with an eccentricity of 0.60 and a length of 36 inches.

a. What is the height of the sign?

b. Write an equation for the ellipse if the origin is located at the center of the sign.

SOLUTION:

a. The length of the sign is 36 inches. So, $2a = 36$, $a = 18$, and $a^2 = 324$. Because the eccentricity is given, you can

use the eccentricity equation to find c.

$$e = \frac{c}{a}$$

$$0.60 = \frac{c}{18}$$

$$10.8 = c$$

Use the values of a and c to find b.

$$b^2 = c^2 - a^2$$

$$b^2 = 18^2 - 10.8^2$$

$$b^2 = 180$$

$$b = \sqrt{180}$$

$$b \approx 14.4$$

So, the height of the sign is $2(14.4)$ or about 28.8 inches.

$$b = 14.4$$

b. Use the values of a and b to write the equation of the ellipse. $b = 14.4$.

$$+ = 1$$

$$+ = 1$$

Write each equation in standard form. Identify the related conic.

$$24. x^2 + y^2 + 6x - 4y - 3 = 0$$

SOLUTION:

$$x^2 + y^2 + 6x - 4y - 3 = 0$$

$$x^2 + y^2 + 6x - 4y - 3 = 0$$

$$x^2 + y^2 + 6x - 4y - 3 = 0$$

$$x^2 + y^2 + 6x - 4y - 3 = 0$$

$$x^2 + y^2 + 6x - 4y - 3 = 0$$

$$x^2 + y^2 + 6x - 4y - 3 = 0 + 9 + 4$$

$$x^2 + y^2 + 6x - 4y - 3 = 0 + 9 + 4$$

$$(x + 3)^2 + (y - 2)^2 = 16$$

$$(x + 3)^2 + (y - 2)^2 = 16$$

Because $a = b$ and the graph is of the form $(x - h)^2 + (y - k)^2 = r^2$, the conic is a circle.

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$$25. 4x^2 + 8y^2 - 8x + 48y + 44 = 0$$

SOLUTION:

$$4x^2 + 8y^2 - 8x + 48y + 44 = 0$$

$$4x^2 + 8y^2 - 8x + 48y + 44 = 0$$

$$(x + 3)^2 + (y - 2)^2 = 16$$

7-2 Because

Ellipses and Circles

$a \neq b$ and

$$4x^2 + 8y^2 - 8x + 48y + 44 = 0$$

the graph is of the form $(x - h)^2 + (y - k)^2 = r^2$, the conic is a circle.

$$25. 4x^2 + 8y^2 - 8x + 48y + 44 = 0$$

SOLUTION:

$$4x^2 + 8y^2 - 8x + 48y + 44 = 0$$

$$4x^2 + 8y^2 - 8x + 48y + 44 = 0$$

$$4x^2 + 8y^2 - 8x + 48y + 44 = 0$$

$$4x^2 + 8y^2 - 8x + 48y + 44 = 0$$

$$4x^2 + 8y^2 - 8x + 48y + 44 = 0$$

$$4(x - 1)^2 + 8(y + 3)^2 = 32$$

$$4(x - 1)^2 + 8(y + 3)^2 = 32$$

$$4(x - 1)^2 + 8(y + 3)^2 = 32$$

$$+ = 1$$

The related conic is an ellipse because $a \neq b$ and the graph is of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

$$26. x^2 - 8x - 8y - 40 = 0$$

SOLUTION:

2

$$x^2 - 8x - 8y - 40 = 0$$

2

$$x^2 - 8x + 16 - 8y - 40 = 0 + 16$$

2

$$(x - 4)^2 = 8y + 56$$

2

$$(x - 4)^2 = 8(y + 7)$$

Because only one term is squared, the graph is a parabola.

$$27. y^2 - 12x + 18y + 153 = 0$$

SOLUTION:

2

$$y^2 - 12x + 18y + 153 = 0$$

2

$$y^2 + 18y + 153 = 12x$$

2

$$y^2 + 18y + 81 + 153 = 12x + 81$$

2

$$(y + 9)^2 = 12x - 72$$

2

$$(y + 9)^2 = 12(x - 6)$$

Because only one term is squared, the graph is a parabola.

$$28. x^2 + y^2 - 8x - 6y - 39 = 0$$

SOLUTION:

2 2

$$x^2 + y^2 - 8x - 6y - 39 = 0$$

2 2

$$x^2 - 8x + y^2 - 6y - 39 = 0$$

2 2

$$x^2 - 8x + 16 + y^2 - 6y + 9 - 39 = 0 + 16 + 9$$

2 2

$$(x - 4)^2 + (y - 3)^2 = 64$$

2 2 2

Because $a = b$, and the graph is of the form $(x - h)^2 + (y - k)^2 = r^2$, the conic is a circle.

$$29. 3x^2 + y^2 - 42x + 4y + 142 = 0$$

SOLUTION:

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2 2

$$3x^2 + y^2 - 42x + 4y + 142 = 0$$

2 2

$$3x^2 - 42x + y^2 + 4y + 142 = 0$$

2 2

$$x^2 - 8x + 16 + y^2 - 6y + 9 - 39 = 0 + 16 + 9$$

2 2

$$(x - 4)^2 + (y - 3)^2 = 64$$

7-2 Ellipses and Circles 2 2 2

Because $a = b$, and the graph is of the form $(x - h)^2 + (y - k)^2 = r^2$, the conic is a circle.

$$29. 3x^2 + y^2 - 42x + 4y + 142 = 0$$

SOLUTION:

2 2

$$3x^2 + y^2 - 42x + 4y + 142 = 0$$

2 2

$$3x^2 - 42x + y^2 + 4y + 142 = 0$$

2 2

$$3(x^2 - 14x + 49) + y^2 + 4y + 4 + 142 = 0 + 147 + 4$$

2 2

$$3(x - 7)^2 + (y + 2)^2 = 9$$

$\div 9 = 1$

The related conic is an ellipse because $a \neq b$ and the graph is of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

$$30. 5x^2 + 2y^2 + 30x - 16y + 27 = 0$$

SOLUTION:

2 2

$$5x^2 + 2y^2 + 30x - 16y + 27 = 0$$

2 2

$$5x^2 + 30x + 2y^2 - 16y + 27 = 0$$

2 2

$$5(x^2 + 6x + 9) + 2(y^2 - 8y + 16) + 27 = 0 + 45 + 32$$

2 2

$$5(x + 3)^2 + 2(y - 4)^2 = 50$$

$\div 50 = 1$

The related conic is an ellipse because $a \neq b$, and the graph is of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

$$31. 2x^2 + 7y^2 + 24x + 84y + 310 = 0$$

SOLUTION:

The related conic is an ellipse because $a \neq b$, and the graph is of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

32. HISTORY The United States Capitol has a room with an elliptical ceiling. This type of room is called a whispering

gallery because sound that is projected from one focus of an ellipse reflects off the ceiling and back to the other

focus. The room in the Capitol is 96 feet in length, 45 feet wide, and has a ceiling that is 23 feet high.

a. Write an equation modeling the shape of the room. Assume that it is centered at the origin and that the major axis

is horizontal.

b. Find the location of the two foci.

c. How far from one focus would one have to stand to be able to hear the sound reflecting from the other focus?

SOLUTION:

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2

a. The length of the room is 96 feet. So, $2a = 96$, $a = 48$, and $a^2 = 2304$. The width of the room is 45 feet. So, $2b =$

2

45, $b = 22.5$, and $b^2 = 506.25$.

7-2 The

Ellipses

related and

conic Circles

is an ellipse because $a \neq b$, and the graph is of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

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is horizontal.

b. Find the location of the two foci.

c. How far from one focus would one have to stand to be able to hear the sound reflecting from the other focus?

SOLUTION:

2

a. The length of the room is 96 feet. So, $2a = 96$, $a = 48$, and $a^2 = 2304$. The width of the room is 45 feet. So, $2b =$

2

45, $b = 22.5$, and $b^2 = 506.25$.

Use the values of a and b to write the equation of the ellipse.

$\frac{x^2}{2304} + \frac{y^2}{506.25} = 1$

$\frac{x^2}{2304} + \frac{y^2}{506.25} = 1$

b. To find the location of the foci, find c .

$$2^2 + 2^2$$

$$c = a - b$$

$$2$$

$$c = 2304 - 506.25$$

$$c =$$

$$c \approx 42.4$$

So, the foci are about 42 ft on either side of the center along the major axis.

c. The length of the segment between the foci is $2c$ units. So, $2(42)$ is about 84 feet.

Write an equation for a circle that satisfies each set of conditions. Then graph the circle.

33. center at $(3, 0)$, radius 2

SOLUTION:

$$2$$

The center is located at $(3, 0)$, so $h = 3$ and $k = 0$. The radius is 2, so $r = 4$.

Use the values of h , k , and r to write the equation of the circle.

$$2^2 + 2^2$$

$$(x - h)^2 + (y - k)^2 = r^2$$

$$2^2 + 2^2$$

$$(x - 3)^2 + y^2 = 4$$

34. center at $(-1, 7)$, diameter 6

SOLUTION:

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$$2$$

The center is located at $(-1, 7)$, so $h = -1$ and $k = 7$. The diameter is 6, so $r = 3$, and $r = 9$.

7-2 Ellipses and Circles

34. center at $(-1, 7)$, diameter 6

SOLUTION:

$$2$$

The center is located at $(-1, 7)$, so $h = -1$ and $k = 7$. The diameter is 6, so $r = 3$, and $r = 9$.

Use the values of h , k , and r to write the equation of the circle.

$$2^2 + 2^2$$

$$(x - h)^2 + (y - k)^2 = r^2$$

$$2^2 + 2^2 = 9$$

$$(x + 1)^2 + (y - 7)^2 = 9$$

35. center at $(-4, -3)$, tangent to $y = 3$

SOLUTION:

The center is located at $(-4, -3)$, so $h = -4$ and $k = -3$. The line $y = 3$ is a horizontal line. Because the circle is

tangent to this line and its center is at $(-4, -3)$, one endpoint of the circle is at $(-4, 3)$. So, the radius is $3 - (-3) = 6$,

$$2$$

and $r = 36$.

36. center at $(2, 0)$, endpoints of diameter at $(-5, 0)$ and $(9, 0)$

SOLUTION:

2

The center is located at $(2, 0)$, so $h = 2$ and $k = 0$. The diameter is $|-5 - 9|$ or 14, so $r = 7$, and $r = 49$.

Use the values of h , k , and r to write the equation of the circle.

2 2 2

$$(x - h)^2 + (y - k)^2 = r^2$$

2 2

$$(x - 2)^2 + y^2 = 49$$

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37. FORMULA

7-2 Ellipses and Circles

36. center at $(2, 0)$, endpoints of diameter at $(-5, 0)$ and $(9, 0)$

SOLUTION:

2

The center is located at $(2, 0)$, so $h = 2$ and $k = 0$. The diameter is $|-5 - 9|$ or 14, so $r = 7$, and $r = 49$.

Use the values of h , k , and r to write the equation of the circle.

2 2 2

$$(x - h)^2 + (y - k)^2 = r^2$$

2 2

$$(x - 2)^2 + y^2 = 49$$

37. FORMULA Derive the general form of the equation for an ellipse with a vertical major axis centered at the origin.

SOLUTION:

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7-2 Ellipses and Circles

37. FORMULA Derive the general form of the equation for an ellipse with a vertical major axis centered at the origin.

SOLUTION:

38. MEDICAL TECHNOLOGY Indoor Positioning Systems (IPS) use ultrasound waves to detect tags that are

linked to digital files containing information regarding a person or item being monitored. Hospitals often use IPS to

detect the location of moveable equipment and patients.

a. If the tracking system receiver must be centrally located for optimal functioning, where should a receiver be

situated in a hospital complex that is 800 meters by 942 meters?

b. Write an equation that models the sonar range of the IPS.

SOLUTION:

a. The receiver should be situated at the central point of the hospital, which is 471 meters within the long side of the building.

b. The building is shaped like a rectangle. In order to reach every part of the building, a circle must be drawn with a center at the central point of the building. From the central point, the farthest point of the building is at a corner of

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the rectangle. Use the Pythagorean Theorem to find this distance.

2 2 2

7-2 Ellipses and Circles

38. MEDICAL TECHNOLOGY Indoor Positioning Systems (IPS) use ultrasound waves to detect tags that are

linked to digital files containing information regarding a person or item being monitored. Hospitals often use IPS to

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SOLUTION:

a. The receiver should be situated at the central point of the hospital, which is 471 meters within the long side of the building.

b. The building is shaped like a rectangle. In order to reach every part of the building, a circle must be drawn with a center at the central point of the building. From the central point, the farthest point of the building is at a corner of

the rectangle. Use the Pythagorean Theorem to find this distance.

2 2 2

$a + b = c$

2 2 2

$400 + 471 = c$

$381,841 = c^2$

This distance is the radius of the circle that encloses the entire building. So, an equation that models the sonar range

2 2

of the IPS is $x + y = 381,841$.

Write an equation for each ellipse.

39.

SOLUTION:

From the graph, you can see that the vertices are located at $(-3, 3)$ and $(5, 3)$ and the co-vertices are located at $(1,$

$4)$ and $(1, 2)$. The major axis is horizontal, so the standard form of the equation is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

The center is the midpoint of the segment between the vertices, or $(1, 3)$. So, $h = 1$ and $k = 3$.

2

The distance between the vertices is equal to $2a$ units. So, $2a = 8$, $a = 4$, and $a^2 = 16$. The distance between the co-

2

vertices is equal to $2b$ units. So, $2b = 2$, $b = 1$, and $b^2 = 1$.

Using the values of h , k , a , and b , the equation for the ellipse is $\frac{(x-1)^2}{16} + \frac{(y-3)^2}{1} = 1$.

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2

vertices is equal to $2b$ units. So, $2b = 2$, $b = 1$, and $b^2 = 1$.

the values of h , k , a , and b , the equation for the ellipse is $\frac{(x-1)^2}{16} + \frac{(y-3)^2}{1} = 1$.

7-2 Using

Ellipses and Circles

40.

SOLUTION:

From the graph, you can see that the vertices are located at $(4, 6)$ and $(4, -4)$ and the co-vertices are located at $(1,$

$1)$ and $(7, 1)$. The major axis is vertical, so the standard form of the equation is $\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$.

The center is the midpoint of the segment between the vertices, or $(4, 1)$. So, $h = 4$ and $k = 1$.

2

The distance between the vertices is equal to $2a$ units. So, $2a = 10$, $a = 5$, and $a^2 = 25$. The distance between the

2

co-vertices is equal to $2b$ units. So, $2b = 6$, $b = 3$, and $b^2 = 9$.

Using the values of h , k , a , and b , the equation for the ellipse is $\frac{(x-4)^2}{9} + \frac{(y-1)^2}{25} = 1$.

41.

SOLUTION:

From the graph, you can see that the vertices are located at $(2, 9)$ and $(2, -5)$ and the co-vertices are located at $(-2,$

$2)$ and $(6, 2)$. The major axis is vertical, so the standard form of the equation is $\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$.

The center is the midpoint of the segment between the vertices, or $(2, 2)$. So, $h = 2$ and $k = 2$.

2

The distance between the vertices is equal to $2a$ units. So, $2a = 14$, $a = 7$, and $a^2 = 49$. The distance between the

2

co-vertices is equal to $2b$ units. So, $2b = 8$, $b = 4$, and $b = 16$.

Using the values of h , k , a , and b , the equation for the ellipse is $+ = 1$.

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42.

SOLUTION:

2

co-vertices is equal to $2b$ units. So, $2b = 8$, $b = 4$, and $b = 16$.

7-2 Using

Ellipses and ofCircles

the values h , k , a , and b , the equation for the ellipse is $+ = 1$.

42.

SOLUTION:

From the graph, you can see that the vertices are located at $(-9, 5)$ and $(7, 5)$ and the co-vertices are located at $(-1,$

$11)$ and $(-1, 1)$. The major axis is horizontal, so the standard form of the equation is $+ = 1$.

The center is the midpoint of the segment between the vertices, or $(-1, 5)$. So, $h = -1$ and $k = 5$.

2

The distance between the vertices is equal to $2a$ units. So, $2a = 16$, $a = 8$, and $a = 64$. The distance between the

2

co-vertices is equal to $2b$ units. So, $2b = 12$, $b = 6$, and $b = 36$.

Using the values of h , k , a , and b , the equation for the ellipse is $+ = 1$.

43. PLANETARY MOTION Each of the planets in the solar system move around the Sun in an elliptical orbit, where

the Sun is one focus of the ellipse. Mercury is 43.4 million miles from the Sun at its farthest point and 28.6 million

miles at its closest, as shown below. The diameter of the Sun is 870,000 miles.

- Find the length of the minor axis.
- Find the eccentricity of the elliptical orbit.

SOLUTION:

a. The length of the major axis is $43.4 + 28.6 + 0.87 = 72.87$.

The value of a is $72.87 \div 2 = 36.435$. The distance from the focus (the sun) to the vertex is $28.6 + = 29.035$.

Therefore, the value of c , the focus to the center, is $36.435 - 29.035 = 7.4$.

2 2 2

In an ellipse, $c = a - b$. Use the values of a and c to find b .

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2

co-vertices is equal to $2b$ units. So, $2b = 12$, $b = 6$, and $b = 36$.

7-2 Using

Ellipses and ofCircles

the values h , k , a , and b , the equation for the ellipse is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

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the Sun is one focus of the ellipse. Mercury is 43.4 million miles from the Sun at its farthest point and 28.6 million

miles at its closest, as shown below. The diameter of the Sun is 870,000 miles.

- Find the length of the minor axis.
- Find the eccentricity of the elliptical orbit.

SOLUTION:

- The length of the major axis is $43.4 + 28.6 + 0.87 = 72.87$.

The value of a is $72.87 \div 2 = 36.435$. The distance from the focus (the sun) to the vertex is $28.6 + 0.87 = 29.035$.

Therefore, the value of c , the focus to the center, is $36.435 - 29.035 = 7.4$.

$2 \cdot 2 \cdot 2$

In an ellipse, $c^2 = a^2 - b^2$. Use the values of a and c to find b .

The value of $2b$, the length of the minor axis, is about $35.676 \cdot 2 \approx 71.35$ million miles.

- Use the values of a and c to find the eccentricity of the orbit.

$e =$

≈ 0.203

So, the orbit has an eccentricity of about 0.203.

Find the center, foci, and vertices of each ellipse.

44. $\frac{(x+5)^2}{16} + \frac{(y-0)^2}{4} = 1$

SOLUTION:

The ellipse is in standard form, where $h = -5$ and $k = 0$. So, the center is located at $(-5, 0)$. The ellipse has a

$2 \cdot 2$

horizontal orientation, so $a = 4$, $a^2 = 16$, and $b = 2$.

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Use the values of a and b to find c .

$2 \cdot 2 \cdot 2$

$c^2 = a^2 - b^2$

≈ 0.203

7-2 Ellipses and Circles

So, the orbit has an eccentricity of about 0.203.

Find the center, foci, and vertices of each ellipse.

44. $\frac{(x+5)^2}{16} + \frac{(y-0)^2}{4} = 1$

SOLUTION:

The ellipse is in standard form, where $h = -5$ and $k = 0$. So, the center is located at $(-5, 0)$. The ellipse has a

2^2

horizontal orientation, so $a = 16$, $a = 4$, and $b = 7$.

Use the values of a and b to find c .

2^2

$$c^2 = a^2 - b^2$$

2

$$c^2 = 16 - 7$$

2

$$c^2 = 9$$

$$c = 3$$

The foci are c units from the center, so they are located at $(-8, 0)$ and $(-2, 0)$. The vertices are a units from the

center, so they are located at $(-9, 0)$ and $(-1, 0)$.

$$45. x^2 + y^2 = 1$$

SOLUTION:

The ellipse is in standard form, where $h = 0$ and $k = -6$. So, the center is located at $(0, -6)$. The ellipse has a

2^2

horizontal orientation, so $a = 100$, $a = 10$, and $b = 25$.

Use the values of a and b to find c .

2^2

$$c^2 = a^2 - b^2$$

2

$$c^2 = 100 - 25$$

2

$$c^2 = 75$$

$$c = 5 \text{ or } 5$$

The foci are c units from the center, so they are located at $(\pm 5, -6)$. The vertices are a units from the center, so

they are located at $(\pm 10, -6)$.

$$46. 9y^2 - 18y + 25x^2 + 100x - 116 = 0$$

SOLUTION:

First, write the equation in standard form.

The equation is now in standard form, where $k = 1$ and $h = -2$. So, the center is located at $(-2, 1)$.

The ellipse has a

2^2

vertical orientation, so $a = 25$, $a = 5$, and $b = 9$.

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Use the values of a and b to find c .

$$2^2 + 2^2$$

$$c = a - b$$

$$c = \text{or } 5$$

7-2 The foci are and

Ellipses c units from the center, so they are located at $(\pm 5$

Circles, $-6)$. The vertices are a units from the center, so they are located at $(\pm 10, -6)$.

$$46. 9y^2 - 18y + 25x^2 + 100x - 116 = 0$$

SOLUTION:

First, write the equation in standard form.

The equation is now in standard form, where $k = 1$ and $h = -2$. So, the center is located at $(-2, 1)$.

The ellipse has a

$$2^2$$

vertical orientation, so $a = 25$, $a = 5$, and $b = 9$.

Use the values of a and b to find c .

$$2^2 + 2^2$$

$$c = a - b$$

$$2$$

$$c = 25 - 9$$

$$2$$

$$c = 16$$

$$c = 4$$

The foci are c units from the center, so they are located at $(-2, 5)$, $(-2, -3)$. The vertices are a units from the

center, so they are located at $(-2, 6)$, $(-2, -4)$.

$$47. 65x^2 + 16y^2 + 130x - 975 = 0$$

SOLUTION:

First, write the equation in standard form.

The equation is now in standard form, where $h = -1$ and $k = 0$. So, the center is located at $(-1, 0)$.

The ellipse has a

$$2^2$$

vertical orientation, so $a = 65$, $a =$, and $b = 16$.

Use the values of a and b to find c .

$$2^2 + 2^2$$

$$c = a - b$$

$$2$$

$$c = 65 - 16$$

$$2$$

$$c = 49$$

$$c = 7$$

The foci are c units from the center, so they are located at $(-1, \pm 7)$. The vertices are a units from the center, so

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they are located at $(-1, \pm 4)$.

48. TRUCKS Elliptical tanker trucks like the one shown are often used to transport liquids because they are more

$$c = 16$$

$$c = 4$$

7-2 The foci are and

Ellipses c units from the center, so they are located at $(-2, 5)$, $(-2, -3)$. The vertices are a units from the

Circles

center, so they are located at $(-2, 6)$, $(-2, -4)$.

$$47. 65x^2 + 16y^2 + 130x - 975 = 0$$

SOLUTION:

First, write the equation in standard form.

The equation is now in standard form, where $h = -1$ and $k = 0$. So, the center is located at $(-1, 0)$. The ellipse has a

$$2^2$$

vertical orientation, so $a = 65$, $a = 16$, and $b = 16$.

Use the values of a and b to find c .

$$2^2$$

$$c = a - b$$

$$2$$

$$c = 65 - 16$$

$$2$$

$$c = 49$$

$$c = 7$$

The foci are c units from the center, so they are located at $(-1, \pm 7)$. The vertices are a units from the center, so

they are located at $(-1, \pm 4)$.

48. TRUCKS Elliptical tanker trucks like the one shown are often used to transport liquids because they are more

stable than circular tanks and the movement of the fluid is minimized.

- Draw and label the elliptical cross-section of the tank on a coordinate plane.
- Write an equation to represent the elliptical shape of the tank.
- Find the eccentricity of the ellipse.

SOLUTION:

a. The major axis of the ellipse is 6.8 ft, so $a = 3.4$. The minor axis is 4.2 ft, so $b = 2.1$. The ellipse is centered about

the origin, so $h = k = 0$. The major axis is along the x -axis, so the ellipse is horizontal. The vertices are located at $(0$

$\pm 3.4, 0)$ and the co-vertices are located at $(0, 0 \pm 2.1)$.

Plot the vertices and co-vertices to sketch the ellipse.

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a. The major axis of the ellipse is 6.8 ft, so $a = 3.4$. The minor axis is 4.2 ft, so $b = 2.1$. The ellipse is centered about

the origin, so $h = k = 0$. The major axis is along the x -axis, so the ellipse is horizontal. The vertices are located at $(0$

$\pm 3.4, 0)$ and the co-vertices are located at $(0, 0 \pm 2.1)$.

7-2 Ellipses and Circles

Plot the vertices and co-vertices to sketch the ellipse.

b. The orientation of the ellipse is horizontal, so the standard form of the equation is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Substitute the values of a and b into the equation.

c. Use the values of a and b to find c .

$$c^2 = a^2 - b^2$$

$$c^2 = 11.56 - 4.41$$

$$c^2 = 7.15$$

$$c = \sqrt{7.15}$$

$$c = 2.67$$

$$c \approx 2.67$$

$$c \approx 2.67$$

Substitute a and c into the eccentricity equation.

$$e = \frac{c}{a}$$

$$e \approx 0.79$$

So, the eccentricity of the ellipse is about 0.79.

Write the standard form of the equation for each ellipse.

49. The vertices are at $(-10, 0)$ and $(10, 0)$ and the eccentricity e is $\frac{3}{5}$.

SOLUTION:

$$2a = 20$$

The vertices are located at $(-10, 0)$ and $(10, 0)$, so $2a = 20$, $a = 10$, and $a^2 = 100$.

The eccentricity of the ellipse is given, so you can use the eccentricity equation to find c .

$$e = \frac{c}{a}$$

$$\frac{3}{5} = \frac{c}{10}$$

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Use the values of a and c to find b

$$c^2 = a^2 - b^2$$

$$9 = 100 - b^2$$

$$b^2 = 91$$

7-2 Ellipses and Circles

So, the eccentricity of the ellipse is about 0.79.

Write the standard form of the equation for each ellipse.

49. The vertices are at $(-10, 0)$ and $(10, 0)$ and the eccentricity e is .

SOLUTION:

2

The vertices are located at $(-10, 0)$ and $(10, 0)$, so $2a = 20$, $a = 10$, and $a^2 = 100$.

The eccentricity of the ellipse is given, so you can use the eccentricity equation to find c .

$e =$

$\frac{6}{10} = \frac{c}{10}$

Use the values of a and c to find b

$a^2 = b^2 + c^2$

$10^2 = b^2 + 6^2$

$100 = b^2 + 36$

$64 = b^2$

2

$36 = 100 - b^2$

2

$64 = b^2$

The y -term is the same for each vertex, so the ellipse is oriented horizontally. The center is the midpoint of the

vertices or $(0, 0)$. So, $h = 0$ and $k = 0$. Therefore, the equation of the ellipse is $\frac{x^2}{100} + \frac{y^2}{64} = 1$.

50. The co-vertices are at $(0, 1)$ and $(6, 1)$ and the eccentricity e is .

SOLUTION:

2

The co-vertices are located at $(0, 1)$ and $(6, 1)$, so $2b = 6$, $b = 3$, and $b^2 = 9$. The center is located at the midpoint of

the co-vertices, or $(3, 1)$. So, $h = 3$ and $k = 1$. The y -term is the same for each co-vertex, so the ellipse is oriented

vertically and the standard form of the equation is $\frac{(y-1)^2}{9} + \frac{(x-3)^2}{a^2} = 1$.

Substitute $e = \frac{4}{5}$ into the eccentricity equation, and solve for c .

$e =$

$\frac{4}{5} = \frac{c}{a}$

$a^2 = b^2 + c^2$

Substitute b and c into $c^2 = a^2 - b^2$ to find a .

$a^2 = b^2 + c^2$

$c^2 = a^2 - b^2$

2

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2

$$=a -9$$

The y-term is the same for each vertex, so the ellipse is oriented horizontally. The center is the midpoint of the

$$+ = 1.$$

7-2 vertices

Ellipses or (0, 0). So, $h = 0$ and $k = 0$. Therefore, the equation of the ellipse is and Circles

50. The co-vertices are at (0, 1) and (6, 1) and the eccentricity e is .

SOLUTION:

2

The co-vertices are located at (0, 1) and (6, 1), so $2b = 6$, $b = 3$, and $b = 9$. The center is located at the midpoint of

the co-vertices, or (3, 1). So, $h = 3$ and $k = 1$. The y-term is the same for each co-vertex, so the ellipse is oriented

vertically and the standard form of the equation is $+ = 1$.

Substitute $e =$ into the eccentricity equation, and solve for c .

$$e =$$

$$a =c$$

$$2 \ 2 \ 2$$

Substitute b and c into $c = a - b$ to find a .

$$2 \ 2 \ 2$$

$$c =a -b$$

$$2$$

$$=a -9$$

$$2$$

$$=a -9$$

$$9 =$$

$$2$$

$$25 =a$$

Therefore, the equation of the ellipse is $+ =1$

51. The center is at (2, -4), one focus is at (2, $-4 + 2$), and the eccentricity e is .

SOLUTION:

The center is located at (2, -4), so $h = 2$ and $k = -4$. The distance from a focus to the center is equal to c . Since one

focus is located at (2, $-4 + 2$), $c = 2$. Since the x-coordinates of the center and a focus are the same, the

ellipse is oriented horizontally and the standard form of the equation is $+ = 1$.

Find a.

$e =$

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$a = 6$

2

$25 = a^2$

7-2 Therefore,

Ellipses the

and Circles

equation of the ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

51. The center is at $(2, -4)$, one focus is at $(2, -4 + 2)$, and the eccentricity e is .

SOLUTION:

The center is located at $(2, -4)$, so $h = 2$ and $k = -4$. The distance from a focus to the center is equal to c . Since one

focus is located at $(2, -4 + 2)$, $c = 2$. Since the x -coordinates of the center and a focus are the same, the

ellipse is oriented horizontally and the standard form of the equation is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

Find a.

$e =$

$a = 6$

2

So, $a = 6$ and $a^2 = 36$.

Find b.

$c^2 = a^2 - b^2$

$2^2 = 6^2 - b^2$

$4 = 36 - b^2$

$(2)^2 = 6^2 - b^2$

$20 = 36 - b^2$

2

$b^2 = 16$

Using the values of h , k , a , and b , the equation for the ellipse is $\frac{(x-2)^2}{36} + \frac{(y+4)^2}{16} = 1$.

52. ROLLER COASTERS The shape of a roller coaster loop in an amusement park can be modeled by $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

a. What is the width of the loop along the horizontal axis?

b. Determine the height of the roller coaster from the ground when it reaches the top of the loop, if the lower rail is

20 feet from ground level.

c. Find the eccentricity of the ellipse.

SOLUTION:

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

a. The equation is in standard form and 2025 is associated with the x -term and is less than 3306.25. So, $b = 2025$

and $b = 45$. The width of the loop along the horizontal line is $2 \cdot 45$ or 90 ft.

b. Find a .

2

$a = 3306.25$

$a =$ or about 57.5

So, $a = 57.5$ and $2a = 115$. Since the lower rail is 20 ft above ground level, the maximum height is $115 + 20$ or 135 ft.

c. Find c .

2 2 2

$c = a - b$

2

$c = 3306.25 - 2025$

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$c =$ or about 35.79

2 = 16

b

7-2 Using

Ellipses and of Circles

the values h , k , a , and b , the equation for the ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

52. ROLLER COASTERS The shape of a roller coaster loop in an amusement park can be modeled by $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

a. What is the width of the loop along the horizontal axis?

b. Determine the height of the roller coaster from the ground when it reaches the top of the loop, if the lower rail is

20 feet from ground level.

c. Find the eccentricity of the ellipse.

SOLUTION:

2 2

a. The equation is in standard form and 2025 is associated with the x -term and is less than 3306.25. So, $b = 2025$

and $b = 45$. The width of the loop along the horizontal line is $2 \cdot 45$ or 90 ft.

b. Find a .

2

$a = 3306.25$

$a =$ or about 57.5

So, $a = 57.5$ and $2a = 115$. Since the lower rail is 20 ft above ground level, the maximum height is $115 + 20$ or 135 ft.

c. Find c .

2 2 2

$c = a - b$

2

$$c = 3306.25 - 2025$$

$$c = \text{or about } 35.79$$

Now you can find the eccentricity of the ellipse.

$$e =$$

$$\approx 0.62$$

53. FOREST FIRES The radius of a forest fire is expanding at a rate of 4 miles per day. The current state of the fire

is shown below, where a city is located 20 miles southeast of the fire.

a. Write the equation of the circle at the current time and the equation of the circle at the time the fire reaches the

city.

b. Graph both circles.

c. If the fire continues to spread at the same rate, how many days will it take to reach the city?

SOLUTION:

2 2

a. The radius of the small circle is 8. So, the equation is $x^2 + y^2 = 64$. When the fire reaches the city, the radius will

2 2

be $8 + 20$ or 28. The equation for the circle is $x^2 + y^2 = 784$.

b. Use the center of $(0, 0)$, $r = 8$, and $r = 28$ to graph both circles.

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$$e =$$

7-2 Ellipses and Circles

$$\approx 0.62$$

53. FOREST FIRES The radius of a forest fire is expanding at a rate of 4 miles per day. The current state of the fire

is shown below, where a city is located 20 miles southeast of the fire.

a. Write the equation of the circle at the current time and the equation of the circle at the time the fire reaches the

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b. Graph both circles.

c. If the fire continues to spread at the same rate, how many days will it take to reach the city?

SOLUTION:

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a. The radius of the small circle is 8. So, the equation is $x^2 + y^2 = 64$. When the fire reaches the city, the radius will

2 2

be $8 + 20$ or 28. The equation for the circle is $x^2 + y^2 = 784$.

b. Use the center of $(0, 0)$, $r = 8$, and $r = 28$ to graph both circles.

c. The radius of the fire is growing 4 miles per day. It will reach the city if $20 \div 4 = 5$ days.

54. The latus rectum of an ellipse is a line segment that passes through a focus, is perpendicular to the major axis of the ellipse, and has endpoints on the ellipse. The length of each latus rectum is units, where a is half the length of the major axis and b is half the length of the minor axis.

Write the equation of a horizontal ellipse with center at $(3, 2)$, major axis is 16 units long, and latus rectum 12 units long.

SOLUTION:

2

The length of the major axis is equal to 16. So, $2a = 16$ and $a = 8$, and $a^2 = 64$. The length of the latus rectum is 12, so $12 = \frac{2b^2}{a}$.

2

Find b .

12Manual

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12 =

7-2 Ellipses and Circles

c. The radius of the fire is growing 4 miles per day. It will reach the city if $20 \div 4 = 5$ days.

54. The latus rectum of an ellipse is a line segment that passes through a focus, is perpendicular to the major axis of the ellipse, and has endpoints on the ellipse. The length of each latus rectum is units, where a is half the length of the major axis and b is half the length of the minor axis.

Write the equation of a horizontal ellipse with center at $(3, 2)$, major axis is 16 units long, and latus rectum 12 units long.

SOLUTION:

2

The length of the major axis is equal to 16. So, $2a = 16$ and $a = 8$, and $a^2 = 64$. The length of the latus rectum is 12, so $12 = \frac{2b^2}{a}$.

2

Find b .

12 =

12 =

2

48 = b^2

The ellipse is oriented horizontally, so the standard form of the equation is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$. Using the values of h , k , a , and b , the equation for the ellipse is $\frac{(x-3)^2}{64} + \frac{(y-2)^2}{48} = 1$.

Find the coordinates of points where a line intersects a circle.

$$55. y = x - 8, (x - 7)^2 + (y + 5)^2 = 16$$

SOLUTION:

Solve the system of equations.

$$2 \quad 2$$

$$(x - 7)^2 + (y + 5)^2 = 16$$

$$2 \quad 2$$

$$(x - 7)^2 + (x - 8 + 5)^2 = 16$$

$$2 \quad 2$$

$$x^2 - 14x + 49 + x^2 - 6x + 9 = 16$$

$$2$$

$$2x^2 - 20x + 42 = 0$$

$$2$$

$$x^2 - 10x + 21 = 0$$

$$(x - 7)(x - 3) = 0$$

$$x = 7 \text{ or } 3$$

Substitute $x = 3$ and $x = 7$ back into the first equation to find the y-coordinate for each point of intersection.

$$y = x - 8$$

$$= 7 - 8 \text{ or } -1$$

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$$y = x - 8$$

$$= 3 - 8 \text{ or } -5$$

The ellipse is oriented horizontally, so the standard form of the equation is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$. Using the

7-2 values

Ellipses

of h, and Circles

k, a, and b, the equation for the ellipse is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

Find the coordinates of points where a line intersects a circle.

$$55. y = x - 8, (x - 7)^2 + (y + 5)^2 = 16$$

SOLUTION:

Solve the system of equations.

$$2 \quad 2$$

$$(x - 7)^2 + (y + 5)^2 = 16$$

$$2 \quad 2$$

$$(x - 7)^2 + (x - 8 + 5)^2 = 16$$

$$2 \quad 2$$

$$x^2 - 14x + 49 + x^2 - 6x + 9 = 16$$

$$2$$

$$2x^2 - 20x + 42 = 0$$

$$2$$

$$x^2 - 10x + 21 = 0$$

$$(x - 7)(x - 3) = 0$$

$$x = 7 \text{ or } 3$$

Substitute $x = 3$ and $x = 7$ back into the first equation to find the y-coordinate for each point of intersection.

$$y = x - 8$$

$$= 7 - 8 \text{ or } -1$$

$$y = x - 8$$

$$= 3 - 8 \text{ or } -5$$

So, the coordinates of the intersection are $(7, -1)$ and $(3, -5)$.

$$56. y = x + 9, (x - 3)^2 + (y + 5)^2 = 169$$

SOLUTION:

Solve the system of equations.

$$2 \quad 2$$

$$(x - 3)^2 + (y + 5)^2 = 169$$

$$2 \quad 2$$

$$(x - 3)^2 + (x + 9 + 5)^2 = 169$$

$$2 \quad 2$$

$$x^2 - 6x + 9 + x^2 + 28x + 196 = 169$$

$$2$$

$$2x^2 + 22x + 36 = 0$$

$$2$$

$$x^2 + 11x + 18 = 0$$

$$(x + 9)(x + 2) = 0$$

$$x = -9 \text{ or } -2$$

Substitute $x = -9$ and $x = -2$ back into the first equation to find the y-coordinate for each point of intersection.

$$y = x + 9$$

$$= -9 + 9 \text{ or } 0$$

$$y = x + 9$$

$$= -2 + 9 \text{ or } 7$$

So, the coordinates of the intersection are $(-2, 7)$ and $(-9, 0)$.

$$57. y = -x + 1, (x - 5)^2 + (y - 2)^2 = 50$$

SOLUTION:

Solve the system of equations.

$$2 \quad 2$$

$$(x - 5)^2 + (y - 2)^2 = 50$$

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$$(x - 5)^2 + (-x + 1 - 2)^2 = 50$$

2 2

$$x^2 - 10x + 25 + x + 2x + 1 = 50$$

2

$$y = x + 9$$

$$= -2 + 9 \text{ or } 7$$

7-2 Ellipses and Circles

So, the coordinates of the intersection are $(-2, 7)$ and $(-9, 0)$.

$$57. y = -x + 1, (x - 5)^2 + (y - 2)^2 = 50$$

SOLUTION:

Solve the system of equations.

2 2

$$(x - 5)^2 + (y - 2)^2 = 50$$

2 2

$$(x - 5)^2 + (-x + 1 - 2)^2 = 50$$

2 2

$$x^2 - 10x + 25 + x + 2x + 1 = 50$$

2

$$2x^2 - 8x - 24 = 0$$

2

$$x^2 - 4x - 12 = 0$$

$$(x - 6)(x + 2) = 0$$

$$x = 6 \text{ or } -2$$

Substitute $x = 6$ and $x = -2$ back into the first equation to find the y -coordinate for each point of intersection.

$$y = -x + 1$$

$$= -6 + 1 \text{ or } -5$$

$$y = -x + 1$$

$$= -(-2) + 1 \text{ or } 3$$

So, the coordinates of the intersection are $(-2, 3)$ and $(6, -5)$.

2 2

$$58. (x + 3)^2 + (y - 3)^2 = 25$$

SOLUTION:

Solve the system of equations.

2 2

$$(x + 3)^2 + (y - 3)^2 = 25$$

2

$$(x + 3)^2 + = 25$$

2 2

$$x^2 + 6x + 9 + x + 4x + 36 = 25$$

2

$$x^2 + 10x + 20 = 0$$

2

$$10x^2 + 90x + 180 = 0$$

2

$$x^2 + 9x + 18 = 0$$

$$(x + 6)(x + 3) = 0$$

$$x = -6 \text{ or } -3$$

Substitute $x = -6$ and $x = -3$ back into the first equation to find the y-coordinate for each point of intersection.

$$y =$$

$$= 2 - 3 \text{ or } -1$$

$$y =$$

$$= 1 - 3 \text{ or } -2$$

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So, the coordinates of the intersection are $(-3, -2)$ and $(-6, -1)$.

59. REFLECTION Silvering is the process of coating glass with a reflective substance. The interior of an ellipse can

$$y = -x + 1$$

$$= -(-2) + 1 \text{ or } 3$$

7-2 Ellipses and Circles

So, the coordinates of the intersection are $(-2, 3)$ and $(6, -5)$.

2 2

$$58. \quad (x + 3)^2 + (y - 3)^2 = 25$$

SOLUTION:

Solve the system of equations.

2 2

$$(x + 3)^2 + (y - 3)^2 = 25$$

2

$$(x + 3)^2 = 25$$

2 2

$$x^2 + 6x + 9 + x^2 + 4x + 36 = 25$$

2

$$x^2 + 10x + 20 = 0$$

2

$$10x^2 + 90x + 180 = 0$$

2

$$x^2 + 9x + 18 = 0$$

$$(x + 6)(x + 3) = 0$$

$$x = -6 \text{ or } -3$$

Substitute $x = -6$ and $x = -3$ back into the first equation to find the y -coordinate for each point of intersection.

$$y =$$

$$= 2 - 3 \text{ or } -1$$

$$y =$$

$$= 1 - 3 \text{ or } -2$$

So, the coordinates of the intersection are $(-3, -2)$ and $(-6, -1)$.

59. REFLECTION Silvering is the process of coating glass with a reflective substance. The interior of an ellipse can

be silvered to produce a mirror with rays that originate at the ellipse's focus and then reflect to the other focus, as

shown. If the segment V_1F_1 is 2 cm long and the eccentricity of the mirror is 0.5, find the equation of the ellipse in

standard form.

SOLUTION:

If V_1F_1 is 2 cm, then the distance from the vertex to the center is 2 cm greater than the distance from the focus to

the center, and $a = c + 2$. If the eccentricity is 0.5, then $e = 0.5$.

Solve the system of equations.

$$e = 0.5$$

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$$c = 0.5c + 1$$

$$0.5c = 1$$

$$c = 2$$

$$y =$$

$$= 1 - 3 \text{ or } -2$$

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So, the coordinates of the intersection are $(-3, -2)$ and $(-6, -1)$.

59. REFLECTION Silvering is the process of coating glass with a reflective substance. The interior of an ellipse can

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SOLUTION:

If V_1F_1 is 2 cm, then the distance from the vertex to the center is 2 cm greater than the distance from the focus to

the center, and $a = c + 2$. If the eccentricity is 0.5, then $e = 0.5$.

Solve the system of equations.

$$e = 0.5$$

$$c = 0.5c + 1$$

$$0.5c = 1$$

$$c = 2$$

By substituting $c = 2$ back into one of the original equations, you can find that $a = 4$. Use the values of a and c to

find b .

$$2^2 + 2^2 + b^2 = 16$$

$$c = a - b$$

$$2^2 + 2^2 + b^2 = 16$$

$$2 = 4 - b$$

$$4 = 16 - b^2$$

$$b^2 = 12$$

$$b = \sqrt{12}$$

So, the equation of the ellipse is $\frac{x^2}{16} + \frac{y^2}{12} = 1$.

60. CHEMISTRY Distillation columns are used to separate chemical substances based on the differences in their

rates of evaporation. The columns may contain plates with bubble caps or small circular openings.

a. Write an equation for the plate shown, assuming that the center is at $(-4, -1)$.

b. What is the surface area of the plate not covered by bubble caps if each cap is 2 inches in diameter?

SOLUTION:

$$2^2 + 2^2 + b^2 = 16$$

a. The plate is circular with center $(-4, -1)$. The diameter is 18 inches, so the radius r is 9 and r^2 is 81. The equation

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$$2^2 + 2^2 + b^2 = 16$$

of the circle is $(x + 4)^2 + (y + 1)^2 = 81$.

$$2^2 + 2^2 + b^2 = 16$$

b. The area of the plate is $\pi \cdot 9^2$ or 81π . The area of a cap is $\pi \cdot 1^2$ or π . There are 27 caps, so the area not

$$2 = 12$$

$$b = \sqrt{12}$$

7-2 So,

Ellipses and Circles

the equation the ellipse is $\frac{x^2}{16} + \frac{y^2}{12} = 1$.

60. CHEMISTRY Distillation columns are used to separate chemical substances based on the differences in their

rates of evaporation. The columns may contain plates with bubble caps or small circular openings.

a. Write an equation for the plate shown, assuming that the center is at $(-4, -1)$.

b. What is the surface area of the plate not covered by bubble caps if each cap is 2 inches in diameter?

SOLUTION:

2

a. The plate is circular with center $(-4, -1)$. The diameter is 18 inches, so the radius r is 9 and r is 81. The equation

2 2

of the circle is $(x + 4)^2 + (y + 1)^2 = 81$.

2 2

b. The area of the plate is $\pi \cdot 9^2$ or 81π . The area of a cap is $\pi \cdot 1^2$ or π . There are 27 caps, so the area not

2

covered by caps is $81\pi - 27\pi$ or about 169.6 in².

61. GEOMETRY The graphs of $x - 5y = -3$, $2x + 3y = 7$, and $4x - 7y = 27$ contain the sides of a triangle. Write the

equation of a circle that circumscribes the triangle.

SOLUTION:

Graph the triangle.

The equation of the circle that circumscribes the triangle is found by locating the circumcenter of the triangle.

Locate the perpendicular bisectors of each side of the triangle.

Find the midpoints of each segment

$(2, 1)$ and $(12, 3)$

$(5, -1)$ and $(12, 3)$

$(2, 1)$ and $(5, -1)$

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7-2 Ellipses and Circles

$(2, 1)$ and $(5, -1)$

Thus, the midpoints of the segments are $(8.5, 1)$, $(3.5, 0)$, and $(7, 2)$.

Find the equations for the perpendicular bisectors.

Find the slope of segment through points $(2, 1)$ and $(12, 3)$.

$m_{\perp}$ is -5 .

Find equation of line perpendicular through $(7, 2)$

Find the slope of segment through points $(5, -1)$ and $(12, 3)$.

$m_{\perp}$ is .

Find equation of line perpendicular through $(8.5, 1)$.

Find the slope of segment through points $(2, 1)$ and $(5, -1)$.

$m_{\perp}$ is .

Find equation of line perpendicular through $(3.5, 0)$.

Thus, the equations of the perpendicular bisectors are:

$y = -x +$

$$y = x -$$

$$y = -5x + 37$$

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Find the intersection of the perpendicular bisectors to find the circumcenter.

$$y = -x +$$

$$y = x -$$

7-2 Ellipses and Circles

$$y = -5x + 37$$

Find the intersection of the perpendicular bisectors to find the circumcenter.

$$y = -x + \text{ \& } y = x -$$

$$y = x - \text{ \& } y = -5x + 37$$

$$y = -x + \text{ and } y = -5x + 37.$$

Thus, these three lines intersect at the circumcenter of the triangle, (6.5, 4.5).

(6.5, 4.5) is the center of the circle.

The radius of the circle is the distance between this point and any vertex

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formula to find the radius. Page 42

(6.5, 4.5) and (12, 3)

7-2 (6.5,

Ellipses and Circles

4.5) is the center of the circle.

The radius of the circle is the distance between this point and any vertex

Use the distance formula to find the radius.

(6.5, 4.5) and (12, 3)

(6.5, 4.5) and (5, -1)

(6.5, 4.5) and (2, 1)

2 2 2

Thus, the radius is and $r = 32.5$. Then equation of the circle is $(x - 6.5)^2 + (y - 4.5)^2 = 32.5$.

Write the standard form of the equation of a circle that passes through each set of points. Then identify

the center and radius of the circle.

62. (2, 3), (8, 3), (5, 6)

SOLUTION:

2 2 2

The standard form of the equation for a circle is $(x - h)^2 + (y - k)^2 = r^2$. Substitute each point into the equation to

obtain three equations.

2 2 2

$$(2 - h) + (3 - k) = r$$

2 2 2

$$(8 - h) + (3 - k) = r$$

2 2 2

$$(5 - h) + (6 - k) = r$$

2 2 2 2 2 2

Since $(2 - h) + (3 - k) = r$ and $(8 - h) + (3 - k) = r$ both have the same y-coordinate, you can solve that

system of equations to find h.

Next, substitute $h = 5$ into two of the equations to get a system of two equations with k and r.

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7-2 Ellipses and Circles

Next, substitute $h = 5$ into two of the equations to get a system of two equations with k and r.

Solve these two new equations for k.

Substitute $h = 5$ and $k = 3$ into any of the original equations to find r.

Since the radius must be positive, $r = 3$.

2 2

The equation of the circle is $(x - 5) + (y - 3) = 9$, with center (5, 3) and radius $r = 3$.

63. (1, -11), (-3, -7), (5, -7)

SOLUTION:

2 2 2

The standard form of the equation for a circle is $(x - h) + (y - k) = r$. Substitute each point into the equation to

obtain three equations.

2 2 2

$$(1 - h) + (-11 - k) = r$$

2 2 2

$$(-3 - h) + (-7 - k) = r$$

2 2 2

$$(5 - h) + (-7 - k) = r$$

2 2 2 2 2 2

Since $(-3 - h) + (-7 - k) = r$ and $(5 - h) + (-7 - k) = r$ both have the same y-coordinate, you can solve that

system of equations to find h.

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Next, Substitute $h = 1$ into tow equations to get a system of two equations with k and r.

2 2 2

$$(5 - h) + (-7 - k) = r$$

2 2 2 2 2 2

$(-3 - h) + (-7 - k) = r$ and $(5 - h) + (-7 - k) = r$ both have the same y-coordinate, you can solve that

7-2 Since

Ellipses and Circles

system of equations to find h.

Next, Substitute $h = 1$ into tow equations to get a system of two equations with k and r.

Solve these two new equations for k.

Substitute $h = 1$ and $k = -7$ into any of the equations to find r.

Since the radius must be positive, $r = 4$.

2 2

So, the equation of the circle is $(x - 1) + (y + 7) = 16$, with center $(1, -7)$ and radius $r = 4$.

64. $(0, 9)$, $(0, 3)$, $(-3, 6)$

SOLUTION:

2 2 2

The standard form of the equation for a circle is $(x - h) + (y - k) = r$. Substitute each point into the equation to

obtain three equations.

2 2 2 2 2 2

$(0 - h) + (9 - k) = r$ or $h + (9 - k) = r$

2 2 2 2 2 2

$(0 - h) + (3 - k) = r$ or $h + (3 - k) = r$

2 2 2

$(-3 - h) + (6 - k) = r$

2 2 2 2 2 2

Since $h + (9 - k) = r$ and $h + (9 - k) = r$ both have 0 as a y-coordinate, you can solve that system of

equations

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Since the radius must be positive, $r = 4$.

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So, the equation of the circle is $(x - 1) + (y + 7) = 16$, with center $(1, -7)$ and radius $r = 4$.

64. $(0, 9)$, $(0, 3)$, $(-3, 6)$

SOLUTION:

2 2 2

The standard form of the equation for a circle is $(x - h) + (y - k) = r$. Substitute each point into the equation to

obtain three equations.

2 2 2 2 2 2

$$(0 - h) + (9 - k) = r \text{ or } h + (9 - k) = r$$

2 2 2 2 2 2

$$(0 - h) + (3 - k) = r \text{ or } h + (3 - k) = r$$

2 2 2

$$(-3 - h) + (6 - k) = r$$

2 2 2 2 2 2

Since $h + (9 - k) = r$ and $h + (9 - k) = r$ both have 0 as a y-coordinate, you can solve that system of

equations to find k.

Next, substitute $k = 6$ into two of the equations to get a system of two equations in h and r.

Solve these two new equations for h.

Substitute $h = 0$ and $k = 6$ into any of the original equations to find r.

Since the radius must be positive, $r = 3$.

2 2

So, the equation of the circle is $x + (y - 6) = 9$, with center $(0, 6)$ and radius $r = 3$.

65. $(7, 4), (-1, 12), (-9, 4)$

SOLUTION:

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2 2 2

The standard form of the equation for a circle is $(x - h) + (y - k) = r$. Substitute each point into the equation to

obtain three equations.

Since the radius must be positive, $r = 3$.

2 2

So, the equation of the circle is $x + (y - 6) = 9$, with center $(0, 6)$ and radius $r = 3$.

7-2 Ellipses and Circles

65. $(7, 4), (-1, 12), (-9, 4)$

SOLUTION:

2 2 2

The standard form of the equation for a circle is $(x - h) + (y - k) = r$. Substitute each point into the equation to

obtain three equations.

2 2 2

$$(7 - h) + (4 - k) = r$$

2 2 2

$$(-1 - h) + (12 - k) = r$$

2 2 2

$$(-9 - h) + (4 - k) = r$$

2 2 2 2 2 2

Since $(7 - h) + (4 - k) = r$ and $(-9 - h) + (4 - k) = r$ both have the same y -coordinate, you can solve that

system of equations to find h .

Next, substitute $h = -1$ into two equations to get a system of two equations with k and r .

Solve these two new equations for k .

Substitute $h = -1$ and $k = 4$ into any of the original equations to find r .

Since the radius must be positive, $r = 8$.

2 2

So, the equation of the circle is $(x + 1)^2 + (y - 4)^2 = 64$, with center $(-1, 4)$ and radius $r = 8$.

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66. ERROR ANALYSIS Yori and Chandra are graphing an ellipse that has a center at $(-1, 3)$, a major axis of length

8, and a minor axis of length 4. Is either of them correct? Explain your reasoning.

Since the radius must be positive, $r = 8$.

7-2 Ellipses and Circles

2 2

So, the equation of the circle is $(x + 1)^2 + (y - 4)^2 = 64$, with center $(-1, 4)$ and radius $r = 8$.

66. ERROR ANALYSIS Yori and Chandra are graphing an ellipse that has a center at $(-1, 3)$, a major axis of length

8, and a minor axis of length 4. Is either of them correct? Explain your reasoning.

SOLUTION:

Both Yori and Chandra are correct. In Yori's graph, the major axis is horizontal. In Chandra's graph, the major axis is vertical.

67. REASONING Determine whether an ellipse represented by $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $r > 0$, will have the same

foci as the ellipse represented by $\frac{x^2}{p} + \frac{y^2}{r} = 1$. Explain your reasoning.

SOLUTION:

No, they will not have the same foci.

Sample answer:

For $\frac{x^2}{p} + \frac{y^2}{r} = 1$, $p > 0$ and $r > 0$, so $p + r > p$. Then $\frac{x^2}{p} + \frac{y^2}{r} = 1$ is of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and is elongated vertically with $a^2 = p + r$ and $b^2 = p$. Use these values to find c and the foci.

So, the foci are found up and down from the center $(0, 0)$. Therefore, the foci are $(0, \pm c)$.

For $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $p > 0$ and $r > 0$, so $p + r > p$. Then $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is of the form $\frac{x^2}{p} + \frac{y^2}{r} = 1$ and is elongated horizontally with $a^2 = p + r$ and $b^2 = p$. Use these values to find c and the foci.

So, the foci are found left and right from the center $(0, 0)$. Therefore, the foci are $(\pm c, 0)$.

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CHALLENGE The area A of an ellipse of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $A = \pi ab$. Write an equation of an ellipse with

each of the following characteristics.

7-2 Ellipses and Circles

So, the foci are found left and right from the center (0, 0). Therefore, the foci are .

CHALLENGE The area A of an ellipse of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $A = \pi ab$. Write an equation of an ellipse with

each of the following characteristics.

68. $b + a = 12$, $A = 35\pi$

SOLUTION:

The area of the ellipse is 35π , so $35 = ab$. Now, you have a system of two equations, $35 = ab$ and $b = 12 - a$.

Solve the system of equations.

$$35 = ab$$

$$35 = a(12 - a)$$

$$35 = 12a - a^2$$

$$0 = -a^2 + 12a - 35$$

$$0 = (-a + 7)(a - 5)$$

$$2$$

Since a cannot be negative, $a = 5$ and $a = 25$.

Substitute $a = 5$ into one of the equations to find b .

$$b + a = 12$$

$$b + 5 = 12$$

$$b = 7$$

$$2$$

So, $b = 7$ and $b = 49$.

Therefore, an equation of the ellipse is $\frac{x^2}{25} + \frac{y^2}{49} = 1$.

69. $a - b = -5$, $A = 24\pi$

SOLUTION:

The area of the ellipse is 24π , so $24 = ab$. Now, you have a system of two equations, $24 = ab$ and $a = 5 + b$.

Solve the system of equations.

$$24 = ab$$

$$24 = b(b + 5)$$

$$24 = b^2 + 5b$$

$$0 = b^2 + 5b - 24$$

$$0 = (b + 8)(b - 3)$$

$$2$$

Since b cannot be negative, $b = 3$ and $b = 9$.

Substitute $b = 8$ into one of the equations to find a .

$$a - b = 5$$

$$a - 3 = 5$$

$$a = 8$$

$$2$$

So, aManual

eSolutions = 8 and a = 64.

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Therefore, an equation of the ellipse is $\frac{x^2}{64} + \frac{y^2}{49} = 1$.

$$b = 7$$

$$2$$

So, b = 7 and b = 49.

$$\frac{x^2}{64} + \frac{y^2}{49} = 1.$$

7-2 Therefore,

Ellipses an equation of the ellipse is

and Circles

$$69. a - b = -5, A = 24\pi$$

SOLUTION:

The area of the ellipse is 24π , so $24 = ab$. Now, you have a system of two equations, $24 = ab$ and $a = 5 + b$.

Solve the system of equations.

$$24 = ab$$

$$24 = b(b + 5)$$

$$24 = b^2 + 5b$$

$$0 = b^2 + 5b - 24$$

$$0 = (b + 8)(b - 3)$$

$$2$$

Since b cannot be negative, $b = 3$ and $b = 9$.

Substitute $b = 8$ into one of the equations to find a.

$$a - b = 5$$

$$a - 3 = 5$$

$$a = 8$$

$$2$$

So, a = 8 and a = 64.

Therefore, an equation of the ellipse is $\frac{x^2}{64} + \frac{y^2}{49} = 1$.

70. Writing in Math Explain how to find the foci and vertices of an ellipse if you are given the standard form of the equation.

SOLUTION:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Sample answer: First, use the equation to find the values of a, b, h, and k. Use $c = \sqrt{a^2 - b^2}$ to find the value of c.

When the equation is in standard form, the square root of the larger denominator is a and the square root of the

smaller denominator is b . h is associated with x and k is associated with y in the numerators.

Then determine whether the major axis is horizontal or vertical. If x corresponds with the larger denominator (a),

then the orientation is horizontal. Otherwise, it is vertical.

If the major axis is horizontal, the foci are at $(h \pm c, k)$, the vertices are at $(h \pm a, k)$, and the co-vertices are at $(h,$

$k \pm b)$. If the major axis is vertical, the foci are at $(h, k \pm c)$, the vertices are at $(h, k \pm a)$, and the co-vertices are

at $(h \pm b, k)$.

71. REASONING Is the ellipse $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$ symmetric with respect to the origin? Explain your reasoning.

SOLUTION:

Yes; sample answer: If (x, y) is a point on the ellipse, then $(-x, -y)$ must also be on the ellipse. This can be

determined because the x - and y -terms are squared. When the center is not at the origin, then we will be squaring x

2 2

$+ h$ and/or $y + k$ instead of x and y . We know that $(x + h)^2 \neq x^2$ when $h \neq 0$, so the ellipse will not be symmetric

with respect to the origin.

$+ = 1$

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$+ = 1$ Page 50

$= 1$

then the orientation is horizontal. Otherwise, it is vertical.

If the major axis is horizontal, the foci are at $(h \pm c, k)$, the vertices are at $(h \pm a, k)$, and the co-vertices are at $(h,$

$\pm b)$. If theand

7-2 kEllipses major axis is vertical, the foci are at $(h, k \pm c)$, the vertices are at $(h, k \pm a)$, and the co-vertices are

Circles

at $(h \pm b, k)$.

71. REASONING Is the ellipse $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$ symmetric with respect to the origin? Explain your reasoning.

SOLUTION:

Yes; sample answer: If (x, y) is a point on the ellipse, then $(-x, -y)$ must also be on the ellipse. This can be

determined because the x - and y -terms are squared. When the center is not at the origin, then we will be squaring x

2 2

$+ h$ and/or $y + k$ instead of x and y . We know that $(x + h)^2 \neq x^2$ when $h \neq 0$, so the ellipse will not be symmetric

with respect to the origin.

$$+ = 1$$

$$+ = 1$$

$$+ = 1$$

Thus, $(-x, -y)$ is also a point on the ellipse and the ellipse is symmetric with respect to the origin.

72. OPEN ENDED If the equation of a circle is $(x - h)^2 + (y - k)^2 = r^2$, where $h > 0$ and $k < 0$, what is the domain of

the circle? Verify your answer with an example, both algebraically and graphically.

SOLUTION:

The value of k does not affect the domain. It only shifts the graph up and down.

The value of h does affect the domain as it shifts the graph left and right. The value of r determines the range of the domain.

$$D: [h - r, h + r]$$

$$2 \ 2 \ 2$$

Sample answer: $(x - 4)^2 + (y + 5)^2 = 8$, with domain $[4 - 8, 4 + 8]$ or $[-4, 12]$.

73. Writing in Math Explain why an ellipse becomes circular as the value of b approaches the value of a .

SOLUTION:

$$2 \ 2 \ 2 \ 2 \ 2$$

Sample answer: As the value of b approaches the value of a , $a - b$ approaches 0. In the relationship $c = a - b$,

$$2 \ 2 \ 2$$

if $a - b$ approaches 0, so does c . This means that c approaches 0. If c approaches 0, then the eccentricity of an

ellipse, defined as $e = \frac{c}{a}$, also approaches 0. As e approaches 0, an ellipse becomes less elongated and more

circular. When $e = 0$, the resulting figure is a circle.

For each equation, identify the vertex, focus, axis of symmetry, and directrix. Then graph the parabola.

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$$74. y = 3x^2 - 24x + 50$$

SOLUTION:

$$2 \ 2 \ 2 \ 2 \ 2$$

Sample answer: As the value of b approaches the value of a , $a - b$ approaches 0. In the relationship $c = a - b$,

$$2 \ 2 \ 2$$

if $a - b$ approaches 0, so does c . This means that c approaches 0. If c approaches 0, then the eccentricity of an

7-2 ellipse,

Ellipses defined

and as e approaches 0, an ellipse becomes less elongated and more circular. When $e = 0$, the resulting figure is a circle.

For each equation, identify the vertex, focus, axis of symmetry, and directrix. Then graph the parabola.

74. $y = 3x^2 - 24x + 50$

SOLUTION:

Write the standard form of the equation.

$$y = 3x^2 - 24x + 50$$

$$y = 3(x^2 - 8x + 16) + 2$$

$$y - 2 = 3(x - 4)^2$$

$$(y - 2) = (x - 4)^2$$

Because the x -term is squared and $p = 1$, the graph opens up. Use the standard form to determine the

characteristics of the parabola.

vertex: $(h, k) = (4, 2)$

focus: $(h + p, k) =$

directrix: $y = k - p$ or 1

axis of symmetry: $x = h$ or 4

Graph the vertex, focus, axis, and directrix of the parabola. Then make a table of values to graph the curve. The

curve should be symmetric about the axis of symmetry.

75. $y = -2x^2 + 5x - 10$

SOLUTION:

Write the standard form of the equation.

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7-2 Ellipses and Circles

75. $y = -2x^2 + 5x - 10$

SOLUTION:

Write the standard form of the equation.

Because the x -term is squared and $p = -\frac{1}{4}$, the graph opens down. Use the standard form to determine the

characteristics of the parabola.

vertex: $(h, k) =$

focus: $(h, k + p) =$

directrix: $y = k - p$ or

axis of symmetry: $x = h$ or

Graph the vertex, focus, axis, and directrix of the parabola. Then make a table of values to graph the curve. The

curve should be symmetric about the axis of symmetry.

$$76. x = 5y^2 - 10y + 9$$

SOLUTION:

Write the standard form of the equation.

$$2$$

$$x = 5y^2 - 10y + 9$$

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$$x = 5(y^2 - 2y + 1) + 4$$

$$2$$

$$x - 4 = 5(y - 1)^2$$

7-2 Ellipses and Circles

$$76. x = 5y^2 - 10y + 9$$

SOLUTION:

Write the standard form of the equation.

$$2$$

$$x = 5y^2 - 10y + 9$$

$$2$$

$$x = 5(y^2 - 2y + 1) + 4$$

$$2$$

$$x - 4 = 5(y - 1)^2$$

$$2$$

$$(x - 4) = 5(y - 1)^2$$

Because the y -term is squared and $p = 1$, the graph opens right. Use the standard form to determine the

characteristics of the parabola.

vertex: $(h, k) = (4, 1)$

focus: $(h + p, k) =$

directrix: $x = h - p$ or

axis of symmetry: $y = k$ or 1

Graph the vertex, focus, axis, and directrix of the parabola. Then make a table of values to graph the curve. The

curve should be symmetric about the axis of symmetry.

77. MANUFACTURING A toy company is introducing two new dolls to its customers: My First Baby, which talks,

laughs, and cries; and My Real Baby, which uses a bottle and crawls. In one hour, the company can produce 8 First

Babies or 20 Real Babies. Because of the demand, the company must produce at least twice as many First Babies

as Real Babies. The company spends no more than 48 hours per week making these two dolls. Find the number and

type of dolls that should be produced to maximize the profit.

SOLUTION:

Let F represent the number of First Babies produced and R represent the number of Real Babies produced.

At least twice as many First Babies must be produced, so $F \geq 2R$. They can produce one First Baby in hour and

one Real Baby in hour. They have at most 48 hours, so $F + R \leq 48$.

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Graph both equations.

7-2 Ellipses and Circles

77. MANUFACTURING A toy company is introducing two new dolls to its customers: My First Baby, which talks,

laughs, and cries; and My Real Baby, which uses a bottle and crawls. In one hour, the company can produce 8 First

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as Real Babies. The company spends no more than 48 hours per week making these two dolls. Find the number and

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SOLUTION:

Let F represent the number of First Babies produced and R represent the number of Real Babies produced.

At least twice as many First Babies must be produced, so $F \geq 2R$. They can produce one First Baby in hour and

one Real Baby in hour. They have at most 48 hours, so $F + R \leq 48$.

Graph both equations.

The vertices of the intersections are (160, 320), (0, 0), and (0, 384).

Because f is greatest at (160, 320), they need to produce 160 My Real Babies and 320 My First Babies.

Verify each identity.

$$78. \sin(\theta + 30^\circ) + \cos(\theta + 60^\circ) = \cos \theta$$

SOLUTION:

$$79. \sin - \cos = \sin \theta$$

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SOLUTION:

7-2 Ellipses and Circles

$$79. \sin - \cos = \sin \theta$$

SOLUTION:

$$80. \sin(3\pi - x) = \sin x$$

SOLUTION:

$$\begin{aligned}
 \sin(3\pi - x) &= \sin 3\pi \cos x - \cos 3\pi \sin x \\
 &= 0 \cos x - (-1) \sin x \\
 &= 0 + \sin x \\
 &= \sin x
 \end{aligned}$$

Find all solutions to each equation on the interval $[0, 2\pi)$.

81. $\sin \theta = \cos \theta$

SOLUTION:

Since squaring each side of a trigonometric equation can produce extraneous results, check each solution in the original equation.

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7-2 Since squaring

Ellipses and each side of a trigonometric equation can produce extraneous results, check each solution in the

Circles

original equation.

Therefore, the only solutions to this equation on the interval $[0, 2\pi)$ are and .

CHECK Graph $y = \sin \theta$ and $y = \cos \theta$ on the interval $[0, 2\pi)$.

The graphs intersect at and , so the solutions are and .

82. $\sin \theta = 1 + \cos \theta$

SOLUTION:

Since squaring each side of a trigonometric equation can produce extraneous results, check each solution in the original equation.

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7-2 The graphs and

Ellipses intersect at and

Circles , so the solutions are and .

82. $\sin \theta = 1 + \cos \theta$

SOLUTION:

Since squaring each side of a trigonometric equation can produce extraneous results, check each solution in the original equation.

Therefore, the only solutions to this equation on the interval $[0, 2\pi)$ are π and .

CHECK Graph $y = \sin \theta$ and $y = 1 + \cos \theta$ on the interval $[0, 2\pi)$.

The graphs intersect at π and , so the solutions are π and .

83. $2 \sin^2 x + 3 \sin x + 1 = 0$

SOLUTION:

Let $u = \sin x$. Then factor the equation.

2

$$2 \sin x + 3 \sin x + 1 = 0$$

2

$$2u + 3u + 1 = 0$$

$$(2u + 1)(u + 1) = 0$$

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Substitute $\sin x$ back into the equation for u .

$$(2u + 1)(u + 1) = 0$$

7-2 The

Ellipses

graphs and Circles

intersect at π and , so the solutions are π and .

$$83. 2 \sin^2 x + 3 \sin x + 1 = 0$$

SOLUTION:

Let $u = \sin x$. Then factor the equation.

2

$$2 \sin x + 3 \sin x + 1 = 0$$

2

$$2u + 3u + 1 = 0$$

$$(2u + 1)(u + 1) = 0$$

Substitute $\sin x$ back into the equation for u .

$$(2u + 1)(u + 1) = 0$$

$$(2 \sin x + 1)(\sin x + 1) = 0$$

Solve for x .

$$2 \sin x + 1 = 0$$

$$\sin x =$$

$$x = \text{or}$$

$$\sin x + 1 = 0$$

$$\sin x = -1$$

$$x =$$

So, the solutions are , , and .

Solve each inequality.

$$84. x^2 - 5x - 24 > 0$$

SOLUTION:

2

Let $f(x) = x^2 - 5x - 24$. Find the real zeros of this function.

2

$$x^2 - 5x - 24 = 0$$

$$(x - 8)(x + 3) = 0$$

$$x = 8 \text{ or } x = -3$$

$f(x)$ has real zeros at $x = -3$ and $x = 8$. Set up a sign chart. Substitute an x -value in each test interval into the polynomial to determine if $f(x)$ is positive or negative at that point.

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$$\sin x + 1 = 0$$

$$\sin x = -1$$

$$x =$$

7-2 So,

Ellipses andare

the solutions Circles

, , and .

Solve each inequality.

$$84. x^2 - 5x - 24 > 0$$

SOLUTION:

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$f(x)$ has real zeros at $x = -3$ and $x = 8$. Set up a sign chart. Substitute an x -value in each test interval into the

polynomial to determine if $f(x)$ is positive or negative at that point.

2

The solutions of $x^2 - 5x - 24 > 0$ are x -values such that $f(x)$ is positive. From the sign chart, you can see that the

solution set is $(-\infty, -3) \cup (8, \infty)$.

$$85. x^2 + 2x - 35 \leq 0$$

SOLUTION:

2

Let $f(x) = x^2 + 2x - 35$. Find the real zeros of this function.

2

$$x^2 + 2x - 35 = 0$$

$$(x + 7)(x - 5) = 0$$

$$x = -7 \text{ or } x = 5$$

$f(x)$ has real zeros at $x = -7$ and $x = 5$. Set up a sign chart. Substitute an x -value in each test interval into the

polynomial to determine

eSolutions Manual - Powered if $f(x)$ is positive or negative at that point.

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2

The solutions of $x^2 - 5x - 24 > 0$ are x-values such that $f(x)$ is positive. From the sign chart, you can see that the

7-2 solution

Ellipses set and Circles

is $(-\infty, -3) \cup (8, \infty)$.

$$85. x^2 + 2x - 35 \leq 0$$

SOLUTION:

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Let $f(x) = x^2 + 2x - 35$. Find the real zeros of this function.

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$$(x + 7)(x - 5) = 0$$

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$f(x)$ has real zeros at $x = -7$ and $x = 5$. Set up a sign chart. Substitute an x-value in each test interval into the

polynomial to determine if $f(x)$ is positive or negative at that point.

2

The solutions of $x^2 + 2x - 35 \leq 0$ are x-values such that $f(x)$ is negative or zero. From the sign chart, you can see

that the solution set is $[-7, 5]$.

$$86. -2x^2 + 7x + 4 < 0$$

SOLUTION:

2

Let $f(x) = -2x^2 + 7x + 4$. Find the real zeros of this function.

2

$$-2x^2 + 7x + 4 = 0$$

2

$$2x^2 - 7x - 4 = 0$$

$$(2x + 1)(x - 4) = 0$$

$$2x + 1 = 0 \text{ or } x - 4 = 0$$

$$x = -\frac{1}{2} \text{ or } x = 4$$

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$f(x)$ has real zeros at $x = -\frac{1}{2}$ and $x = 4$. Set up a sign chart. Substitute an x-value in each test interval into the

2

7-2 The

Ellipses and

solutions of x Circles

$-2x^2 + 7x + 4 \leq 0$ are x-values such that $f(x)$ is negative or zero. From the sign chart, you can see that the solution set is $[-\frac{1}{2}, 4]$.

$$86. -2x^2 + 7x + 4 < 0$$

SOLUTION:

2

Let $f(x) = -2x^2 + 7x + 4$. Find the real zeros of this function.

2

$$-2x^2 + 7x + 4 = 0$$

2

$$2x^2 - 7x - 4 = 0$$

$$(2x + 1)(x - 4) = 0$$

$$2x + 1 = 0 \text{ or } x - 4 = 0$$

$$x = -\frac{1}{2} \text{ or } x = 4$$

$f(x)$ has real zeros at $x = -\frac{1}{2}$ and $x = 4$. Set up a sign chart. Substitute an x -value in each test interval into the

polynomial to determine if $f(x)$ is positive or negative at that point.

2

The solutions of $-2x^2 + 7x + 4 > 0$ are x -values such that $f(x)$ is positive. From the sign chart, you can see that the

solution set is $-\frac{1}{2} < x < 4$.

State the number of possible real zeros and turning points of each function. Then determine all of the

real zeros by factoring.

$$87. f(x) = 3x^4 + 18x^3 + 24x^2$$

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SOLUTION: Page 62

The degree of the polynomial is 4, so there are possibly 4 real zeros and 3 turning points.

4 3 2

$$3x^4 + 18x^3 + 24x^2 = 0$$

2

The solutions of $-2x^2 + 7x + 4 > 0$ are x -values such that $f(x)$ is positive. From the sign chart, you can see that the

7-2 solution

Ellipses set and

is Circles

State the number of possible real zeros and turning points of each function. Then determine all of the

real zeros by factoring.

$$87. f(x) = 3x^4 + 18x^3 + 24x^2$$

SOLUTION:

The degree of the polynomial is 4, so there are possibly 4 real zeros and 3 turning points.

4 3 2

$$3x^4 + 18x^3 + 24x^2 = 0$$

$$2 \ 2$$

$$3x(x + 6x + 8) = 0$$

$$2$$

$$3x(x + 4)(x + 2) = 0$$

The zeros of the function are -4, -2, and 0.

$$88. f(x) = 8x^6 + 48x^5 + 40x^4$$

SOLUTION:

The degree of the polynomial is 6, so there are possibly 6 real zeros and 5 turning points.

$$6 \ 5 \ 4$$

$$8x^6 + 48x^5 + 40x^4 = 0$$

$$4 \ 2$$

$$8x^4(x + 6x + 5) = 0$$

$$4$$

$$8x^4(x + 1)(x + 5) = 0$$

The zeros of the function are -5, -1, and 0.

$$89. f(x) = 5x^5 - 15x^4 - 50x^3$$

SOLUTION:

The degree of the polynomial is 5, so there are possibly 5 real zeros and 4 turning points.

$$5 \ 4 \ 3$$

$$5x^5 - 15x^4 - 50x^3 = 0$$

$$3 \ 2$$

$$5x^3(x^2 - 3x - 10) = 0$$

$$3$$

$$5x^3(x - 5)(x + 2) = 0$$

The zeros of the function are -2, 0, and 5.

Simplify.

$$90. (2 + 4i) + (-1 + 5i)$$

SOLUTION:

$$(2 + 4i) + (-1 + 5i)$$

$$= 2 + (-1) + 4i + 5i$$

$$= 1 + 9i$$

$$91. (-2 - i)^2$$

SOLUTION:

$$2$$

$$(-2 - i)$$

$$2$$

$$= 4 + 4i + i$$

$$= 4 + 4i - 1$$

$$= 3 + 4i$$

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92.

SOLUTION:

2

$(-2 - i)$

2

$= 4 + 4i + i$

7-2 =Ellipses

4 + 4i – 1and Circles

$= 3 + 4i$

92.

SOLUTION:

93. SAT/ACT Point B lies 10 units from point A, which is the center of a circle of radius 6. If a tangent line is drawn

from B to the circle, what is the distance from B to the point of tangency?

A 6

B 8

C 10

D

E

SOLUTION:

Draw the circle and tangent line.

The tangent line forms a right triangle.

Solving the right triangle shows the distance from point B to the point of tangency to be 8 units.

94. REVIEW What is the standard form of the equation of the conic given below?

$F + = 1$

$G + = 1$

$H + = 1$

$J + = 1$

SOLUTION:

Write the equation in standard form.

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7-2 The

Ellipses

tangentand Circles

line forms a right triangle.

Solving the right triangle shows the distance from point B to the point of tangency to be 8 units.

94. REVIEW What is the standard form of the equation of the conic given below?

$F + = 1$

$$G + = 1$$

$$H + = 1$$

$$J + = 1$$

SOLUTION:

Write the equation in standard form.

95. Ruben is making an elliptical target for throwing darts. He wants the target to be 27 inches wide and 15 inches high.

Which equation should Ruben use to draw the target?

$$A + = 1$$

$$B + = 1$$

$$C + = 1$$

$$D + = 1$$

SOLUTION:

2

The target is to be 27 inches wide, so $2a = 27$, $a = 13.5$, and $a^2 = 182.25$. The target is to be 15 inches wide, so $2b =$

2 2 2

15, $b = 7.5$, and $b^2 = 56.25$. The ellipse is horizontally oriented, so the a^2 -term is with the x^2 -term. So, the equation

is $+ = 1$.

96. REVIEW If $m =$, $n = 7m$, $p =$, $q = 14p$, and $r =$, find x .

eSolutions

F Manual - Powered by Cognero Page 65

r

G q

H p

2

The target is to be 27 inches wide, so $2a = 27$, $a = 13.5$, and $a^2 = 182.25$. The target is to be 15 inches wide, so $2b =$

2 2 2

15, $b = 7.5$, and $b^2 = 56.25$. The ellipse is horizontally oriented, so the a^2 -term is with the x^2 -term. So, the equation

7-2 is Ellipses and

+ Circles

$= 1$.

96. REVIEW If $m =$, $n = 7m$, $p =$, $q = 14p$, and $r =$, find x .

F r

G q

H p

J

SOLUTION:

First, find r in terms of p .

Find r in terms of n .

Find r in terms of m .

Find x in terms of r .

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7-3 Hyperbolas

Graph the hyperbola given by each equation.

1. $\frac{x^2}{16} - \frac{y^2}{9} = 1$

SOLUTION:

The equation is in standard form, where $h = 0$, $k = 0$, $a = 4$, $b = 3$, and $c = 5$.

In the standard form of the equation, the y -term is being subtracted. Therefore, the orientation of the hyperbola is

horizontal.

center: $(h, k) = (0, 0)$

vertices: $(h \pm a, k) = (4, 0)$ and $(-4, 0)$

foci: $(h \pm c, k) = (5, 0)$ and $(-5, 0)$

Graph the center, vertices, foci, and asymptotes. Then make a table of values to sketch the hyperbola.

3. $\frac{x^2}{49} - \frac{y^2}{89} = 1$

SOLUTION:

The equation is in standard form, where $h = 0$, $k = 0$, $a = 7$, $b = \text{about } 5.48$, and $c = \text{about } 8.89$. In the standard form of the equation, the y -term is being subtracted. Therefore, the orientation of the

hyperbola is horizontal.

center: $(h, k) = (0, 0)$

vertices: $(h \pm a, k) = (7, 0)$ and $(-7, 0)$

foci: $(h \pm c, k) = (8.89, 0)$ and $(0, -8.89)$

Graph the center, vertices, foci, and asymptotes. Then use a table of values to sketch the hyperbola.

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7-3 Hyperbolas

3. $\frac{x^2}{49} - \frac{y^2}{89} = 1$

SOLUTION:

The equation is in standard form, where $h = 0$, $k = 0$, $a = 7$, $b = \text{about } 5.48$, and $c = \text{about } 8.89$. In the standard form of the equation, the y -term is being subtracted. Therefore, the orientation of the

hyperbola is horizontal.

center: $(h, k) = (0, 0)$

vertices: $(h \pm a, k) = (7, 0)$ and $(-7, 0)$

foci: $(h \pm c, k) = (8.89, 0)$ and $(0, -8.89)$

Graph the center, vertices, foci, and asymptotes. Then use a table of values to sketch the hyperbola.

5. $- = 1$

SOLUTION:

The equation is in standard form, where $h = 0$, $k = 0$, $a =$ or 3 , $b =$ or about 4.58 , and $c =$ or about 5.48 . In the standard form of the equation, the y -term is being subtracted. Therefore, the orientation of the hyperbola is horizontal.

center: $(h, k) = (0, 0)$

vertices: $(h \pm a, k) = (3, 0)$ and $(-3, 0)$

foci: $(h \pm c, k) = (5.48, 0)$ and $(-5.48, 0)$

Graph the center, vertices, foci, and asymptotes. Then use a table of values to sketch the hyperbola.

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7-3 Hyperbolas

5. $- = 1$

SOLUTION:

The equation is in standard form, where $h = 0$, $k = 0$, $a =$ or 3 , $b =$ or about 4.58 , and $c =$ or about 5.48 . In the standard form of the equation, the y -term is being subtracted. Therefore, the orientation of the hyperbola is horizontal.

center: $(h, k) = (0, 0)$

vertices: $(h \pm a, k) = (3, 0)$ and $(-3, 0)$

foci: $(h \pm c, k) = (5.48, 0)$ and $(-5.48, 0)$

Graph the center, vertices, foci, and asymptotes. Then use a table of values to sketch the hyperbola.

7. $- = 1$

SOLUTION:

The equation is in standard form, where $h = 0$, $k = 0$, $a =$ or 9 , $b =$ or about 2.83 , and $c =$ or about 9.43 . In the standard form of the equation, the x -term is being subtracted. Therefore, the orientation of the hyperbola is vertical.

center: $(h, k) = (0, 0)$

vertices: $(h, k \pm a) = (0, 9)$ and $(0, -9)$

foci: $(h, k \pm c) = (0, 9.43)$ and $(0, -9.43)$

Graph the center, vertices, foci, and asymptotes. Then use a table of values to sketch the hyperbola.

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7-3 Hyperbolas

$$7. - = 1$$

SOLUTION:

The equation is in standard form, where $h = 0$, $k = 0$, $a =$ or 9 , $b =$ or about 2.83 , and $c =$ or about 9.43 . In the standard form of the equation, the x -term is being subtracted. Therefore, the orientation of the hyperbola is vertical.

center: $(h, k) = (0, 0)$

vertices: $(h, k \pm a) = (0, 9)$ and $(0, -9)$

foci: $(h, k \pm c) = (0, 9.43)$ and $(0, -9.43)$

Graph the center, vertices, foci, and asymptotes. Then use a table of values to sketch the hyperbola.

$$9. 3x^2 - 2y^2 = 12$$

SOLUTION:

First, divide each side by 12 to write the equation in standard form.

The equation is now in standard form, where $h = 0$, $k = 0$, $a =$ or 2 , $b =$ or about 2.45 , and $c =$ or about 3.16 . In the standard form of the equation, the y -term is being subtracted. Therefore, the orientation of the hyperbola is horizontal.

center: $(h, k) = (0, 0)$

vertices: $(h \pm a, k) = (2, 0)$ and $(-2, 0)$

foci: $(h \pm c, k) = (3.16, 0)$ and $(-3.16, 0)$

Graph the center, vertices, foci, and asymptotes. Then use a table of values to sketch the hyperbola.

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7-3 Hyperbolas

$$9. 3x^2 - 2y^2 = 12$$

SOLUTION:

First, divide each side by 12 to write the equation in standard form.

The equation is now in standard form, where $h = 0$, $k = 0$, $a =$ or 2 , $b =$ or about 2.45 , and $c =$ or about 3.16 . In the standard form of the equation, the y -term is being subtracted. Therefore, the orientation of the hyperbola is horizontal.

center: $(h, k) = (0, 0)$

vertices: $(h \pm a, k) = (2, 0)$ and $(-2, 0)$

foci: $(h \pm c, k) = (3.16, 0)$ and $(-3.16, 0)$

Graph the center, vertices, foci, and asymptotes. Then use a table of values to sketch the hyperbola.

11. LIGHTING The light projected on a wall by a table lamp can be represented by a hyperbola. The light from a certain table lamp can be modeled by $- = 1$. Graph the hyperbola. Refer to the photo on Page 449.

SOLUTION:

The equation is in standard form, where $h = 0$, $k = 0$, $a =$ or 15 , $b =$ or 9 , and $c =$ or about 17.49 . In the standard form of the equation, the x -term is being subtracted. Therefore, the orientation of the hyperbola is vertical.

center: $(h, k) = (0, 0)$

vertices: $(h, k \pm a) = (0, 15)$ and $(0, -15)$

foci: $(h, k \pm c) = (0, 17.49)$ and $(0, -17.49)$

Graph the center, vertices, foci, and asymptotes. Then use a table of values to sketch the hyperbola.

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7-3 Hyperbolas

11. LIGHTING The light projected on a wall by a table lamp can be represented by a hyperbola. The light from a certain table lamp can be modeled by $- = 1$. Graph the hyperbola. Refer to the photo on Page 449.

SOLUTION:

The equation is in standard form, where $h = 0$, $k = 0$, $a =$ or 15 , $b =$ or 9 , and $c =$ or about 17.49 . In the standard form of the equation, the x -term is being subtracted. Therefore, the orientation of the hyperbola is vertical.

center: $(h, k) = (0, 0)$

vertices: $(h, k \pm a) = (0, 15)$ and $(0, -15)$

foci: $(h, k \pm c) = (0, 17.49)$ and $(0, -17.49)$

Graph the center, vertices, foci, and asymptotes. Then use a table of values to sketch the hyperbola.

Graph the hyperbola given by each equation.

13. $- = 1$

SOLUTION:

The equation is in standard form, where $h = 0$, $k = 7$, $a =$ or 2 , $b =$ or about 5.74 , and $c =$ or about 6.08 . In the standard form of the equation, the x -term is being subtracted. Therefore, the orientation of the hyperbola is vertical.

center: $(h, k) = (0, 7)$

vertices: $(h, k \pm a) = (0, 9)$ and $(0, 5)$

foci: $(h, k \pm c) = (0, 13.08)$ and $(0, 0.92)$

asymptote:

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7-3 Hyperbolas

Graph the hyperbola given by each equation.

$$13. \frac{y^2}{16} - \frac{x^2}{9} = 1$$

SOLUTION:

The equation is in standard form, where $h = 0$, $k = 7$, $a = 4$, $b = 3$, and $c = 5$. In the standard form of the equation, the y -term is being subtracted. Therefore, the orientation of the hyperbola is vertical.

center: $(h, k) = (0, 7)$

vertices: $(h, k \pm a) = (0, 11)$ and $(0, 3)$

foci: $(h, k \pm c) = (0, 12)$ and $(0, 2)$

asymptote:

Graph the center, vertices, foci, and asymptotes. Then use a table of values to sketch the hyperbola.

$$15. \frac{y^2}{16} - \frac{x^2}{9} = 1$$

SOLUTION:

The equation is in standard form, where $h = 5$, $k = 1$, $a = 4$, $b = 3$, and $c = 5$. In the standard form of the equation, the y -term is being subtracted. Therefore, the orientation of the hyperbola is horizontal.

center: $(h, k) = (5, 1)$

vertices: $(h \pm a, k) = (9, 1)$ and $(1, 1)$

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foci: $(h \pm c, k)$

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$k) = (10, 1)$ and $(4, 1)$ Page 7

asymptotes:

7-3 Hyperbolas

$$15. \frac{y^2}{16} - \frac{x^2}{9} = 1$$

SOLUTION:

The equation is in standard form, where $h = 5$, $k = 1$, $a = 4$, $b = 3$, and $c = 5$. In the standard form of the equation, the y -term is being subtracted. Therefore, the orientation of the hyperbola is horizontal.

center: $(h, k) = (5, 1)$

vertices: $(h \pm a, k) = (9, 1)$ and $(1, 1)$

foci: $(h \pm c, k) = (10, 1)$ and $(4, 1)$

asymptotes:

Graph the center, vertices, foci, and asymptotes. Then use a table of values to sketch the hyperbola.

$$17. - = 1$$

SOLUTION:

The equation is in standard form, where $h = -6$, $k = -5$, $a =$ or 8, $b =$ or about 7.62, and $c =$ or about 11.05. In the standard form of the equation, the y -term is being subtracted. Therefore, the

orientation of the hyperbola is horizontal.

center: $(h, k) = (-6, -5)$

vertices: $(h \pm a, k) = (2, -5)$ and $(-14, -5)$

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foci: $(h \pm c, k) = (5.05, -5)$ and $(-17.05, -5)$

asymptotes:

7-3 Hyperbolas

$$17. - = 1$$

SOLUTION:

The equation is in standard form, where $h = -6$, $k = -5$, $a =$ or 8, $b =$ or about 7.62, and $c =$ or about 11.05. In the standard form of the equation, the y -term is being subtracted. Therefore, the

orientation of the hyperbola is horizontal.

center: $(h, k) = (-6, -5)$

vertices: $(h \pm a, k) = (2, -5)$ and $(-14, -5)$

foci: $(h \pm c, k) = (5.05, -5)$ and $(-17.05, -5)$

asymptotes:

Graph the center, vertices, foci, and asymptotes. Then use a table of values to sketch the hyperbola.

$$19. -x^2 + 3y^2 - 4x + 6y = 28$$

SOLUTION:

First,

eSolutions write

Manual the equation

- Powered in standard form.

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2 2

19. Hyperbolas

$$7-3 -x^2 + 3y^2 - 4x + 6y = 28$$

SOLUTION:

First, write the equation in standard form.

The equation is now in standard form, where $h = -2$, $k = -1$, $a =$ or 3, $b =$ or about 5.2, and $c =$ or 6. In the standard form of the equation, the x -term is being subtracted. Therefore, the orientation of

the hyperbola is vertical.

center: $(h, k) = (-2, -1)$

vertices: $(h, k \pm a) = (-2, 2)$ and $(-2, -4)$

foci: $(h, k \pm c) = (-2, 5)$ and $(-2, -7)$

asymptotes:

Graph the center, vertices, foci, and asymptotes. Then use a table of values to sketch the hyperbola.

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$$21. -5x^2 + 2y^2 - 70x - 8y = 287$$

SOLUTION:

7-3 Hyperbolas

$$21. -5x^2 + 2y^2 - 70x - 8y = 287$$

SOLUTION:

First, write the equation in standard form.

The equation is now in standard form, where $h = -7$, $k = 2$, $a = 5$, $b =$ or about 3.2, and $c =$ or about 5.92. In the standard form of the equation, the x -term is being subtracted. Therefore, the orientation of the hyperbola is vertical.

center: $(h, k) = (-7, 2)$

vertices: $(h, k \pm a) = (-7, 7)$ and $(-7, -3)$

foci: $(h, k \pm c) = (-7, 7.92)$ and $(-7, -3.92)$

asymptotes:

Graph

eSolutions the- Powered

Manual center, by

vertices,

Cognerofoci, and asymptotes. Then use a table of values to sketch the hyperbola. Page 11

7-3 Hyperbolas

Graph the center, vertices, foci, and asymptotes. Then use a table of values to sketch the hyperbola.

Write an equation for the hyperbola with the given characteristics.

23. foci $(-1, 9)$, $(-1, -7)$; conjugate axis length of 14 units

SOLUTION:

Because the x -coordinates of the foci are the same, the transverse axis is vertical, and the standard form of the

equation is $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$.

The center is the midpoint of the segment between the foci, or $(-1, 1)$. So, $h = -1$ and $k = 1$. The length of the

2

conjugate axis of a hyperbola is $2b$. So, $2b = 14$, $b = 7$, and $b^2 = 49$. You can find c by determining the distance

from a focus to the center. One focus is located at $(-1, 9)$, which is 8 units from $(-1, 1)$. So, $c = 8$.

Now you can use the values of b and c to find a .

$$2^2 - \frac{y^2}{2^2} = 1$$

$$a^2 = c^2 - b^2$$

$$2^2 - 2^2$$

$$a^2 = 8 - 7$$

$$2$$

$$a^2 = 64 - 49$$

$$2$$

$$a^2 = 15$$

$$a =$$

Using the values of h , k , a , and b , the equation for the hyperbola is $\frac{(x-3)^2}{15} - \frac{(y+1)^2}{4} = 1$.

25. foci $(9, -1)$, $(-3, -1)$; conjugate axis length of 6 units

SOLUTION:

Because the y -coordinates of the foci are the same, the transverse axis is horizontal, and the standard form of the

equation is $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$.

The center is the midpoint of the segment between the foci, or $(3, -1)$. So, $h = 3$ and $k = -1$. The length of the

$$2$$

conjugate axis of a hyperbola is $2b$. So, $2b = 6$, $b = 3$, and $b^2 = 9$. You can find c by determining the distance from

a focus to the center. One focus is located at $(9, -1)$, which is 6 units from $(3, -1)$. So, $c = 6$.

Now you can use the values of b and c to find a .

$$2^2 - \frac{y^2}{2^2} = 1$$

$$a^2 = c^2 - b^2$$

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$$2^2 - 2^2$$

$$a^2 = 6^2 - 3^2$$

$$2$$

$$a^2 = 36 - 9$$

$$a =$$

Using the values of h , k , a , and b , the equation for the hyperbola is $\frac{(x-3)^2}{15} - \frac{(y+1)^2}{4} = 1$.

7-3 Hyperbolas

25. foci $(9, -1)$, $(-3, -1)$; conjugate axis length of 6 units

SOLUTION:

Because the y -coordinates of the foci are the same, the transverse axis is horizontal, and the standard form of the

equation is $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$.

The center is the midpoint of the segment between the foci, or $(3, -1)$. So, $h = 3$ and $k = -1$. The length of the

2

conjugate axis of a hyperbola is $2b$. So, $2b = 6$, $b = 3$, and $b = 9$. You can find c by determining the distance from

a focus to the center. One focus is located at $(9, -1)$, which is 6 units from $(3, -1)$. So, $c = 6$.

Now you can use the values of b and c to find a .

$$2^2 - 2^2 = 2^2$$

$$a^2 = c^2 - b^2$$

$$2^2 - 2^2 = 2^2$$

$$a^2 = 6^2 - 3^2$$

2

$$a^2 = 36 - 9$$

2

$$a^2 = 27$$

$$a =$$

Using the values of h , k , a , and b , the equation for the hyperbola is $- = 1$.

27. vertices $(-3, -12)$, $(-3, -4)$; foci $(-3, -15)$, $(-3, -1)$

SOLUTION:

Because the x -coordinates of the vertices are the same, the transverse axis is vertical, and the standard form of

the equation is $- = 1$.

The center is the midpoint of the segment between the foci, or $(-3, -8)$. So, $h = -3$ and $k = -8$.

You can find c by

determining the distance from a focus to the center. One focus is located at $(-3, -15)$ which is 7 units from $(-3,$

$-8)$. So, $c = 7$. You can find a by determining the distance from a vertex to the center. One vertex is located at

2

$(-3, -12)$ which is 4 units from $(-3, -8)$. So, $a = 4$ and $a = 16$.

Now you can use the values of c and a to find b .

$$2^2 - 2^2 = 2^2$$

$$b^2 = c^2 - a^2$$

$$2^2 - 2^2 = 2^2$$

$$b^2 = 7^2 - 4^2$$

2

$$b^2 = 49 - 16$$

2

$$b^2 = 33$$

$$b =$$

Using the values of h , k , a , and b , the equation for the hyperbola is $- = 1$.

29. center $(-7, 2)$; asymptotes $y = \pm x +$, transverse axis length of 10 units

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SOLUTION:

$b =$

7-3 Hyperbolas

Using the values of h , k , a , and b , the equation for the hyperbola is $- = 1$.

29. center $(-7, 2)$; asymptotes $y = \pm x +$, transverse axis length of 10 units

SOLUTION:

The center is $(-7, 2)$. Therefore, $h = -7$ and $k = 2$.

2

The transverse axis has a length of $2a$ units. So, $10 = 2a$, $a = 5$, and $a = 25$. The slopes of the asymptotes are ,

and the standard form of the equation is $- = 1$. So, the positive slope is equal to ,

2

where $b = 7$ and $b = 49$.

Using the values of h , k , a , and b , the equation for the hyperbola is $- = 1$.

31. vertices $(0, -3)$, $(-4, -3)$; conjugate axis length of 12 units

SOLUTION:

Because the y -coordinates of the vertices are the same, the transverse axis is horizontal, and the standard form of

the equation is $- = 1$.

The center is located at the midpoint of the vertices, or $(-2, -3)$. So, $h = -2$ and $k = -3$. Because the vertices are 4

2 2

units apart, $2a = 4$, $a = 2$, and $a = 4$. The length of the conjugate axis is 12 units, so $2b = 12$, $b = 6$, and $b = 36$.

Using the values of h , k , a , and b , the equation for the hyperbola is $- = 1$.

33. ARCHITECTURE The floor plan for an office building is shown below.

a. Write an equation that could model the curved sides of the building.

b. Each unit on the coordinate plane represents 15 feet. What is the narrowest width of the building?

SOLUTION:

a. From the graph, you can see that the transverse axis is vertical, so the standard form of the equation is

$- = 1$.

The center is located at the midpoint of the vertices, or $(5, 4)$. So, $h = 5$ and $k = 4$. Because the vertices are 6 units

2

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apart, $2a = 6$, $a = 3$, and $a = 9$. The length of the conjugate axis is $2b = 10$. So, $b = 5$ and $b = 25$.

2 2

units apart, $2a = 4$, $a = 2$, and $a = 4$. The length of the conjugate axis is 12 units, so $2b = 12$, $b = 6$, and $b = 36$.

7-3 Hyperbolas

Using the values of h , k , a , and b , the equation for the hyperbola is $- = 1$.

33. ARCHITECTURE The floor plan for an office building is shown below.

- Write an equation that could model the curved sides of the building.
- Each unit on the coordinate plane represents 15 feet. What is the narrowest width of the building?

SOLUTION:

a. From the graph, you can see that the transverse axis is vertical, so the standard form of the equation is

$$- = 1.$$

The center is located at the midpoint of the vertices, or $(5, 4)$. So, $h = 5$ and $k = 4$. Because the vertices are 6 units

$$2^2$$

apart, $2a = 6$, $a = 3$, and $a = 9$. The length of the conjugate axis is $2b = 10$. So, $b = 5$ and $b = 25$.

Using the values of h , k , a , and b , the equation for the hyperbola is $- = 1$.

b. The length of the narrowest width of the building is 6 units. Since each unit represents 15 feet, the length of the

narrowest width is $(15)(6)$ or 90 feet.

Determine the eccentricity of the hyperbola given by each equation.

$$35. - = 1$$

SOLUTION:

First, find c .

$$2^2 + 2^2$$

$$c = a^2 + b^2$$

$$2$$

$$c = 24 + 15$$

$$c =$$

Use the eccentricity equation and the values of c and a to determine the eccentricity of the hyperbola.

$$e =$$

$$\approx 1.27$$

$$37. - = 1$$

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SOLUTION:

First, find c .

7-3 Hyperbolas

$$\approx 1.27$$

$$37. - = 1$$

SOLUTION:

First, find c .

$$c^2 = a^2 + b^2$$

$$c = \sqrt{a^2 + b^2}$$

$$c = \sqrt{32 + 25}$$

$$c = \sqrt{57}$$

$$c = \sqrt{57}$$

Use the eccentricity equation and the values of c and a to determine the eccentricity of the hyperbola.

$$e = \frac{c}{a}$$

$$e = \frac{\sqrt{57}}{4}$$

$$e \approx 1.33$$

39. $- = 1$

SOLUTION:

First, find c .

$$c^2 = a^2 + b^2$$

$$c = \sqrt{a^2 + b^2}$$

$$c = \sqrt{16 + 29}$$

$$c = \sqrt{45}$$

$$c = \sqrt{45}$$

Use the eccentricity equation and the values of c and a to determine the eccentricity of the hyperbola.

$$e = \frac{c}{a}$$

$$e = \frac{\sqrt{45}}{2.7}$$

Determine the eccentricity of the hyperbola given by each equation.

$$41. -4x^2 + 3y^2 + 72x - 18y = 321$$

SOLUTION:

First, write the equation in standard form.

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7-3 Hyperbolas

$$e \approx 1.68$$

Determine the eccentricity of the hyperbola given by each equation.

$$41. -4x^2 + 3y^2 + 72x - 18y = 321$$

SOLUTION:

First, write the equation in standard form.

Next, find c .

$$c^2 = a^2 + b^2$$

$$c = \sqrt{a^2 + b^2}$$

$$c = \sqrt{8 + 6}$$

$$c = \sqrt{14}$$

$$c =$$

Use the eccentricity equation and the values of c and a to determine the eccentricity of the hyperbola.

$$e =$$

$$\approx 1.32$$

$$43. -x^2 + 7y^2 + 24x + 70y = -24$$

SOLUTION:

First, write the equation in standard form.

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$$\approx 1.32$$

7-3 Hyperbolas

$$43. -x^2 + 7y^2 + 24x + 70y = -24$$

SOLUTION:

First, write the equation in standard form.

Next, find c .

$$c^2 = a^2 + b^2$$

$$c^2 = 1 + 7$$

$$c^2 =$$

$$c^2 = 8$$

$$c =$$

Use the eccentricity equation and the values of c and a to determine the eccentricity of the hyperbola.

$$e =$$

$$\approx 2.83$$

Use the discriminant to identify each conic section.

$$45. 18x^2 - 3x^2 + 4 = -8y^2 + 32y$$

SOLUTION:

$$18x^2 - 3x^2 + 4 = -8y^2 + 32y$$

First, write the equation in the general form $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$.

$$18x^2 - 3x^2 + 4 = -8y^2 + 32y$$

$$18x^2 - 3x^2 + 4 = -8y^2 + 32y$$

$$18x^2 - 3x^2 + 4 = -8y^2 + 32y$$

$$-3x^2 + 8y^2 + 18x - 32y + 4 = 0$$

Next, find the discriminant.

$$B^2 - 4AC = 0 - 4(-3)(8)$$

$$B^2 - 4AC = 0 - 4(-3)(8)$$

$$= 96$$

The discriminant is greater than 0, so the conic is a hyperbola.

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$$47. 12y^2 - 76x - x^2 = 16x$$

SOLUTION:

$$2^2$$

$$B^2 - 4AC = 0 - 4(-3)(8) \\ = 96$$

7-3 Hyperbolas

The discriminant is greater than 0, so the conic is a hyperbola.

$$47. 12y^2 - 76x - x^2 = 16x$$

SOLUTION:

$$2^2$$

First, write the equation in the general form $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$.

$$2^2$$

$$12y^2 - 76x - x^2 = 16x$$

$$2^2$$

$$-x^2 - 16x + 12y^2 - 76 = 0$$

Next, find the discriminant.

$$2^2$$

$$B^2 - 4AC = 0 - 4(-1)(0) \\ = 0$$

The discriminant is 0, so the conic is a parabola.

$$49. 5y^2 - 6x + 3x^2 - 50y = -3x^2 - 113$$

SOLUTION:

$$2^2$$

First, write the equation in the general form $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$.

$$2^2$$

$$5y^2 - 6x + 3x^2 - 50y = -3x^2 - 113$$

$$2^2$$

$$6x + 5y^2 - 6x - 50y + 113 = 0$$

Next, find the discriminant.

$$2^2$$

$$B^2 - 4AC = 0 - 4(6)(5) \\ = -120$$

The discriminant is less than 0, so the conic must be either a circle or an ellipse. Because $A \neq C$, the conic is an ellipse.

$$51. -56y + 5x^2 = 211 + 4y^2 + 10x$$

SOLUTION:

$$2^2$$

First, write the equation in the general form $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$.

2 2

$$-56y + 5x = 211 + 4y + 10x$$

2 2

$$5x - 4y - 10x - 56y - 211 = 0$$

Next, find the discriminant.

2 2

$$B - 4AC = 0 - 4(5)(-4)$$

$$= 80$$

The discriminant is greater than 0, so the conic must be a hyperbola.

$$53. x^2 - 4x = -y^2 + 12y - 31$$

SOLUTION:

2 2

First, write the equation in the general form $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$.

2 2

eSolutions Manual - Powered by $-4x$

x Cognero = $-y^2 + 12y - 31$ Page 19

$$x^2 - 4x = 0$$

$$x^2 + y^2 - 4x - 12y + 31$$

2 2

$$B - 4AC = 0 - 4(5)(-4)$$

$$= 80$$

7-3 Hyperbolas

The discriminant is greater than 0, so the conic must be a hyperbola.

$$53. x^2 - 4x = -y^2 + 12y - 31$$

SOLUTION:

2 2

First, write the equation in the general form $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$.

2 2

$$x^2 - 4x = -y^2 + 12y - 31$$

$$x^2 + y^2 - 4x - 12y + 31$$

$$x^2 + y^2 - 4x - 12y + 31$$

Next, find the discriminant.

2 2

$$B - 4AC = 0 - 4(1)(1)$$

$$= -4$$

The discriminant is less than 0, so the conic must be either a circle or an ellipse. Because $A = C$, the conic is a circle.

55. AVIATION The Federal Aviation Administration performs flight trials to test new technology in aircraft. When

one of the test aircraft collected its data, it was 18 kilometers farther from Airport B than Airport A. The two airports are 72 kilometers apart along the same highway, with Airport B due south of Airport A.

a. Write an equation for the hyperbola centered at the origin on which the aircraft was located when the data were collected.

b. Graph the equation, indicating on which branch of the hyperbola the plane was located.

c. When the data were collected, the plane was 40 miles from the highway. Find the coordinates of the plane.

SOLUTION:

a. The common difference is 18 kilometers and the absolute value of the difference of the distances from any point

2

on a hyperbola to the foci is $2a$, so $2a = 18$, $a = 9$, and $a = 81$. The two airports are the foci of the hyperbola and

2 2 2 2 2

are 72 kilometers apart, so $2c = 72$, $c = 36$, and $c = 1296$. $c = a + b$, so $b = 1296 - 81$ or 1215. Airport B is

2 2

due south of airport A, so the transverse axis is vertical and a $-$ term goes with the y $-$ term. The equation for the

hyperbola is $- = 1$.

b.

The plane is on the top branch because it is closer to airport A.

c. Because the highway is the transverse axis and the plane is 40 kilometers from the highway, $x = 40$. Substitute

40 for x in the equation.

$- = 1$

$- 1.317 = 1$

2

$y = 187.67$

$y \approx 13.7$

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The coordinates of the plane are (40, 13.7)

Derive the general form of the equation for a hyperbola with each of the following characteristics.

$= -4$

The discriminant is less than 0, so the conic must be either a circle or an ellipse. Because $A = C$, the conic is a

7-3 Hyperbolas

circle.

55. AVIATION The Federal Aviation Administration performs flight trials to test new technology in aircraft. When

one of the test aircraft collected its data, it was 18 kilometers farther from Airport B than Airport A. The two airports are 72 kilometers apart along the same highway, with Airport B due south of Airport A.

a. Write an equation for the hyperbola centered at the origin on which the aircraft was located when the data were collected.

b. Graph the equation, indicating on which branch of the hyperbola the plane was located.

c. When the data were collected, the plane was 40 miles from the highway. Find the coordinates of the plane.

SOLUTION:

a. The common difference is 18 kilometers and the absolute value of the difference of the distances from any point

2

on a hyperbola to the foci is $2a$, so $2a = 18$, $a = 9$, and $a = 81$. The two airports are the foci of the hyperbola and

2 2 2 2 2

are 72 kilometers apart, so $2c = 72$, $c = 36$, and $c = 1296$. $c = a + b$, so $b = 1296 - 81$ or 1215. Airport B is

2 2

due south of airport A, so the transverse axis is vertical and a $-$ term goes with the y $-$ term. The equation for the

hyperbola is $- = 1$.

b.

The plane is on the top branch because it is closer to airport A.

c. Because the highway is the transverse axis and the plane is 40 kilometers from the highway, $x = 40$. Substitute

40 for x in the equation.

$- = 1$

$- 1.317 = 1$

2

$y = 187.67$

$y \approx 13.7$

The coordinates of the plane are (40, 13.7)

Derive the general form of the equation for a hyperbola with each of the following characteristics.

57. vertical transverse axis centered at the origin

SOLUTION:

From the definition of a hyperbola, you know that the absolute value of the difference of distances from any point P

(x, y) on a hyperbola to the foci is constant, so $|PF_1 - PF_2| = 2a$. Since you want to determine the equation for a

hyperbola with a vertical transverse axis centered at the origin, use the distance formula $d =$ and let $F_1 = (0, c)$ and $F_2 = (0, -c)$.

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SOLUTION:

From the definition of a hyperbola, you know that the absolute value of the difference of distances from any point P

(x, y) on a hyperbola to the foci is constant, so $|PF_1 - PF_2| = 2a$. Since you want to determine the equation for a

7-3 Hyperbolas

hyperbola with a vertical transverse axis centered at the origin, use the distance formula $d =$ and let $F_1 = (0, c)$ and $F_2 = (0, -c)$.

Use the definition of a hyperbola.

Isolate one radical and then square both sides.

2

Square both sides again. Simplify, and then substitute for c .

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Solve each system of equations. Round to the nearest tenth if necessary.

7-3 Hyperbolas

Solve each system of equations. Round to the nearest tenth if necessary.

59. $2y = x - 10$ and $- = 1$

SOLUTION:

Solve the first equation for y .

Substitute Equation 1 into Equation 2 and simplify.

Use the quadratic formula to solve for x .

Substitute the values for x back into Equation 1 to find that the solutions of the system.

The solutions are $(-1.3, -5.7)$ and $(7, -1.5)$.

61. $y = 2x$ and $- = 1$

SOLUTION:

Substitute Equation 1 into Equation 2 and simplify.

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7-3 Hyperbolas

The solutions are $(-1.3, -5.7)$ and $(7, -1.5)$.

61. $y = 2x$ and $- = 1$

SOLUTION:

Substitute Equation 1 into Equation 2 and simplify.

Use the quadratic formula to solve for x .

$x =$

$= -5, 6.879$

Substitute the values for x back into Equation 1 to find that the solutions of the system.

The solutions are $(-5, -10)$ and $(6.9, 13.8)$.

63. $+ = 1$ and $- = 1$

SOLUTION:

Add the two equations together.

Substitute and solve for x .

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7-3 Hyperbolas

The solutions are $(-5, -10)$ and $(6.9, 13.8)$.

63. $+ = 1$ and $- = 1$

SOLUTION:

Add the two equations together.

Substitute and solve for x .

$x = 0$ when $y = -6$ as well.

The solutions of the system are $(0, 6)$ and $(0, -6)$.

65. FIREWORKS A fireworks grand finale is heard by Carson and Emmett, who are 3 miles apart talking on their

cell phones. Emmett hears the finale about 1 second before Carson does. Assume that sound travels at 1100 feet

per second.

a. Write an equation for the hyperbola on which the fireworks were located. Place the locations of Carson and

Emmett on the x -axis, with Carson on the left and the midpoint between them at the origin.

b. Describe the branch of the hyperbola on which the fireworks display was located.

SOLUTION:

a. Because Emmett and Carson are located on the x -axis, the standard form of the equation for the hyperbola is

$- = 1$. The distance between Emmett and Carson is 3 miles, which is equivalent to 15,840

feet. They are located at the foci of the hyperbola, so $c = (15840)$ or 7920. The absolute value of the difference

of the distances from any point on a hyperbola to the foci is $2a$. Since Emmett hears the finale about 1 second

before Carson does, and sound travels 1100 feet per second, $2a = 1100$ and $a = 550$.

Use the values of a and c to find b .

2 2 2

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= c- Powered Page 25

2 2 2

$b = 7920 - 550$

$2 = 62726400 - 302500$

$x = 0$ when $y = -6$ as well.

7-3 Hyperbolas

The solutions of the system are $(0, 6)$ and $(0, -6)$.

65. FIREWORKS A fireworks grand finale is heard by Carson and Emmett, who are 3 miles apart talking on their

cell phones. Emmett hears the finale about 1 second before Carson does. Assume that sound travels at 1100 feet

per second.

a. Write an equation for the hyperbola on which the fireworks were located. Place the locations of Carson and

Emmett on the x -axis, with Carson on the left and the midpoint between them at the origin.

b. Describe the branch of the hyperbola on which the fireworks display was located.

SOLUTION:

a. Because Emmett and Carson are located on the x -axis, the standard form of the equation for the hyperbola is

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. The distance between Emmett and Carson is 3 miles, which is equivalent to 15,840

feet. They are located at the foci of the hyperbola, so $c = (15840) \text{ or } 7920$. The absolute value of the difference

of the distances from any point on a hyperbola to the foci is $2a$. Since Emmett hears the finale about 1 second

before Carson does, and sound travels 1100 feet per second, $2a = 1100$ and $a = 550$.

Use the values of a and c to find b .

$$c^2 = a^2 + b^2$$

$$b^2 = c^2 - a^2$$

$$b^2 = 7920^2 - 550^2$$

$$b^2 = 62726400 - 302500$$

$$b^2 = 62726400 - 302500$$

$$b^2 = 62423900$$

$$b = \sqrt{62423900}$$

$$b = 7900.88$$

Using the values of h , k , a , and b , the equation for the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

b. Since Emmett heard the finale before Carson did and Emmett's location is to the right of the origin on the x -axis,

the fireworks display was located on the right branch of the hyperbola.

Write an equation for each hyperbola.

67.

SOLUTION:

From the vertices, you can determine that the center is located at $(0, 0)$ and the length of the transverse axis is 10

$$2a = 10$$

units. So, $2a = 10$, $a = 5$, and $a^2 = 25$.

From the definition of a hyperbola, you know that the absolute value of the difference of distances from any point

on the hyperbola to the foci is constant. So, $|PF_1 - PF_2| = 2a$. Let $F_1 = (-c, 0)$, $F_2 = (c, 0)$, and $P = (x, y)$.

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Using the values of h , k , a , and b , the equation for the hyperbola is $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$.

b. Since Emmett heard the finale before Carson did and Emmett's location is to the right of the origin on the x -axis,

7-3 Hyperbolas

the fireworks display was located on the right branch of the hyperbola.

Write an equation for each hyperbola.

67.

SOLUTION:

From the vertices, you can determine that the center is located at $(0, 0)$ and the length of the transverse axis is 10

2

units. So, $2a = 10$, $a = 5$, and $a^2 = 25$.

From the definition of a hyperbola, you know that the absolute value of the difference of distances from any point

on the hyperbola to the foci is constant. So, $|PF_1 - PF_2| = 2a$. Let $F_1 = (-c, 0)$, $F_2 = (c, 0)$, and $P = (x, y)$.

2

Use the values of a and c to find b .

Using the values of a and b , the equation for the hyperbola is $\frac{x^2}{25} - \frac{y^2}{16} = 1$.

69. SOUND

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tornado

Cognero siren goes off, three people are located at J, K, and O, as shown on the graph below.

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Using the values of a and b , the equation for the hyperbola is $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$.

69. SOUND When a tornado siren goes off, three people are located at J, K, and O, as shown on the graph below.

The person at J hears the siren 2 seconds before the person at O. The person at K hears the siren 1 second before

the person at O. Find each possible location of the tornado siren. Assume that sound travels at 1100 feet per

second. (Hint: Each location of the siren is a point of intersection between a hyperbola that has foci at O and J and

a hyperbola that has foci at O and K.)

SOLUTION:

First, find the equation that corresponds to each hyperbola. Let S represent the location of the siren and t be the

time it takes for the sound to travel from point S to point O .

The distance between points J and O is 3500 feet. So, the center of the hyperbola with foci at J and O is located at

$(0, 1750)$. Thus, $h = 0$, $k = 1750$, and $c = 1750$. Since the person at J hears the siren 2 seconds before the person at

O , the distance from the siren to point O is $1100t$ and the distance from the siren to point J is $1100(t - 2)$. For the

hyperbola, $PF_1 - PF_2 = 2a$. Substitute and find the value of a .

2

Use the values of a and c to find b .

$c^2 = a^2 + b^2$

$b^2 = c^2 - a^2$

$b^2 = 1750^2 - 1100^2$

$b^2 = 1750^2 - 1100^2$

2

$b^2 = 3062500 - 1210000$

$b^2 = 1852500$

b

Since the foci are on a vertical axis, the equation of the hyperbola will have the form .

$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$

Using the values of h , k , a , and b , the equation for the hyperbola with foci at points J and O is

The distance between points K and O is 2600 feet. So, the center is located at $(1300, 0)$. Thus, $h = 1300$, $k = 0$, and

$c = 1300$. Since the person at K hears the siren 1 second before the person at O , the distance from the siren to

point O is $1100t$ and the distance from the siren to point K is $1100(t - 1)$. For the hyperbola, $PF_1 - PF_2 = 2a$.

Substitute and find the value of a .

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2

Use the values of a and c to find b .

$c^2 = a^2 + b^2$

The distance between points K and O is 2600 feet. So, the center is located at $(1300, 0)$. Thus, $h = 1300$, $k = 0$, and

$c = 1300$. Since the person at K hears the siren 1 second before the person at O , the distance from the siren to

7-3 Hyperbolas

point O is $1100t$ and the distance from the siren to point K is $1100(t - 1)$. For the hyperbola, $PF_1 - PF_2 = 2a$.

Substitute and find the value of a .

2

Use the values of a and c to find b .

$$b^2 = c^2 - a^2$$

$$b^2 = 1300^2 - 550^2$$

2

$$b^2 = 1690000 - 302500$$

2

$$b^2 = 1387500$$

Since the foci are on a horizontal axis, the equation of the hyperbola will have the form

2 2

Using the values of h , k , a , and b , the equation for the hyperbola with foci at points K and O is

Solving each of these equations for y will give you the following two equations.

and

Separate the $\pm$ and enter the equations in $Y1 - Y4$ on a graphing calculator and graph the two hyperbolas. Use

the intersect function to find the four points of intersections. So, the possible locations of the tornado siren, are at

$$(-142.5, 2856.0), (3653.6, 4901.0), (700.1, 513.0), (1852.3, -107.7).$$

Write an equation for the hyperbola with the given characteristics.

71. The hyperbola has its center at $(-4, 3)$ and a vertex at $(1, 3)$. The equation of one of its asymptotes is $7x + 5y =$

$$-13.$$

SOLUTION:

2

The center is located at $(-4, 3)$, so $h = -4$ and $k = 3$. One vertex is located at $(1, 3)$, so $a = 5$ and $a^2 = 25$. Because

the y -coordinates of the center and a vertex are the same, the transverse axis is horizontal, and the standard form

of the equation is $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$. The slopes of the asymptotes are $\pm \frac{b}{a}$. An equation of an

2

asymptote is $7x + 5y = -13$ or $y = -\frac{7}{5}x - \frac{13}{5}$. So, the slope of the other asymptote is $\frac{7}{5}$, $b = 7$, and $b^2 = 49$.

Using the values of h , k , a , and b , the equation for the hyperbola is $\frac{(x+4)^2}{25} - \frac{(y-3)^2}{49} = 1$.

73. The eccentricity of the hyperbola is $\frac{5}{4}$ and the foci are at $(-1, -2)$ and $(13, -2)$.

SOLUTION:

Because the y -coordinates of the foci are the same, the transverse axis is horizontal, and the standard form of the

equation is $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$. The center is the midpoint of the segment between the foci, or $(6, -2)$.

So, Manual

eSolutions $h = 6$ and $k = -2$. Cогnero

$k = -2$

Powered The distance from a focus to the center is 7 units, so $c = 7$. Page 29

Substitute c into the eccentricity equation to find a .

Using the values of h , k , a , and b , the equation for the hyperbola is $- = 1$.

7-3 Hyperbolas

73. The eccentricity of the hyperbola is and the foci are at $(-1, -2)$ and $(13, -2)$.

SOLUTION:

Because the y -coordinates of the foci are the same, the transverse axis is horizontal, and the standard form of the

equation is $- = 1$. The center is the midpoint of the segment between the foci, or $(6, -2)$.

So, $h = 6$ and $k = -2$. The distance from a focus to the center is 7 units, so $c = 7$.

Substitute c into the eccentricity equation to find a .

$e =$

$a = 6$

2

So, $a = 36$.

Now you can use the values of a and c to find b .

$2 \ 2 \ 2$

$b = c - a$

$2 \ 2$

$= 7 - 6$

$= 49 - 36$ or 13

Using the values of h , k , a , and b , the equation for the hyperbola is $- = 1$.

75. For an equilateral hyperbola, $a = b$ when the equation of the hyperbola is written in standard form. The

asymptotes of an equilateral hyperbola are perpendicular. Write an equation for the equilateral hyperbola shown

below.

SOLUTION:

From the graph, you can see that the center of the hyperbola is $(0, 0)$, so $h = 0$ and $k = 0$. The distance from a

focus to the center is 11 units, so $c = 11$.

$2 \ 2 \ 2$

Substitute $c = 11$ and $a = b$ into $c = a + b$ to find a .

$2 \ 2 \ 2$

$c = a + b$

$2 \ 2 \ 2$

$11 = a + a$

$121 = 2a^2$

2

$= a$

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Using the values of h , k , a , and b , the equation for the hyperbola is $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$.

$$= 49 - 36 \text{ or } 13$$

7-3 Hyperbolas

Using the values of h , k , a , and b , the equation for the hyperbola is $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$.

75. For an equilateral hyperbola, $a = b$ when the equation of the hyperbola is written in standard form. The

asymptotes of an equilateral hyperbola are perpendicular. Write an equation for the equilateral hyperbola shown

below.

SOLUTION:

From the graph, you can see that the center of the hyperbola is $(0, 0)$, so $h = 0$ and $k = 0$. The distance from a

focus to the center is 11 units, so $c = 11$.

$$c^2 = a^2 + b^2$$

Substitute $c = 11$ and $a = b$ into $c^2 = a^2 + b^2$ to find a .

$$c^2 = a^2 + b^2$$

$$c^2 = a^2 + a^2$$

$$c^2 = 2a^2$$

$$11^2 = 2a^2$$

$$121 = 2a^2$$

$$\frac{121}{2} = a^2$$

$$a = \pm \sqrt{\frac{121}{2}}$$

Using the values of h , k , a , and b , the equation for the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

77. OPEN ENDED Write an equation for a hyperbola where the distance between the foci is twice the length of the transverse axis.

SOLUTION:

Sample answer: Let the location of the center of the hyperbola be the origin, so that $h = 0$ and $k = 0$.

The distance between the foci of a hyperbola is equal to $2c$, and the length of the transverse axis is equal to $2a$. So,

if the distance between the foci is twice the length of the transverse axis, $2c = 2(2a)$ or $c = 2a$.

$$c^2 = a^2 + b^2$$

Substitute $c = 2a$ into $c^2 = a^2 + b^2$.

$$c^2 = a^2 + b^2$$

$$c^2 = a^2 + b^2$$

$$c^2 = a^2 + b^2$$

$$(2a)^2 = a^2 + b^2$$

$$4a^2 = a^2 + b^2$$

$$4a^2 = a^2 + b^2$$

$$2^2$$

$$3a = b$$

Let $a = 4$.

$$2^2$$

$$3a = b$$

$$2^2$$

$$3(4) = b$$

$$48 = b^2$$

So, one equation for a hyperbola in which the distance between the foci is twice the length of the transverse axis is

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$$- = 1.$$

$$2$$

$$=a$$

7-3 Hyperbolas

Using the values of h , k , a , and b , the equation for the hyperbola is $- = 1$.

77. OPEN ENDED Write an equation for a hyperbola where the distance between the foci is twice the length of the transverse axis.

SOLUTION:

Sample answer: Let the location of the center of the hyperbola be the origin, so that $h = 0$ and $k = 0$.

The distance between the foci of a hyperbola is equal to $2c$, and the length of the transverse axis is equal to $2a$. So,

if the distance between the foci is twice the length of the transverse axis, $2c = 2(2a)$ or $c = 2a$.

$$2^2$$

Substitute $c = 2a$ into $c^2 = a^2 + b^2$.

$$2^2$$

$$c^2 = a^2 + b^2$$

$$2^2$$

$$(2a)^2 = a^2 + b^2$$

$$2^2$$

$$4a^2 = a^2 + b^2$$

$$2^2$$

$$3a^2 = b^2$$

Let $a = 4$.

$$2^2$$

$$3a^2 = b^2$$

$$2^2$$

$$3(4)^2 = b^2$$

$$48 = b^2$$

So, one equation for a hyperbola in which the distance between the foci is twice the length of the transverse axis is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

79. Writing in Math Explain why the equation for the asymptotes of a hyperbola changes from $\pm$ to $\pm$ depending on the location of the transverse axis.

SOLUTION:

Sample answer: If the transverse axis is horizontal, the distance between the two vertices a is a horizontal distance

and b is a vertical distance. So, the slopes of the asymptotes are equal to the rise over the run or $\pm \frac{b}{a}$. If the

transverse axis is vertical, then a is a vertical distance and b is a horizontal distance. So, the slopes of the asymptotes are $\pm \frac{a}{b}$.

81. CHALLENGE A hyperbola has foci at $F_1(0, 9)$ and $F_2(0, -9)$ and contains point P . The distance between P and

F_1 is 6 units greater than the distance between P and F_2 . Write the equation of the hyperbola in standard form.

SOLUTION:

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Let $P(x, y)$ be the point on the hyperbola. From the definition of a hyperbola, you know that the absolute value of

the difference of distances from any point on the hyperbola to the foci is constant, so $|PF_1 - PF_2| = 2a$. In the

and b is a vertical distance. So, the slopes of the asymptotes are equal to the rise over the run or $\pm \frac{b}{a}$. If the

transverse axis is vertical, then a is a vertical distance and b is a horizontal distance. So, the slopes of the

7-3 Hyperbolas

asymptotes are $\pm \frac{b}{a}$.

81. CHALLENGE A hyperbola has foci at $F_1(0, 9)$ and $F_2(0, -9)$ and contains point P . The distance between P and

F_1 is 6 units greater than the distance between P and F_2 . Write the equation of the hyperbola in standard form.

SOLUTION:

Let $P(x, y)$ be the point on the hyperbola. From the definition of a hyperbola, you know that the absolute value of

the difference of distances from any point on the hyperbola to the foci is constant, so $|PF_1 - PF_2| = 2a$. In the

problem, you are told that the distance between P and F_1 is 6 units greater than the distance between P and F_2 , so

2

$PF_1 - PF_2 = 6$. If you substitute $PF_1 - PF_2 = 6$ into $|PF_1 - PF_2| = 2a$, you get $2a = 6$, $a = 3$, and $a = 9$.

The center is the midpoint of the segment between the foci, or $(0, 0)$. So, $h = 0$ and $k = 0$. You can find c by

determining the distance from a focus to the center. One focus is located at $(0, 9)$ which is 9 units from $(0, 0)$. So, c

$= 9$.

You can use the values of a and c to solve for b .

$$c^2 = a^2 + b^2$$

$$b^2 = c^2 - a^2$$

$$b^2 = 9^2 - 3^2$$

$$b^2 = 81 - 9$$

$$b^2 = 72$$

$$b = \sqrt{72}$$

$$b = 6\sqrt{2}$$

$$b = 72$$

$$b =$$

Using the values of h , k , a , and b , the equation for the hyperbola is $\frac{y^2}{81} - \frac{x^2}{72} = 1$.

83. Writing in Math Describe the steps for finding the equation of a hyperbola if the foci and length of the transverse axis are given.

SOLUTION:

Sample answer: First, determine whether the orientation of the hyperbola is vertical or horizontal. Then use the foci

to

locate the center of the hyperbola and determine the values of h and k . Use the transverse axis length to find a .

$$c^2 = a^2 + b^2$$

Find c , the distance from the center to a focus. Use the equation $b^2 = c^2 - a^2$ to find b . Finally, use the correct

standard form to write the equation, depending on whether the transverse axis is parallel to the x -axis or to the

y -axis.

Graph the ellipse given by each equation.

$$85. \frac{x^2}{16} + \frac{y^2}{9} = 1$$

SOLUTION:

The ellipse is in standard form, where $h = 0$, $k = -5$, $a = 4$ or 8 , $b = 3$ or 7 , and $c = 5$. The

orientation is horizontal because the x -term contains a^2 .

center: $(h, k) = (0, -5)$

foci: $(h \pm c, k) = (3.87, -5)$ and $(-3.87, -5)$

vertices: $(h \pm a, k) = (8, -5)$ and $(-8, -5)$

covertices: $(h, k \pm b) = (0, 2)$ and $(0, -12)$

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Graph the center, vertices, foci, and axes. Then use a table of values to sketch the ellipse.

2

to locate the center of the hyperbola and determine the values of h and k . Use the transverse axis length to find a .

2 2 2 2

Find c , the distance from the center to a focus. Use the equation $b = c - a$ to find b . Finally, use the correct

standard form to write the equation, depending on whether the transverse axis is parallel to the x -axis or to the

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y -axis.

Graph the ellipse given by each equation.

85. $\frac{x^2}{25} + \frac{y^2}{16} = 1$

SOLUTION:

The ellipse is in standard form, where $h = 0$, $k = -5$, $a = 5$ or 8 , $b = 4$ or 7 , and $c = 9$. The

2

orientation is horizontal because the x -term contains a 25 .

center: $(h, k) = (0, -5)$

foci: $(h \pm c, k) = (9, -5)$ and $(-9, -5)$

vertices: $(h \pm a, k) = (8, -5)$ and $(-8, -5)$

covertices: $(h, k \pm b) = (0, 2)$ and $(0, -12)$

Graph the center, vertices, foci, and axes. Then use a table of values to sketch the ellipse.

87. PROJECTILE MOTION The height of a baseball hit by a batter with an initial speed of 80 feet per second

2

can be modeled by $h = -16t^2 + 80t + 5$, where t is the time in seconds.

a. How high above the ground is the vertex located?

b. If an outfielder's catching height is the same as the initial height of the ball, about how long after the ball is hit

will the player catch the ball?

SOLUTION:

a. Find the x -coordinate of the axis of symmetry.

Substitute 2.5 for t in the equation.

The vertex is located 105 feet above the ground.

b. First, find the initial height of the ball.

$$h = -16t^2 + 80t + 5$$

$$= -16(0) + 80(0) + 5$$

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$$=5$$

Substitute $h = 5$ into the original equation.

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87. PROJECTILE MOTION The height of a baseball hit by a batter with an initial speed of 80 feet per second

2

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a. How high above the ground is the vertex located?

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SOLUTION:

a. Find the x-coordinate of the axis of symmetry.

Substitute 2.5 for t in the equation.

The vertex is located 105 feet above the ground.

b. First, find the initial height of the ball.

$$h = -16t^2 + 80t + 5$$

$$= -16(0) + 80(0) + 5$$

$$=5$$

Substitute $h = 5$ into the original equation.

$$h = -16t^2 + 80t + 5$$

$$5 = -16t^2 + 80t + 5$$

$$0 = -16t^2 + 80t$$

$$0 = t(-16t + 80)$$

Solve for t .

$$-16t + 80 = 0$$

$$-16t = -80$$

$$t = 5$$

Therefore, the player will catch the ball 5 seconds after it was hit.

Write each system of equations as a matrix equation, $AX = B$. Then use Gauss–Jordan elimination on

the augmented matrix to solve the system.

$$89. \begin{cases} x_1 - 7x_2 + 8x_3 = -3 \\ 6x_1 + 5x_2 - 2x_3 = 2 \\ 3x_1 - 4x_2 + 9x_3 = 26 \end{cases}$$

$$6x_1 + 5x_2 - 2x_3 = 2$$

$$3x_1 - 4x_2 + 9x_3 = 26$$

SOLUTION:

Write the system matrix in form $AX = B$. Make sure you align the variables. For the first matrix, the first column

should include x_1 , the second column x_2 , and the third column x_3 . The column matrix and the matrix of constant

terms should be listed in order.

$$89. x_1 - 7x_2 + 8x_3 = -3$$

$$6x_1 + 5x_2 - 2x_3 = 2$$

$$13x - 4x + 9x = 26$$

2 3

7-3 Hyperbolas

SOLUTION:

Write the system matrix in form $AX = B$. Make sure you align the variables. For the first matrix, the first column

should include x_1 , the second column x_2 , and the third column x_3 . The column matrix and the matrix of constant

terms should be listed in order.

Write the augmented matrix . Attach the matrix of constant terms to the end of the 3×3 matrix.

Use Gauss-Jordan elimination to solve the system. First, use elementary row operations to transform A into I.

The solution of the system is given by X

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X= =

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The solution of the system is given by X

X= =

Therefore, the solution of the system of equations is $(-5, 10, 9)$.

Solve each equation for all values of .

$$91. \tan \theta = \sec \theta - 1$$

SOLUTION:

$$\tan \theta = \sec \theta - 1$$

$$2 = (\sec \theta - 1)(\sec \theta - 1)$$

$$\tan \theta$$

$$2 2$$

$$\sec \theta - 1 = \sec \theta - 2\sec \theta + 1$$

$$-1 = -2\sec \theta + 1$$

$$-2 = -2\sec \theta$$

$$1 = \sec \theta$$

On the interval $[0, 2\pi)$, $\sec \theta = 1$ when $\theta = 0$ or 2π . Therefore, the solutions on the interval $(-\infty, \infty)$ have the

general form $2n\pi$, where n is an integer.

$$93. \csc \theta - \cot \theta = 0$$

SOLUTION:

$\cos \theta = 1$ when $\theta = 0$. However, when $\theta = 0$, $\csc \theta$ does not exist.

This equation has no solution.

Find the exact values of the six trigonometric functions of .

95.

SOLUTION:

The length of the side opposite θ is 24 and the length of the hypotenuse is 25. Use the Pythagorean Theorem to

find the length of the side adjacent to θ .

$$2^2 + 2^2 = 2^2$$

$$a^2 + b^2 = c^2$$

$$2^2 + 2^2 = 2^2$$

$$a^2 + 24^2 = 25^2$$

$$2^2$$

$$a^2 = 625 - 576$$

$$a = 7 \text{ or } -7$$

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Since the length of each side has been found, you can find the exact values of the trigonometric functions.

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7-3 Hyperbolas

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The length of the side opposite θ is 24 and the length of the hypotenuse is 25. Use the Pythagorean Theorem to

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$$a^2 + b^2 = c^2$$

$$2^2 + 2^2 = 2^2$$

$$a^2 + 24^2 = 25^2$$

$$2^2$$

$$a^2 = 625 - 576$$

$$a = 7 \text{ or } -7$$

Since the length of each side has been found, you can find the exact values of the trigonometric functions.

$$\sin \theta = \frac{24}{25} = \csc \theta = \frac{25}{24}$$

$$\cos \theta = \frac{7}{25} = \sec \theta = \frac{25}{7}$$

$$\tan \theta = \frac{24}{7} = \cot \theta = \frac{7}{24}$$

Use the given zero to find all complex zeros of each function. Then write the linear factorization of the function.

$$97. f(x) = 2x^5 - 9x^4 + 146x^3 + 618x^2 + 752x + 291; 4 + 9i$$

SOLUTION:

Use synthetic substitution to verify that $4 + 9i$ is a zero of $f(x)$.

Because $x = 4 + 9i$ is a zero of f , $x = 4 - 9i$ is also a zero of f . Divide the depressed polynomial by $4 - 9i$.

Using these two zeros and the depressed polynomial from this last division, you can write $f(x) = [x - (4 + 9i)][x -$

$$3 2$$

$$(4 - 9i)](2x + 7x + 8x + 3).$$

Factor the remaining depressed polynomial.

$$3 2$$

$$(2x + 7x + 8x + 3)$$

$$2$$

$$= (2x$$

$$eSolutions + 3)(x$$

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$$- \text{Powered 1) Page 38}$$

$$= (2x + 3)(x + 1)(x + 1)$$

$$\cos \theta = = \sec \theta = =$$

$$\tan \theta = = \tan \theta = =$$

7-3 Hyperbolas

Use the given zero to find all complex zeros of each function. Then write the linear factorization of the function.

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Using these two zeros and the depressed polynomial from this last division, you can write $f(x) = [x - (4 + 9i)][x -$

$$3 2$$

$$(4 - 9i)](2x + 7x + 8x + 3).$$

Factor the remaining depressed polynomial.

$$3 2$$

$$(2x + 7x + 8x + 3)$$

$$2$$

$$= (2x + 3)(x + 2x + 1)$$

$$= (2x + 3)(x + 1)(x + 1)$$

So, remaining zeros are $x =$ and $x = -1$ with multiplicity 2.

2

The linear factorization of f is $f(x) = (2x + 3)(x + 1)(x - 4 + 9i)(x - 4 - 9i)$.

99. REVIEW The graph of $= 1$ is a hyperbola. Which set of equations represents the asymptotes of the

hyperbola's graph?

F $y = x$, $y =$

G $y = x$, $y =$

H $y = x$, $y =$

J $y = x$, $y =$

SOLUTION:

2

The standard form of the equation is $= 1$, where $h = 0$, $k = 0$, $a = 4$, and $b = 5$. Since the y -term is

being subtracted, the orientation of the hyperbola is horizontal, and the asymptotes are
or .

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Therefore, the answer is H.

So, remaining zeros are $x =$ and $x = -1$ with multiplicity 2.

7-3 Hyperbolas 2

The linear factorization of f is $f(x) = (2x + 3)(x + 1)(x - 4 + 9i)(x - 4 - 9i)$.

99. REVIEW The graph of $= 1$ is a hyperbola. Which set of equations represents the asymptotes of the

hyperbola's graph?

F $y = x$, $y =$

G $y = x$, $y =$

H $y = x$, $y =$

J $y = x$, $y =$

SOLUTION:

2

The standard form of the equation is $= 1$, where $h = 0$, $k = 0$, $a = 4$, and $b = 5$. Since the y -term is

being subtracted, the orientation of the hyperbola is horizontal, and the asymptotes are
or .

Therefore, the answer is H.

101. SAT/ACT If $z =$, then what is the effect on the value of z when y is multiplied by 4 and x is doubled?

F z is unchanged.

G z is halved.

H z is doubled.

J z is multiplied by 4.

SOLUTION:

Multiply y by 4 and x by 2.

z =

Because is equal to $\cdot$ or $\cdot$ z, the answer is G.

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7-1 Parabolas

For each equation, identify the vertex, focus, axis of symmetry, and directrix. Then graph the parabola.

1. $(x - 3)^2 = 12(y - 7)$

SOLUTION:

The equation is in standard form and the squared term is x, which means that the parabola opens vertically. The

2

equation is in the form $(x - h) = 4p(y - k)$, so $h = 3$ and $k = 7$. Because $4p = 12$ and $p = 3$, the graph opens

up. Use the values of h, k, and p to determine the characteristics of the parabola.

vertex: $(h, k) = (3, 7)$

focus: $(h, k + p) = (3, 10)$

directrix: $y = k - p$ or 4

axis of symmetry: $x = h$ or 3

Graph the vertex, focus, axis, and directrix of the parabola. Then make a table of values to graph the general shape

of the curve.

3. $(y - 4)^2 = 20(x + 2)$

SOLUTION:

The equation is in standard form and the squared term is y, which means that the parabola opens horizontally. The

2

equation is in the form $(y - k) = 4p(x - h)$, so $h = -2$ and $k = 4$. Because $4p = 20$ and $p = 5$, the graph opens to

the right. Use the values of h, k, and p to determine the characteristics of the parabola.

vertex: $(h, k) = (-2, 4)$

focus: $(h + p, k) = (3, 4)$

directrix: $x = h - p$ or -7

axis of symmetry: $y = k$ or 4

Graph the vertex, focus, axis, and directrix of the parabola. Then make a table of values to graph the general shape

of the curve.

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$$5. (x + 8) = 8(y - 3)$$

SOLUTION:

7-1 Parabolas

$$5. (x + 8)^2 = 8(y - 3)$$

SOLUTION:

The equation is in standard form and the squared term is x , which means that the parabola opens vertically. The

2

equation is in the form $(x - h) = 4p(y - k)$, so $h = -8$ and $k = 3$. Because $4p = 8$ and $p = 2$, the graph opens up.

Use the values of h , k , and p to determine the characteristics of the parabola.

vertex: $(h, k) = (-8, 3)$

focus: $(h, k + p) = (-8, 5)$

directrix: $y = k - p$ or 1

axis of symmetry: $x = h$ or -8

Graph the vertex, focus, axis, and directrix of the parabola. Then make a table of values to graph the general shape

of the curve.

$$7. (y + 5)^2 = 24(x - 1)$$

SOLUTION:

The equation is in standard form and the squared term is y , which means that the parabola opens horizontally. The

2

equation is in the form $(y - k) = 4p(x - h)$, so $h = 1$ and $k = -5$. Because $4p = 24$ and $p = 6$, the graph opens to

the right. Use the values of h , k , and p to determine the characteristics of the parabola.

vertex: $(h, k) = (1, -5)$

focus: $(h + p, k) = (7, -5)$

directrix: $x = h - p$ or -5

axis of symmetry: $y = k$ or -5

Graph the vertex, focus, axis, and directrix of the parabola. Then make a table of values to graph the general shape

of the curve.

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$$-4(y + 2) = (x + 8) \text{ Page 2}$$

SOLUTION:

7-1 Parabolas

9. $-4(y + 2) = (x + 8)^2$

SOLUTION:

The equation is in standard form and the squared term is x , which means that the parabola opens vertically. The

2

equation is in the form $4p(y - k) = (x - h)^2$, so $h = -8$ and $k = -2$. Because $4p = -4$ and $p = -1$, the graph opens

down. Use the values of h , k , and p to determine the characteristics of the parabola.

vertex: $(h, k) = (-8, -2)$

focus: $(h + p, k) = (-8, -3)$

directrix: $y = k - p$ or -1

axis of symmetry: $x = h$ or -8

Graph the vertex, focus, axis, and directrix of the parabola. Then make a table of values to graph the general shape

of the curve.

11. SKATEBOARDING A group of high school students designing a half-pipe have decided that the ramps, or

transitions, could be obtained by splitting a parabola in half. A parabolic cross section of the ramps can be modeled

2

by $x = 8(y - 2)$, where the values of x and y are measured in feet. Where is the focus of the parabola in relation

to the ground if the ground represents the directrix?

SOLUTION:

Because the x -term is squared and p is positive, the parabola opens up and the focus is located at $(h, k + p)$. The

equation is provided in standard form, and $h = 0$ and $k = 2$. Because $4p = 8$, p is 2. So, the location of the focus is

$(0, 2 + 2)$ or $(0, 4)$. Therefore, the focus is 4 feet above the ground.

13. BOATING As a speed boat glides through the water, it creates a wake in the shape of a parabola. The vertex of

this parabola meets with the stern of the boat. A swimmer on a wakeboard, attached by a piece of rope, is being

pulled by the boat. When he is directly behind the boat, he is positioned at the focus of the parabola. The parabola

2

formed by the wake can be modeled using $y - 180x + 10y + 565 = 0$, where x and y are measured in feet.

a. Write the equation in standard form.

b. How long is the length of rope attaching the swimmer to the stern of the boat?

SOLUTION:

a.

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SOLUTION:

Because the x -term is squared and p is positive, the parabola opens up and the focus is located at $(h, k + p)$. The

equation is provided in standard form, and $h = 0$ and $k = 2$. Because $4p = 8$, p is 2. So, the location of the focus is

7-1 Parabolas

$(0, 2 + 2)$ or $(0, 4)$. Therefore, the focus is 4 feet above the ground.

13. BOATING As a speed boat glides through the water, it creates a wake in the shape of a parabola. The vertex of

this parabola meets with the stern of the boat. A swimmer on a wakeboard, attached by a piece of rope, is being

pulled by the boat. When he is directly behind the boat, he is positioned at the focus of the parabola. The parabola

2

formed by the wake can be modeled using $y - 180x + 10y + 565 = 0$, where x and y are measured in feet.

a. Write the equation in standard form.

b. How long is the length of rope attaching the swimmer to the stern of the boat?

SOLUTION:

a.

b. The equation in standard form has y as the squared term, which means that the parabola opens horizontally.

2

Because $4p = 180$, $p = 45$ and the graph opens to the right. The equation is in the form $(y - k) = 4p(x - h)$, so $h =$

3 and $k = -5$. Since the stern is located at the vertex of the parabola formed, it is at the point (h, k) or $(3, -5)$.

The swimmer is at the focus, located at $(h + p, k)$, which is $(3 + 45, -5)$ or $(48, -5)$. The distance the swimmer is

from the stern of the boat represents the length of rope needed to attach the swimmer to the stern.

Using the distance formula, the distance between these two points is or 45.

Therefore, the length of rope attaching the swimmer to the stern of the boat is 45 feet.

Write each equation in standard form. Identify the vertex, focus, axis of symmetry, and directrix. Then

graph the parabola.

15. $x^2 - 17 = 8y + 39$

SOLUTION:

Because the x -term is squared and $p = 2$, the graph opens up. Use the standard form to determine the

characteristics of the parabola.

vertex: $(h, k) = (0, -7)$

focus: $(h, k + p) = (0, -5)$

directrix: $y = k - p$ or -9

axis of symmetry: $x = h$ or 0

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Graph the vertex, focus, axis, and directrix of the parabola. Then make a table of values to graph the curve. The

curve should be symmetric about the axis of symmetry.

from the stern of the boat represents the length of rope needed to attach the swimmer to the stern.

Using the distance formula, the distance between these two points is

7-1 Parabolas or 45.

Therefore, the length of rope attaching the swimmer to the stern of the boat is 45 feet.

Write each equation in standard form. Identify the vertex, focus, axis of symmetry, and directrix. Then

graph the parabola.

15. $x^2 - 17 = 8y + 39$

SOLUTION:

Because the x -term is squared and $p = 2$, the graph opens up. Use the standard form to determine the

characteristics of the parabola.

vertex: $(h, k) = (0, -7)$

focus: $(h, k + p) = (0, -5)$

directrix: $y = k - p$ or -9

axis of symmetry: $x = h$ or 0

Graph the vertex, focus, axis, and directrix of the parabola. Then make a table of values to graph the curve. The

curve should be symmetric about the axis of symmetry.

17. $3x^2 + 72 = -72y$

SOLUTION:

Because the x -term is squared and $p = -6$, the graph opens down. Use the standard form to determine the

characteristics of the parabola.

vertex: $(h, k) = (0, -1)$

focus: $(h, k + p) = (0, -7)$

directrix: $y = k - p$ or 5

axis of symmetry: $x = h$ or 0

Graph the vertex, focus, axis, and directrix of the parabola. Then make a table of values to graph the curve. The

curve should be symmetric about the axis of symmetry.

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$$17. 3x^2 + 72 = -72y$$

SOLUTION:

Because the x-term is squared and $p = -6$, the graph opens down. Use the standard form to determine the

characteristics of the parabola.

vertex: $(h, k) = (0, -1)$

focus: $(h, k + p) = (0, -7)$

directrix: $y = k - p$ or 5

axis of symmetry: $x = h$ or 0

Graph the vertex, focus, axis, and directrix of the parabola. Then make a table of values to graph the curve. The

curve should be symmetric about the axis of symmetry.

$$19. 60x - 80 = 3y^2 + 100$$

SOLUTION:

Because the y-term is squared and $p = 5$, the graph opens to the right. Use the standard form to determine the

characteristics of the parabola.

vertex: $(h, k) = (3, 0)$

focus: $(h + p, k) = (8, 0)$

directrix: $x = h - p$ or -2

axis of symmetry: $y = k$ or 0

Graph the vertex, focus, axis, and directrix of the parabola. Then make a table of values to graph the curve. The

curve

eSolutions should

Manual be symmetric

- Powered by Cognero about the axis of symmetry. Page 6

7-1 Parabolas

$$19. 60x - 80 = 3y^2 + 100$$

SOLUTION:

Because the y-term is squared and $p = 5$, the graph opens to the right. Use the standard form to determine the

characteristics of the parabola.

vertex: $(h, k) = (3, 0)$

focus: $(h + p, k) = (8, 0)$

directrix: $x = h - p$ or -2

axis of symmetry: $y = k$ or 0

Graph the vertex, focus, axis, and directrix of the parabola. Then make a table of values to graph the curve. The

curve should be symmetric about the axis of symmetry.

$$21. -72 = 2y^2 - 16y - 20x$$

SOLUTION:

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Because the y -term is squared and $p = 2.5$, the graph opens to the right. Use the standard form to determine the

characteristics of the parabola.

7-1 Parabolas

$$21. -72 = 2y^2 - 16y - 20x$$

SOLUTION:

Because the y -term is squared and $p = 2.5$, the graph opens to the right. Use the standard form to determine the

characteristics of the parabola.

vertex: $(h, k) = (2, 4)$

focus: $(h + p, k) = (4.5, 4)$

directrix: $x = h - p$ or -0.5

axis of symmetry: $y = k$ or 4

Graph the vertex, focus, axis, and directrix of the parabola. Then make a table of values to graph the curve. The

curve should be symmetric about the axis of symmetry.

$$23. x^2 - 18y + 12x = 126$$

SOLUTION:

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Because the x -term is squared and $p = 4.5$, the graph opens up. Use the standard form to determine the

7-1 Parabolas

$$23. x^2 - 18y + 12x = 126$$

SOLUTION:

Because the x -term is squared and $p = 4.5$, the graph opens up. Use the standard form to determine the

characteristics of the parabola.

vertex: $(h, k) = (-6, -9)$

focus: $(h, k + p) = (-6, -4.5)$

directrix: $y = k - p$ or -13.5

axis of symmetry: $x = h$ or -6

Graph the vertex, focus, axis, and directrix of the parabola. Then make a table of values to graph the curve. The

curve should be symmetric about the axis of symmetry.

25. LIGHTING Stadium lights at an athletic field need to reflect light at maximum intensity. The bulb should be placed

2

at the focal point of the parabolic globe surrounding it. If the shape of the globe is given by $x = 36y$, where x and y

are in inches, how far from the vertex of the globe should the bulb be placed for maximum light?

SOLUTION:

The bulb is placed at the focus of the parabola. Because the x -term is squared and p is positive, the parabola

opens up and the focus is located at $(h, k + p)$. The equation is provided in standard form, and $h = 0$ and $k = 0$.

Because $4p = 36$, p is 9. So, the location of the focus is $(0, 0 + 9)$ or $(0, 9)$. The vertex of the globe is located at $(0,$

$0)$. The distance from the focus to the vertex is $9 - 0$ or 9. Therefore, the bulb should be placed 9 inches from the

vertex of the globe.

Write an equation for and graph a parabola with the given focus F and vertex V .

27. $F(2, -1)$, $V(-4, -1)$

SOLUTION:

Because the focus and vertex share the same y -coordinate, the graph is horizontal. The focus is $(h + p, k)$, so the

value of p is $2 - (-4)$ or 6. Because p is positive, the graph opens to the right.

Write the equation for the parabola in standard form using the values of h , p , and k .

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$$4p(x - h) = (y - k)^2$$

$$4(6)[x - (-4)] = [y - (-1)]^2$$

The bulb is placed at the focus of the parabola. Because the x -term is squared and p is positive, the parabola

opens up and the focus is located at $(h, k + p)$. The equation is provided in standard form, and $h = 0$ and $k = 0$.

Because $4p = 36$, p is 9. So, the location of the focus is $(0, 0 + 9)$ or $(0, 9)$. The vertex of the globe is located at $(0,$

$0)$. The distance from the focus to the vertex is $9 - 0$ or 9. Therefore, the bulb should be placed 9 inches from the

7-1 Parabolas

vertex of the globe.

Write an equation for and graph a parabola with the given focus F and vertex V .

27. $F(2, -1)$, $V(-4, -1)$

SOLUTION:

Because the focus and vertex share the same y -coordinate, the graph is horizontal. The focus is $(h + p, k)$, so the

value of p is $2 - (-4)$ or 6. Because p is positive, the graph opens to the right.

Write the equation for the parabola in standard form using the values of h , p , and k .

$$4p(x - h) = (y - k)^2$$

$$4(6)[x - (-4)] = [y - (-1)]^2$$

$$24(x + 4) = (y + 1)^2$$

2

The standard form of the equation is $(y + 1) = 24(x + 4)$. Graph the vertex and focus. Then make a table of values

to graph the parabola.

29. $F(-3, 4), V(-3, 2)$

SOLUTION:

Because the focus and vertex share the same x-coordinate, the graph is vertical. The focus is $(h, k + p)$, so the

value of p is $4 - 2$ or 2 . Because p is positive, the graph opens up.

Write the equation for the parabola in standard form using the values of h , p , and k .

$$4p(y - k) = (x - h)^2$$

$$4(2)(y - 2) = [x - (-3)]^2$$

$$8(y - 2) = (x + 3)^2$$

2

The standard form of the equation is $(x + 3) = 8(y - 2)$. Graph the vertex and focus. Then make a table of values

to graph the parabola.

31. $F(-1,$

$eSolutions 4), -V(7,$

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SOLUTION:

7-1 Parabolas

31. $F(-1, 4), V(7, 4)$

SOLUTION:

Because the focus and vertex share the same y-coordinate, the graph is horizontal. The focus is $(h + p, k)$, so the

value of p is $-1 - 7$ or -8 . Because p is negative, the graph opens to the left.

Write the equation for the parabola in standard form using the values of h , p , and k .

$$4p(x - h) = (y - k)^2$$

$$4(-8)(x - 7) = (y - 4)^2$$

$$-32(x - 7) = (y - 4)^2$$

2

The standard form of the equation is $(y - 4) = -32(x - 7)$. Graph the vertex and focus. Then make a table of

values to graph the parabola.

33. $F(1, 3), V(1, 6)$

SOLUTION:

Because the focus and vertex share the same x-coordinate, the graph is vertical. The focus is $(h, k + p)$, so the

value of p is $3 - 6$ or -3 . Because p is negative, the graph opens down.

Write the equation for the parabola in standard form using the values of h , p , and k .

$$4p(y - k) = (x - h)^2$$

$$4(-3)(y - 6) = (x - 1)^2$$

$$-12(y - 6) = (x - 1)^2$$

2

The standard form of the equation is $(x - 1)^2 = -12(y - 6)$. Graph the vertex and focus. Then make a table of

values to graph the parabola.

$$35. F(8, -3), V(8, -7)$$

SOLUTION:

Because

eSolutions Manual -the focus by and

Powered vertex share the same x -coordinate, the graph is vertical. The focus is $(h, k + p)$, so the
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value of p is $-3 - (-7)$ or 4 . Because p is positive, the graph opens up.

7-1 Parabolas

$$35. F(8, -3), V(8, -7)$$

SOLUTION:

Because the focus and vertex share the same x -coordinate, the graph is vertical. The focus is $(h, k + p)$, so the

value of p is $-3 - (-7)$ or 4 . Because p is positive, the graph opens up.

Write the equation for the parabola in standard form using the values of h , p , and k .

$$4p(y - k) = (x - h)^2$$

$$4(4)[y - (-7)] = (x - 8)^2$$

$$16(y + 7) = (x - 8)^2$$

2

The standard form of the equation is $(x - 8)^2 = 16(y + 7)$. Graph the vertex and focus. Then make a table of values

to graph the parabola.

Write an equation for and graph each parabola with focus F and the given characteristics.

$$37. F(1, 2); \text{ opens down; contains } (7, 2)$$

SOLUTION:

Because the parabola opens down, the vertex is $(1, 2 - p)$. Use the standard form of the equation of a vertical

parabola and the point $(7, 2)$ to find the equation.

$$4p(y - k) = (x - h)^2$$

$$4p[2 - (2 - p)] = (7 - 1)^2$$

$$4p(p) = 36$$

2

$$p = 9$$

$$p = \pm 3$$

Because the parabola opens down, the value of p must be negative. Therefore, $p = -3$. The vertex is $(1, 5)$, and the

2

standard form of the equation is $(x - 1) = -12(y - 5)$. Use a table of values to graph the parabola.

39. $F(-4, 0)$; opens down; contains $(4, -15)$

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SOLUTION: Page 12

Because the parabola opens down, the vertex is $(-4, -p)$. Use the standard form of the equation of a vertical

parabola and the point $(4, -15)$ to find the equation.

7-1 Parabolas

39. $F(-4, 0)$; opens down; contains $(4, -15)$

SOLUTION:

Because the parabola opens down, the vertex is $(-4, -p)$. Use the standard form of the equation of a vertical

parabola and the point $(4, -15)$ to find the equation.

$$4p(y - k) = (x - h)^2$$

$$4p[-15 - (-p)] = [4 - (-4)]^2$$

$$4p(-15 + p) = 64$$

2

$$4p - 60p = 64$$

2

$$p - 15p - 16 = 0$$

$$(p - 16)(p + 1) = 0$$

$$p = 16 \text{ or } -1$$

Because the parabola opens down, the value of p must be negative. Therefore, $p = -1$. The vertex is $(-4, 1)$, and

2

the standard form of the equation is $(x + 4) = -4(y - 1)$. Use a table of values to graph the parabola.

41. $F(-5, -9)$; opens right; contains $(10, -1)$

SOLUTION:

Because the parabola opens to the right, the vertex is $(-5 - p, -9)$. Use the standard form of the equation of a

horizontal parabola and the point $(10, -1)$ to find the equation.

$$4p(x - h) = (y - k)^2$$

$$4p[10 - (-5 - p)] = [-1 - (-9)]^2$$

$$4p(15 + p) = 64$$

2

$$4p + 60p = 64$$

2

$$p + 15p - 16 = 0$$

$$(p + 16)(p - 1) = 0$$

$$p = 1 \text{ or } -16$$

Because the parabola opens to the right, the value of p must be positive. Therefore, $p = 1$. The vertex is $(-6, -9)$,

2

and the standard form of the equation is $4(x + 6) = (y + 9)$. Use a table of values to graph the parabola.

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7-1 Parabolas

41. $F(-5, -9)$; opens right; contains $(10, -1)$

SOLUTION:

Because the parabola opens to the right, the vertex is $(-5 - p, -9)$. Use the standard form of the equation of a

horizontal parabola and the point $(10, -1)$ to find the equation.

$$4p(x - h) = (y - k)^2$$

$$4p[10 - (-5 - p)] = [-1 - (-9)]^2$$

$$4p(15 + p) = 64$$

2

$$4p + 60p = 64$$

2

$$p + 15p - 16 = 0$$

$$(p + 16)(p - 1) = 0$$

$$p = 1 \text{ or } -16$$

Because the parabola opens to the right, the value of p must be positive. Therefore, $p = 1$. The vertex is $(-6, -9)$,

2

and the standard form of the equation is $4(x + 6) = (y + 9)$. Use a table of values to graph the parabola.

43. $F(-5, -2)$; opens up; contains $(-13, -2)$

SOLUTION:

Because the parabola opens up, the vertex is $(-5, -2 - p)$. Use the standard form of the equation of a horizontal

parabola and the point $(-13, -2)$ to find the equation.

$$4p(y - k) = (x - h)^2$$

$$4p[-2 - (-2 - p)] = [-13 - (-5)]^2$$

$$4p(p) = 64$$

2

$$p = 16$$

$$p = \pm 4$$

Because the parabola opens up, the value of p must be positive. Therefore, $p = 4$. The vertex is $(-5, -6)$, and the

2

standard form of the equation is $(x + 5) = 16(y + 6)$. Use a table of values to graph the parabola.

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7-1 Parabolas

43. $F(-5, -2)$; opens up; contains $(-13, -2)$

SOLUTION:

Because the parabola opens up, the vertex is $(-5, -2 - p)$. Use the standard form of the equation of a horizontal

parabola and the point $(-13, -2)$ to find the equation.

$$4p(y - k) = (x - h)^2$$

$$4p[-2 - (-2 - p)] = [-13 - (-5)]^2$$

$$4p(p) = 64$$

2

$$p = 16$$

$$p = \pm 4$$

Because the parabola opens up, the value of p must be positive. Therefore, $p = 4$. The vertex is $(-5, -6)$, and the

2

standard form of the equation is $(x + 5) = 16(y + 6)$. Use a table of values to graph the parabola.

Write an equation for the line tangent to each parabola at each given point.

45.

SOLUTION:

The graph opens vertically. Determine the vertex and focus. is written in standard form.

Because . The vertex is $(-7, 3)$ and the focus is $(-7, 2.875)$. We need to determine d , the distance between the focus and the point of tangency, C .

This is one leg of the isosceles triangle.

Use d to find A , the endpoint of the other leg of the isosceles triangle. Since p is negative, the parabola opens down

and A will be above the focus.

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Points A and C both lie on the line tangent to the parabola. Find an equation of this line.

7-1 Parabolas

Write an equation for the line tangent to each parabola at each given point.

45.

SOLUTION:

The graph opens vertically. Determine the vertex and focus. is written in standard form.

Because . The vertex is $(-7, 3)$ and the focus is $(-7, 2.875)$. We need to determine d , the

distance between the focus and the point of tangency, C.

This is one leg of the isosceles triangle.

Use d to find A, the endpoint of the other leg of the isosceles triangle. Since p is negative, the parabola opens down

and A will be above the focus.

Points A and C both lie on the line tangent to the parabola. Find an equation of this line.

$$47. (x + 6)^2 = 3(y - 2), (0, 14)$$

SOLUTION:

2

The graph opens vertically. Determine the vertex and focus. $(x + 6) = 3(y - 2)$ is written in standard form.

Because $4p = 3$, $p = \frac{3}{4}$. The vertex is $(-6, 2)$ and the focus is $(-6, 2.75)$. We need to determine d , the distance

between the focus and the point of tangency, C.

This is one leg of the isosceles triangle.

$d =$

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$= 12.75$

7-1 Parabolas

$$47. (x + 6)^2 = 3(y - 2), (0, 14)$$

SOLUTION:

2

The graph opens vertically. Determine the vertex and focus. $(x + 6) = 3(y - 2)$ is written in standard form.

Because $4p = 3$, $p = \frac{3}{4}$. The vertex is $(-6, 2)$ and the focus is $(-6, 2.75)$. We need to determine d , the distance

between the focus and the point of tangency, C.

This is one leg of the isosceles triangle.

$d =$

$= 12.75$

Use d to find A, the endpoint of the other leg of the isosceles triangle. Since p is positive, the parabola opens up and

A will be below the focus.

$$A = (-6, 2.75 - 12.75) \text{ or } (-6, -10)$$

Points A and C both lie on the line tangent to the parabola. Find an equation of this line.

$m =$ or 4

$$y - y_1 = m(x - x_1)$$

$$y - 14 = 4(x - 0)$$

$$y - 14 = 4x$$

$$y = 4x + 14$$

$$49. -0.25(x - 6)^2 = y - 9, (10, 5)$$

SOLUTION:

The graph opens vertically. Determine the vertex and focus.

2

$$-0.25(x - 6) = y - 9$$

2

$$(x - 6) = -4(y - 9)$$

Because $4p = -4$, $p = -1$. The vertex is $(6, 9)$ and the focus is $(6, 8)$. We need to determine d , the distance

between the focus and the point of tangency, C .

This is one leg of the isosceles triangle.

$$d =$$

$$= 5$$

Use d to find A , the endpoint of the other leg of the isosceles triangle. Since p is negative, the parabola opens down

and A will be above the focus.

$$A = (6, 8 + 5) \text{ or } (6, 13)$$

Points A and C both lie on the line tangent to the parabola. Find an equation of this line.

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$$m = \text{or } -2$$

$$y - 14 = 4(x - 0)$$

$$y - 14 = 4x$$

$$y = 4x + 14$$

7-1 Parabolas

$$49. -0.25(x - 6)^2 = y - 9, (10, 5)$$

SOLUTION:

The graph opens vertically. Determine the vertex and focus.

2

$$-0.25(x - 6) = y - 9$$

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$$(x - 6) = -4(y - 9)$$

Because $4p = -4$, $p = -1$. The vertex is $(6, 9)$ and the focus is $(6, 8)$. We need to determine d , the distance

between the focus and the point of tangency, C .

This is one leg of the isosceles triangle.

$$d =$$

$$= 5$$

Use d to find A , the endpoint of the other leg of the isosceles triangle. Since p is negative, the parabola opens down

and A will be above the focus.

$$A = (6, 8 + 5) \text{ or } (6, 13)$$

Points A and C both lie on the line tangent to the parabola. Find an equation of this line.

$$m = \text{or } -2$$

$$y - y_1 = m(x - x_1)$$

$$y - 5 = -2(x - 10)$$

$$y - 5 = -2x + 20$$

$$y = -2x + 25$$

Determine the orientation of each parabola.

51. directrix $y = 4$, $p = -2$

SOLUTION:

Since the directrix is $y = 4$, the graph opens vertically. Since p is negative, the graph will open down.

53. vertex $(-5, 1)$, focus $(-5, 3)$

SOLUTION:

Because the focus and vertex share the same x -coordinate, the graph is vertical. The focus is $(h, k + p)$, so the

value of p is $3 - 1$ or 2 . Because p is positive, the graph opens up.

Write an equation for each parabola.

55.

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SOLUTION:

The focus and vertex share the same x -coordinate. The focus is $(h, k + p)$, so the value of p is $4 - 5$ or -1 .

53. vertex $(-5, 1)$, focus $(-5, 3)$

SOLUTION:

Because the focus and vertex share the same x -coordinate, the graph is vertical. The focus is $(h, k + p)$, so the

7-1 Parabolas

value of p is $3 - 1$ or 2 . Because p is positive, the graph opens up.

Write an equation for each parabola.

55.

SOLUTION:

The focus and vertex share the same x -coordinate. The focus is $(h, k + p)$, so the value of p is $4 - 5$ or -1 .

Write the equation for the parabola in standard form using the values of h , p , and k .

$$4p(y - k) = (x - h)^2$$

$$4(-1)(y - 5) = (x - 3)^2$$

$$-4(y - 5) = (x - 3)^2$$

57.

SOLUTION:

The focus is $(-9, 1)$ and the directrix is $x = -1$. The directrix is $x = h - p$ or $h - p = -1$ and the x -coordinate of the focus is $h + p = -9$.

Solve the system of equations.

$$h - p = -1$$

$$h + p = -9$$

$$2h = -10$$

$$h = -5$$

Substitute $h = -5$ into the equation for the directrix to find that p is -4 .

Write the equation for the parabola in standard form using the values of h , p , and k .

$$4p(x - h) = (y - k)^2$$

$$4(-4)[x - (-5)] = (y - 1)^2$$

$$-16(x + 5) = (y - 1)^2$$

59. BRIDGES The arch of the railroad track bridge below is in the shape of a parabola. The two main support

towers are 208 meters apart and 80 meters tall. The distance from the top of the parabola to the water below is 60 meters.

a. Write an equation that models the shape of the arch. Let the railroad track represent the x -axis.

b. Two vertical supports attached to the arch are equidistant from the center of the parabola as shown in the

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diagram. Powered by lengths

their Cognero if they are 86.4 meters apart. Page 19

$$4p(x - h) = (y - k)^2$$

$$4(-4)[x - (-5)] = (y - 1)^2$$

$$-16(x + 5) = (y - 1)^2$$

7-1 Parabolas

59. BRIDGES The arch of the railroad track bridge below is in the shape of a parabola. The two main support

towers are 208 meters apart and 80 meters tall. The distance from the top of the parabola to the water below is 60 meters.

a. Write an equation that models the shape of the arch. Let the railroad track represent the x -axis.

b. Two vertical supports attached to the arch are equidistant from the center of the parabola as shown in the

diagram. Find their lengths if they are 86.4 meters apart.

SOLUTION:

a. If the railroad track represents the x -axis and we assume that the vertex lies on the y -axis, then the vertex is at

the point $(0, -20)$. The arch meets each support tower 104 meters to the left and to the right of the vertex and 80

feet below the railroad or the x-axis. Thus, two points on the parabola are at $(-104, -80)$ and $(104, -80)$. Find p

using the vertex and a point on the parabola.

$$4p(y - k) = (x - h)^2$$

$$4p[-80 - (-20)] = (104 - 0)^2$$

$$-240p = 10816$$

$$p = -45.066$$

Write the equation for the parabola in standard form using the values of h , p , and k .

$$4p(y - k) = (x - h)^2$$

$$4(-45.066)[y - (-20)] = (x - 0)^2$$

$$-180.27(y + 20) = x^2$$

b. If the supports are 86.4 meters apart and are equidistant from the center of the parabola, then they are located

43.2 meters from the center of the parabola. Therefore, the supports meet the parabola when $x = 43.2$. Find y

when $x = 43.2$.

$$-180.27(y + 20) = x^2$$

$$-180.27(y + 20) = 43.2^2$$

$$-180.27(y + 20) = 1866.24$$

$$y + 20 = -10.35$$

$$y = -30.35$$

Therefore, the supports meet the parabola at the points $(43.2, -30.35)$ and $(-43.2, -30.35)$ and the supports are

30.35 meters in length.

Write an equation for and graph a parabola with each set of characteristics.

61. vertex $(-6, 4)$, contains $(-10, 8)$, opens horizontally

SOLUTION:

2

If the graph opens horizontally, then the standard form is $4p(x - h) = (y - k)^2$. Find p using the vertex and the point

on the parabola.

$$4p(x - h) = (y - k)^2$$

$$4p[-10 - (-6)] = (8 - 4)^2$$

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$$-16p = 16$$

$$p = -1$$

$$y + 20 = -10.35$$

$$y = -30.35$$

Therefore, the supports meet the parabola at the points $(43.2, -30.35)$ and $(-43.2, -30.35)$ and the supports are

7-1 Parabolas

30.35 meters in length.

Write an equation for and graph a parabola with each set of characteristics.

61. vertex $(-6, 4)$, contains $(-10, 8)$, opens horizontally

SOLUTION:

2

If the graph opens horizontally, then the standard form is $4p(x - h) = (y - k)^2$. Find p using the vertex and the point

on the parabola.

$$4p(x - h) = (y - k)^2$$

$$4p[-10 - (-6)] = (8 - 4)^2$$

$$-16p = 16$$

$$p = -1$$

Write the equation for the parabola in standard form using the values of h , p , and k .

$$4p(x - h) = (y - k)^2$$

$$4(-1)[x - (-6)] = (y - 4)^2$$

$$-4(x + 6) = (y - 4)^2$$

63. opens horizontally, passes through points $(-1, -1)$, $(5, 3)$, and $(15, 7)$

SOLUTION:

2

If the graph opens horizontally, then the standard form is $4p(x - h) = (y - k)^2$. Find an equation for the parabola for

each set of points.

Equation 1:

$$4p(x - h) = (y - k)^2$$

$$4p(-1 - h) = (-1 - k)^2$$

$$-4p - 4hp = 1 + 2k + k^2$$

Equation 2:

$$4p(x - h) = (y - k)^2$$

$$4p(5 - h) = (3 - k)^2$$

$$20p - 4hp = 9 - 6k + k^2$$

Equation 3:

$$4p(x - h) = (y - k)^2$$

$$4p(15 - h) = (7 - k)^2$$

$$60p - 4hp = 49 - 14k + k^2$$

Solve this system of equations. Subtract Equation 2 from Equation 1.

2

$$-4p - 4hp$$

$$eSolutions Manual - Powered = 1 + 2k + k^2$$

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$$20p - 4hp = 9 - 6k + k^2$$

$$-24p = -8 + 8k$$

$$4p(x - h) = (y - k)^2$$

$$4p(15 - h) = (7 - k)^2$$

$$60p - 4hp = 49 - 14k + k^2$$

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Solve this system of equations. Subtract Equation 2 from Equation 1.

$$-4p - 4hp = 1 + 2k + k^2$$

$$20p - 4hp = 9 - 6k + k^2$$

$$-24p = -8 + 8k$$

Subtract Equation 2 from Equation 3.

$$60p - 4hp = 49 - 14k + k^2$$

$$20p - 4hp = 9 - 6k + k^2$$

$$40p = 40 - 8k$$

Solve the system consisting of the two new equations.

$$-24p = -8 + 8k$$

$$40p = 40 - 8k$$

$$16p = 32$$

$$p = 2$$

Substitute $p = 2$ into one of the equations and solve for k .

$$40p = 40 - 8k$$

$$40(2) = 40 - 8k$$

$$80 = 40 - 8k$$

$$40 = -8k$$

$$-5 = k$$

Substitute $p = 2$ and $k = -5$ into one of the original equations and solve for h .

$$-4p - 4hp = 1 + 2k + k^2$$

$$-4(2) - 4(2)h = 1 + 2(-5) + (-5)^2$$

$$-8 - 8h = 16$$

$$-8h = 24$$

$$h = -3$$

Write the equation for the parabola in standard form using the values of h , p , and k .

$$4p(x - h) = (y - k)^2$$

$$4(2)[x - (-3)] = [y - (-5)]^2$$

$$8(x + 3) = (y + 5)^2$$

Determine the point of tangency for each equation and line.

$$65. (x + 2)^2 = 2y$$

$$y = 4x$$

SOLUTION:

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The point of tangency occurs at the solution for the system of equations.

Use substitution to solve the system.

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Determine the point of tangency for each equation and line.

$$65. (x + 2)^2 = 2y$$

$$y = 4x$$

SOLUTION:

The point of tangency occurs at the solution for the system of equations.

Use substitution to solve the system.

$$2$$

$$(x + 2)^2 = 2y$$

$$2$$

$$= 2(4x)$$

$$(x + 2)^2$$

$$2$$

$$x + 4x + 4 = 8x$$

$$2$$

$$x - 4x + 4 = 0$$

$$2$$

$$(x - 2) = 0$$

The solution to the system occurs when $x = 2$. To find the y-coordinate, substitute $x = 2$ in one of the original equations.

$$y = 4x$$

$$y = 4(2)$$

$$y = 8$$

The point of tangency occurs at $(2, 8)$.

$$67. (y + 3)^2 = -x + 4$$

SOLUTION:

The point of tangency occurs at the solution for the system of equations.

Use substitution to solve the system.

$$2$$

$$(y + 3)^2 = -x + 4$$

$$= -x + 4$$

$$= -x + 4$$

$$= -x + 4$$

$$= 0$$

The solution to the system occurs when $x = 0$. To find the y-coordinate, substitute $x = 0$ into one of the original equations.

$$y = -1$$

The point of tangency occurs at $(0, -1)$.

69. ILLUMINATION In a searchlight, the bulb is placed at the focus of a parabolic mirror 1.5 feet from the vertex.

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This causes the light rays from the bulb to bounce off the mirror as parallel rays, thus providing a concentrated beam of light.

$$y = -1$$

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The point of tangency occurs at $(0, -1)$.

69. ILLUMINATION In a searchlight, the bulb is placed at the focus of a parabolic mirror 1.5 feet from the vertex.

This causes the light rays from the bulb to bounce off the mirror as parallel rays, thus providing a concentrated beam of light.

a. Write an equation for the parabola if the focal diameter of the bulb is 2 feet, as shown in the diagram.

b. Suppose the focal diameter is increased to 3 feet. If the depth of both searchlights is 3.5 feet, how much greater is the width of the opening of the larger light? Round to the nearest hundredth.

SOLUTION:

a. Sketch a coordinate plane on top of the searchlight with the vertex at the origin. The focus is 1.5 feet from the

vertex located at the point $(1.5, 0)$. Since the focal diameter of the bulb is 2 feet, 1 foot will lie above and below the

x-axis. Therefore, the points $(1.5, 1)$ and $(1.5, -1)$ lie on the parabola. Substitute the values of x , y , h , and k to

solve for p .

$$4p(x - h) = (y - k)^2$$

$$4p(1.5 - 0) = (1 - 0)^2$$

$$6p = 1$$

$$p =$$

Substitute values for h , k , and p to write an equation to model the reflector.

$$4p(x - h) = (y - k)^2$$

$$4(x - 0) = (y - 0)^2$$

$$2$$

$$x = y$$

b. Write an equation in standard form for a parabola with a focal diameter of 3 feet. Sketch a coordinate plane on

top of the new searchlight with the vertex at the origin. The focus is still 1.5 feet from the vertex located at the

point $(1.5, 0)$. Since the focal diameter of the bulb is 3 feet, 1.5 feet will lie above and below the x-axis. Therefore,

the points (1.5, 1.5) and (1.5, -1.5) lie on the parabola. Substitute the values of x, y, h, and k to solve for p .

$$4p(x - h) = (y - k)^2$$

$$4p(1.5 - 0) = (1.5 - 0)^2$$

$$6p = 2.25$$

$$p = 0.375$$

Substitute values for h, k, and p to write an equation to model the reflector.

$$4p(x - h) = (y - k)^2$$

$$4(0.375)(x - 0) = (y - 0)^2$$

$$1.5x = y^2$$

The depth of both searchlights is 3.5 feet. The depth represents the distance the reflector spans horizontally and the

width

eSolutions represents

Manual - Powered the distance the reflector spans vertically. To find the width of the first searchlight, substitute

by Cognero $x = 24$

3.5 into the equation found in part a and solve for y.

$$2$$

$$x = y$$

$$4p(x - h) = (y - k)^2$$

$$4(0.375)(x - 0) = (y - 0)^2$$

$$1.5x = y^2$$

7-1 Parabolas

The depth of both searchlights is 3.5 feet. The depth represents the distance the reflector spans horizontally and the

width represents the distance the reflector spans vertically. To find the width of the first searchlight, substitute $x =$

3.5 into the equation found in part a and solve for y.

$$2$$

$$x = y$$

$$(3.5) = y^2$$

$$2$$

$$= y$$

$$\pm = y$$

$$\pm \text{ or } \pm 1.53 = y$$

At a depth of 3.5 feet, the reflector will span 1.53 feet above and below the x-axis for a total width of $2(1.53)$ or

3.06 feet. Repeat the same process to find the width of the second reflector.

$$1.5x = y^2$$

$$1.5(3.5) = y^2$$

$$5.25 = y^2$$

$$\pm \text{ or } 2.29 = y$$

At a depth of 3.5 feet, the reflector will span 2.29 feet above and below the x-axis for a total width of $2(2.29)$ or

4.58 feet. The difference between the two widths is $4.58 - 3.06$ or 1.52 feet.

71. The latus rectum of a parabola is the line segment that passes through the focus, is perpendicular to the axis of

the parabola, and has endpoints on the parabola. The length of the latus rectum is $|4p|$ units, where p is the distance

from the vertex to the focus.

a. Write an equation for the parabola with vertex at $(-3, 2)$, axis $y = 2$, and latus rectum 8 units long.

b. Prove that the endpoints of the latus rectum and point of intersection of the axis and directrix are the vertices of

a right isosceles triangle.

SOLUTION:

a. Since the parabola has an axis of $y = 2$, the parabola will open horizontally and will have the standard form of $(y - k)^2 = 4p(x - h)$. The latus rectum is 8 units long. This can be used to solve for p .

$$|4p| = 8, \text{ so } p = \pm 2.$$

Substitute values for h , k , and p to write an equation for the parabola.

$$4p(x - h) = (y - k)^2$$

$4(\pm 2)[x - (-3)] = (y - 2)^2$

$$\pm 8(x + 3) = (y - 2)^2$$

$$\pm 8(x + 3) = (y - 2)^2$$

$$\pm 8(x + 3) = (y - 2)^2$$

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b.

$$5.25 = y^2$$

$$\pm \text{ or } 2.29 = y$$

At a depth of 3.5 feet, the reflector will span 2.29 feet above and below the x-axis for a total width of $2(2.29)$ or

4.58 feet. The difference between the two widths is $4.58 - 3.06$ or 1.52 feet.

71. The latus rectum of a parabola is the line segment that passes through the focus, is perpendicular to the axis of

the parabola, and has endpoints on the parabola. The length of the latus rectum is $|4p|$ units, where p is the distance

from the vertex to the focus.

a. Write an equation for the parabola with vertex at $(-3, 2)$, axis $y = 2$, and latus rectum 8 units long.

b. Prove that the endpoints of the latus rectum and point of intersection of the axis and directrix are the vertices of

a right isosceles triangle.

a right isosceles triangle.

SOLUTION:

a. Since the parabola has an axis of $y = 2$, the parabola will open horizontally and will have the standard form of $(y - k)^2 = 4p(x - h)$. The latus rectum is 8 units long. This can be used to solve for p .

2

$-k) = 4p(x - h)$. The latus rectum is 8 units long. This can be used to solve for p .

$|4p| = 8$, so $p = \pm 2$.

Substitute values for h , k , and p to write an equation for the parabola.

$$4p(x - h) = (y - k)^2$$

$$4(\pm 2)[x - (-3)] = (y - 2)^2$$

$$\pm 8(x + 3) = (y - 2)^2$$

b.

To prove that the endpoints of the latus rectum X and W and the point of intersection of the axis and directrix D are

the vertices of a right isosceles triangle $\triangle XDW$, we need to show that $\angle XDW$ is a right angle and that

.

Since $\angle XDF$ and $\angle FWD$ are both right angles, $\triangle XDF \cong \triangle FWD$ by SAS. Thus, $\angle XDF \cong \angle FWD$. To prove that

$\angle XDW$ is a right angle, we can first prove that $\angle XDF$ and $\angle FDW$ are both right angles. Since $\angle XDF$ is a right angle, $\triangle XDF$ is an isosceles right triangle. Therefore, $\angle XDF$ is 45° . This also means that

$\angle FDW$ is 45° . Resultantly, $\angle XDW$ is a right angle. Thus, $\triangle XDW$ is an isosceles right triangle.

Write a possible equation for a parabola with focus F such that the line given is tangent to the parabola.

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$\angle XDW$ is a right angle, we can first prove that $\angle XDF$ and $\angle FDW$ are both right angles. Since $\angle XDF$ is a right angle, $\triangle XDF$ is an isosceles right triangle. Therefore, $\angle XDF$ is 45° . This also means that $\angle FDW$ is

45° . Resultantly, $\angle XDW$ is a right angle. Thus, $\triangle XDW$ is an isosceles right triangle.

Write a possible equation for a parabola with focus F such that the line given is tangent to the parabola.

73.

SOLUTION:

Find point A , the point of intersection of the axis and the line tangent to the parabola.

The equation for the tangent line is $y = 2x - 2$. Since point A lies on the y -axis, the x -coordinate is 0. Substitute $x =$

0 to find y .

$$y = 2x - 2$$

$$y = 2(0) - 2$$

$$y = -2$$

A is at $(0, -2)$. The segment formed by FA represents one of the legs of the isosceles triangle formed by F , A , and

the point of tangency. Find the length of this leg.

$$d = 5$$

The lengths of the legs of the isosceles triangle are 5 units. This is the distance from F to the point of tangency (x,

y). Use the distance formula and the tangent line to solve for the point of tangency.

$$d$$

$$5$$

$$25 = x^2 + (y - 3)^2$$

$$25 = x^2 + y^2 - 6y + 9$$

$$25 = x^2 + (2x - 2)^2 - 6(2x - 2) + 9$$

$$25 = x^2 + 4x^2 - 8x + 4 - 12x + 12 + 9$$

$$25 = 5x^2 - 20x + 25$$

$$0 = 5x^2 - 20x$$

$$0 = 5x(x - 4)$$

The solutions to this equation are $x = 0$ and 4. Since A has an x-coordinate of 0, the point of tangency must have

an x-coordinate of 4. Substitute $x = 4$ to solve for y.

$$y = 2x - 2$$

$$y = 2(4) - 2$$

$$y = 6$$

The point of tangency is located at (4, 6). Since the parabola opens upward, the standard form of the equation is (x

$$2$$

$-h) = 4p(y - k)$ and the focus is located at $(h, k + p)$. The focus is at the point (0, 3); thus, $h = 0$ and $k + p = 3$ or

$k = 3 - p$. Use, h, the standard form of the equation, the point of tangency, and the equation for k to solve for p.

$$2$$

$$(x - h) = 4p(y - k)$$

$$2$$

$$(4 - 0) = 4p(6 - k)$$

$$16 = 24p$$

$$eSolutions Manual - Powered - 4p k$$

$$\text{by Cognero Page 27}$$

$$16 = 24p - 4p(3 - p)$$

$$16 = 24p - 12p + 4p^2$$

The point of tangency is located at (4, 6). Since the parabola opens upward, the standard form of the equation is (x

$$2$$

$-h) = 4p(y - k)$ and the focus is located at $(h, k + p)$. The focus is at the point (0, 3); thus, $h = 0$ and $k + p = 3$ or

$k = 3 - p$. Use, h, the standard form of the equation, the point of tangency, and the equation for k to solve for p.

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$$(x - h) = 4p (y - k)$$

2

$$(4 - 0) = 4p (6 - k)$$

$$16 = 24p - 4p k$$

$$16 = 24p - 4p (3 - p)$$

$$16 = 24p - 12p + 4p^2$$

$$0 = 4p^2 + 12p - 16$$

$$0 = 4(p + 4)(p - 1)$$

The solutions to this equation are $p = -4$ and 1 . Since the parabola opens upward, p is positive; thus, $p = 1$. Use p

to find k . $k = 3 - (1)$ or 2 . The vertex of the parabola is $(0, 2)$. Use the vertex and p to write the standard form of

the equation.

2

$$(x - h) = 4p (y - k)$$

2

$$(x - 0) = 4(1)(y - 2)$$

$$2 = 4(y - 2)$$

x

75.

SOLUTION:

Find point A, the point of intersection of the axis and the line tangent to the parabola.

The equation for the tangent line is $y = x + 6$. Since point A lies on the x-axis, the y-coordinate is 0. Substitute y

$= 0$ to find x .

$$y = x + 6$$

$$0 = x + 6$$

$$-6 = x$$

$$-12 = x$$

A is at $(-12, 0)$. The segment formed by FA represents one of the legs of the isosceles triangle formed by F, A, and

the point of tangency. Find the length of this leg.

d=

d=

$$d = 15$$

The lengths of the legs of the isosceles triangle are 15 units. This is the distance from F to the point of tangency (x,

y). Use the distance formula and the tangent line to solve for the point of tangency.

d =

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$$225 = (x - 3)^2 + y^2$$

$$225 = x^2 - 6x + 9 + y^2$$

$$d = 15$$

The lengths of the legs of the isosceles triangle are 15 units. This is the distance from F to the point of tangency (x,

7-1 Parabolas

y). Use the distance formula and the tangent line to solve for the point of tangency.

$$d =$$

$$15 =$$

$$225 = (x - 3)^2 + y^2$$

$$225 = x^2 - 6x + 9 + y^2$$

$$225 = x^2 - 6x + 9 +$$

$$225 = x^2 - 6x + 9 + 2$$

$$x^2 + 6x + 36$$

$$2$$

$$225 = x^2 + 45$$

$$2$$

$$180 = x^2$$

$$144 = x^2$$

$$\pm 12 = x$$

The solutions to this equation are $x = \pm 12$. Since A has an x-coordinate of -12 , the point of tangency must have an

x-coordinate of 12. Substitute $x = 12$ to solve for y.

$$y = x + 6$$

$$y = (12) + 6$$

$$y = 12$$

The point of tangency is located at (12, 12). Since the parabola opens to the right, the standard form of the equation

$$4p(x - h) = (y - k)^2$$

is $4p(x - h) = (y - k)^2$ and the focus is located at $(h + p, k)$. The focus is at the point (3, 0); thus, $k = 0$ and $h + p = 3$ or $h = 3 - p$. Use, k, the standard form of the equation, the point of tangency, and the

equation for h to solve

for p.

$$4p(x - h) = (y - k)^2$$

$$4p(12 - h) = (12 - 0)^2$$

$$48p - 4ph = 144$$

$$48p - 4p(3 - p) = 144$$

$$2p = 144$$

$$48p - 12p + 4p$$

2

$$4p + 36p - 144 = 0$$

$$4(p + 12)(p - 3) = 0$$

The solutions to this equation are $p = -12$ and 3 . Since the parabola opens to the right, p is positive; thus, $p = 3$.

Use p to find h . $h = 3 - (3)$ or 0 . The vertex of the parabola is $(0, 0)$. Use the vertex and p to write the standard

form of the equation.

$$4p(x - h) = (y - k)^2$$

$$4(3)(x - 0) = (y - 0)^2$$

$$12x = y^2$$

77. MULTIPLE REPRESENTATIONS In this problem, you will investigate how the shape of a parabola changes

as the position of the focus changes.

a. GEOMETRIC Find the distance between the vertex and the focus of each parabola.

2

i. yManual

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= 4(x- Powered Page 29

2

ii. $y = 8(x - 2)$

iii. $y^2 = 16(x - 2)$

=y

77. MULTIPLE REPRESENTATIONS In this problem, you will investigate how the shape of a parabola changes

7-1 Parabolas

as the position of the focus changes.

a. GEOMETRIC Find the distance between the vertex and the focus of each parabola.

i. $y^2 = 4(x - 2)$

2

ii. $y = 8(x - 2)$

iii. $y^2 = 16(x - 2)$

b. GRAPHICAL Graph the parabolas in part a using a different color for each. Label each focus.

c. VERBAL Describe the relationship between a parabola's shape and the distance between its vertex and focus.

2

d. ANALYTICAL Write an equation for a parabola that has the same vertex as $(x + 1) = 20(y + 7)$, but is

narrower.

e. ANALYTICAL Make a conjecture about the graphs of $x^2 = -2(y + 1)$, $x^2 = -12(y + 1)$, and $x^2 = -5(y + 1)$.

Check your conjecture by graphing the parabolas.

SOLUTION:

a.

2 2

i. Written in standard form, $(y - k) = 4p(x - h)$, p is the distance between the vertex and the focus.

$$y = 4(x - 2)$$

2

can be written as $y = 4(1)(x - 2)$, where $p = 1$. Thus, the distance between the vertex and the focus is 1 unit.

2 2

ii. Written in standard form, $(y - k) = 4p(x - h)$, p is the distance between the vertex and the focus. $y = 8(x - 2)$

2

can be written as $y = 4(2)(x - 2)$, where $p = 2$. Thus, the distance between the vertex and the focus is 2 units.

2 2

iii. Written in standard form, $(y - k) = 4p(x - h)$, p is the distance between the vertex and the focus. $y = 16(x - 2)$

2

can be written as $y = 4(4)(x - 2)$, where $p = 4$. Thus, the distance between the vertex and the focus is 4 units.

b. Use a table of values to graph each parabola.

c. As the focus is moved farther away from the vertex, the parabola becomes wider.

2 2

d. The parabola $(x + 1) = 20(y + 7)$ can be written as $(x + 1) = (4)(5)(y + 7)$, where $p = 5$ and the vertex is $(-1,$

$-7)$. A parabola that is more narrow has a smaller value for p . Let $p = 1$ and write the standard form for the

equation of a parabola with the same vertex.

2

$$(x - h) = 4p(y - k)$$

2

$$[x - (-1)] = 4(1)[y - (-7)]$$

2

$$(x + 1) = 4(y + 7)$$

e. Sample answer: The parabolas all have a vertex of $(0, -1)$ and open downward. The first equation produces the

most narrow parabola and the second equation produces the widest parabola.

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2

$$[x - (-1)] = 4(1)[y - (-7)]$$

2

$$(x + 1) = 4(y + 7)$$

7-1 Parabolas

e. Sample answer: The parabolas all have a vertex of $(0, -1)$ and open downward. The first equation produces the

most narrow parabola and the second equation produces the widest parabola.

79. CHALLENGE The area of the parabolic sector shaded in the graph below is given by

$A = xy$. Find the equation of the parabola if the sector area is 2.4 square units and the width of the sector is 3

units.

SOLUTION:

The width of the sector is 3 units, so $y = 1.5$. $A = xy$ and the area of the sector is 2.4 square units. Substitute the

values into the equation and solve for x .

$$2.4 = x(1.5)$$

$$2.4 = 2x$$

$$1.2 = x$$

Therefore, the vertex of the parabola is $(0, 0)$, and two other points on the parabola are $(1.2, 1.5)$ and $(1.2, -1.5)$.

Set up and solve a system of equations using y as the independent variable.

2

$$ay + by + c = x$$

2

$$a(0) + b(0) + c = 0$$

$$2 = 1.2$$

$$a(1.5) + b(1.5) + c$$

$$2 = 1.2$$

$$a(-1.5) + b(-1.5) + c$$

$$c = 0$$

$$2.25a + 1.5b = 1.2$$

$$2.25a - 1.5b = 1.2$$

$$4.5a = 2.4$$

$$a =$$

$$2.25 \cdot + 1.5b = 1.2$$

$$1.2 + 1.5b = 1.2$$

$$1.5b = 0$$

$$b = 0$$

2 2

The

eSolutions equation

Manual of the

- Powered by parabola

Cognero is $y = x$ or $y = -x$. Page 31

81. REASONING Without graphing, determine in which quadrants the graph of $(y - 5)^2 = -8(x + 2)$ will have no

7-1 Parabolas

79. CHALLENGE The area of the parabolic sector shaded in the graph below is given by

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$$2.4 = x(1.5)$$

$$2.4 = 1.5x$$

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2

$$ay + by + c = x$$

2

$$a(0) + b(0) + c = 0$$

$$2 = 1.2$$

$$a(1.5) + b(1.5) + c$$

$$2 = 1.2$$

$$a(-1.5) + b(-1.5) + c$$

$$c = 0$$

$$2.25a + 1.5b = 1.2$$

$$2.25a - 1.5b = 1.2$$

$$4.5a = 2.4$$

$$a =$$

$$2.25 \cdot + 1.5b = 1.2$$

$$1.2 + 1.5b = 1.2$$

$$1.5b = 0$$

$$b = 0$$

2 2

The equation of the parabola is $y = x$ or $y = -x$.

81. REASONING Without graphing, determine in which quadrants the graph of $(y - 5)^2 = -8(x + 2)$ will have no

points. Explain your reasoning.

SOLUTION:

Quadrants I and IV; the vertex is $(-2, 5)$ and $p = -2$. Since the vertex is to the left of the y -axis, and the parabola

opens to the left, no points will be to the right of the y -axis, or in Quadrants I and IV.

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83. PREWRITE Write a letter outlining and explaining the content you have learned from this lesson. Make an

outline that addresses purpose, audience, a controlling idea, logical sequence, and time frame for completion.

$$1.2 + 1.5b = 1.2$$

$$1.5b = 0$$

$$b = 0$$

7-1 Parabolas

The equation of the parabola is

$$y = x^2$$

$$y = x^2 \text{ or } y = -x^2.$$

81. REASONING Without graphing, determine in which quadrants the graph of $(y - 5)^2 = -8(x + 2)$ will have no points. Explain your reasoning.

SOLUTION:

Quadrants I and IV; the vertex is $(-2, 5)$ and $p = -2$. Since the vertex is to the left of the y -axis, and the parabola

opens to the left, no points will be to the right of the y -axis, or in Quadrants I and IV.

83. PREWRITE Write a letter outlining and explaining the content you have learned from this lesson. Make an

outline that addresses purpose, audience, a controlling idea, logical sequence, and time frame for completion.

SOLUTION:

Sample answer:

Purpose: To outline and explain the process of graphing parabolas.

Audience: Parents of PreCalculus students.

Controlling idea: Using characteristics of parabolas to graph them.

Logical sequence:

Find the vertex of the parabola.

Find the focus of the parabola.

Find the directrix of the parabola.

Find the axis of symmetry of the parabola.

Graph those characteristics.

Make a table of values and graph the general shape of the curve.

Timeframe: You have one day to complete the outline.

Find the maximum and minimum values of the objective function $f(x, y)$ and for what values of x and y

they occur, subject to the given constraints.

$$85. y \geq -x + 2$$

$$2 \leq x \leq 7$$

$$y \leq x + 5$$

$$f(x, y) = 8x + 3y$$

SOLUTION:

Begin by graphing the given system of three inequalities. The solution of the system, which makes up the set of

feasible solutions for the objective function, is the shaded region, including its boundary segments.

The polygonal region of feasible solutions has four vertices.

Solve the four systems of two boundary equations below to find the coordinates of the remaining vertices.

System of Boundary $y = x + 5$ $y = x + 5$

Equations

eSolutions Manual - Powered by Cognero $x = 2$ $x = 7$ Page 33

System of Boundary $y = -x + 2$ $y = -x + 2$

Equations $x = 2$ $x = 7$

The polygonal region of feasible solutions has four vertices.

7-1 Parabolas

Solve the four systems of two boundary equations below to find the coordinates of the remaining vertices.

System of Boundary $y = x + 5$ $y = x + 5$

Equations

$$x = 2 \quad x = 7$$

System of Boundary $y = -x + 2$ $y = -x + 2$

Equations $x = 2$ $x = 7$

To solve the first system, substitute $x = 2$ into the boundary equation and solve for y .

$$y = x + 5$$

$$y = (2) + 5$$

$$y = 6$$

A vertex is located at (2, 6).

To solve the second system, substitute $x = 7$ into the boundary equation and solve for y .

$$y = x + 5$$

$$y = (7) + 5$$

$$y = 8.5$$

A vertex is located at (7, 8.5).

To solve the third system, substitute $x = 2$ into the boundary equation and solve for y .

$$y = -x + 2$$

$$y = -(2) + 2$$

$$y = 0$$

A vertex is located at (2, 0).

To solve the fourth system, substitute $x = 7$ into the boundary equation and solve for y .

$$y = -x + 2$$

$$y = -(7) + 2$$

$$y = -5$$

A vertex is located at $(7, -5)$.

Find the value of the objective function $f(x, y) = 8x + 3y$ at each of the four vertices.

$$f(2, 6) = 8(2) + 3(6) \text{ or } 34$$

$$f(7, 8.5) = 8(7) + 3(8.5) \text{ or } 81.5$$

$$f(2, 0) = 8(2) + 3(0) \text{ or } 16$$

$$f(7, -5) = 8(7) + 3(-5) \text{ or } 41$$

So, the maximum value of f is 81.5 when $x = 7$ and $y = 8.5$. The minimum value of f is 16 when $x = 2$ and $y = 0$.

87. Find the partial fraction decomposition of .

SOLUTION:

Rewrite the equation as partial fractions with constant numerators, A and B , and denominators that are the linear

factors of the original denominator.

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$$= +$$

$$f(2, 0) = 8(2) + 3(0) \text{ or } 16$$

$$f(7, -5) = 8(7) + 3(-5) \text{ or } 41$$

7-1 Parabolas

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87. Find the partial fraction decomposition of .

SOLUTION:

Rewrite the equation as partial fractions with constant numerators, A and B , and denominators that are the linear

factors of the original denominator.

$$= +$$

$$2y + 5 = A(y + 1) + B(y + 2)$$

$$2y + 5 = Ay + A + By + 2B$$

$$2y + 5 = (A + B)y + (A + 2B)$$

Equate the coefficients on the left and right side of the equation to obtain a system of two equations.

$$A + B = 2$$

$$A + 2B = 5$$

To solve the system, let $A = 2 - B$ and use substitution.

$$(2 - B) + 2B = 5$$

$$2 + B = 5$$

$$B = 3$$

Thus, $A = 2 - (3)$ or -1 . So, the partial fraction decomposition of .

Find the value of each expression using the given information.

$$89. \cot \theta \text{ and } \csc \theta; \tan \theta = , \sec \theta > 0$$

SOLUTION:

Use the reciprocal identity to find $\cot \theta$.

$$\cot \theta =$$

$$\cot \theta =$$

$$\cot \theta =$$

Use the Pythagorean identity and $\cot \theta$ to find $\csc \theta$.

$$2^2$$

$$\cot^2 \theta + 1 = \csc^2 \theta$$

$$+ 1 = \csc^2 \theta$$

$$2$$

$$+ = \csc^2 \theta$$

$$2$$

$$= \csc^2 \theta$$

$$\pm = \csc \theta$$

$$\text{eSolutions Manual} \pm - \text{Powered by } \csc \theta$$

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Since $\sec \theta > 0$ and $\cot$ is positive, $\csc > 0$. Thus, $\csc =$.

$$(2 - B) + 2B = 5$$

$$2 + B = 5$$

$$B = 3$$

7-1 Parabolas

Thus, $A = 2 - (3)$ or -1 . So, the partial fraction decomposition of .

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$$2$$

$$+ = \csc \theta$$

$$2$$

$$= \csc \theta$$

$$\pm = \csc \theta$$

$$\pm = \csc \theta$$

Since $\sec \theta > 0$ and $\cot$ is positive, $\csc > 0$. Thus, $\csc =$.

Locate the vertical asymptotes, and sketch the graph of each function.

$$91. y = \tan x + 4$$

SOLUTION:

The graph of $y = \tan \theta + 4$ is the graph of $y = \tan x$ shifted 4 units up. The period is or π . Find the location of

two consecutive vertical asymptotes.

and

Create a table listing the coordinates of key points for $y = \tan \theta + 4$ for one period on .

$$y = \tan x +$$

Function $y = \tan x$

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Vertical

Asymptote

$$\pm = \csc \theta$$

Since $\sec \theta > 0$ and $\cot$ is positive, $\csc > 0$. Thus, $\csc =$

7-1 Parabolas .

Locate the vertical asymptotes, and sketch the graph of each function.

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and

Create a table listing the coordinates of key points for $y = \tan \theta + 4$ for one period on .

$$y = \tan x +$$

Function $y = \tan x$

$$4$$

Vertical

Asymptote

Intermediate

Point

x-intercept (0, 0) (0, 4)

Intermediate

Point

Vertical

Asymptote

Sketch the curve through the indicated key points for the function. Then repeat the pattern to sketch more to the

left and right of the first curve.

$$93. y = \csc x -$$

SOLUTION:

The graph of $y = \csc x -$ is the graph of $y = \csc x$ shifted unit down. The period is or π . Find the location of two consecutive vertical asymptotes.

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7-1 Parabolas

$$93. y = \csc x -$$

SOLUTION:

The graph of $y = \csc x -$ is the graph of $y = \csc x$ shifted unit down. The period is or π . Find the location of two consecutive vertical asymptotes.

and

Create a table listing the coordinates of key points for $y = \csc x -$ for one period on .

$$y = \csc x -$$

Function $y = \csc x$

Vertical

Asymptote $x = -\pi$ $x = -\pi$

Intermediate

Point

Vertical

Asymptote $x = 0$

Intermediate

Point

Vertical

$x = \pi$ $x = \pi$

Asymptote

Sketch the curve through the indicated key points for the function. Then repeat the pattern to sketch more to the

left and right of the first curve.

Evaluate each expression.

$$95. \log_4 16x$$

SOLUTION:

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2 x Page 38

$$\log_4 16 = \log_4 (4^2)$$

$$2x$$

$$= \log_4 4$$

7-1 Parabolas

Evaluate each expression.

$$95. \log_4 16x$$

SOLUTION:

$$x \cdot 2x$$

$$\log_4 16 = \log_4 (4^2)$$

$$2x$$

$$= \log_4 4$$

$$= 2x$$

Graph each function.

$$97. f(x) = x^3 - x^2 - 4x + 4$$

SOLUTION:

Use a table of values to graph the function.

$$x \quad f(x)$$

$$-3 \quad -20$$

$$-2 \quad 0$$

$$-1 \quad 6$$

$$0 \quad 4$$

$$1 \quad 0$$

$$2 \quad 0$$

$$3 \quad 10$$

$$99. h(x) = x^4 - 4x^2 + 2$$

SOLUTION:

Use a table of values to graph the function.

$$x \quad f(x)$$

$$-3 \quad 47$$

$$-2 \quad 2$$

$$-1 \quad -1$$

$$0 \quad 2$$

$$1 \quad -1$$

$$2 \quad 2$$

$$3 \quad 47$$

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7-1 Parabolas

99. $h(x) = x^4 - 4x^2 + 2$

SOLUTION:

Use a table of values to graph the function.

x f(x)

−3 47

−2 2

−1 −1

0 2

1 −1

2 2

3 47

101. SAT/ACT If x is a positive number, then

F

G

H

J

SOLUTION:

103. REVIEW What are the x-intercepts of the graph of $y = -2x^2 - 5x + 12$?

F

G −4,

H −2,

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J

7-1 Parabolas

103. REVIEW What are the x-intercepts of the graph of $y = -2x^2 - 5x + 12$?

F

G −4,

H −2,

J

SOLUTION:

2

The x-intercepts of a graph occur at the zeros of the function. Factor to solve $0 = -2x^2 - 5x + 12$ for x.

2

$$0 = -2x^2 - 5x + 12$$

$$0 = (-2x + 3)(x + 4)$$

The solutions to this equation occur when either of the factors is equal to 0.

$$-2x + 3 = 0$$

$$-2x = -3$$

$$x =$$

$$x + 4 = 0$$

$$x = -4$$

The x-intercepts of the graph of y are -4 and $.$

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